Notes on 2D Conformal Field Theory (2024.10.13-)

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Chapter 1

The plane

The subject of this chapter is local. It derives the conformal algebra, which in two dimensions is infinite-dimensional, introduces primary fields by their behaviour under holomorphic maps, and shows that a traceless stress tensor splits into holomorphic and antiholomorphic halves whose modes generate the Virasoro algebra. Nothing here requires the surface to be compact; that is the subject of Chapter 2.

1.1 Conformal algebra and conformal group

1.1.1 Conformal transformation

A *conformal map* or *conformal transformation* is an angle-preserving local diffeomorphism between two Riemannian manifolds (X, g^X) and (Y, g^Y). In particular, when $(X, g^X) = (Y, g^Y)$, the subset of conformal transformations that are global diffeomorphisms forms a group called *conformal group*. On the other hand, the set of infinitesimal conformal transformations (which are not necessarily global diffeomorphisms), equipped with commutators, is called *conformal algebra*.

The angle-preserving property of conformal map can be stated in many ways, just as is the case for rigid transformations.

"Active" and "Passive" conformal transformation

If we have a conformal map $\varphi : (X, g^X) \to (Y, g^Y)$, we can interpret it in two ways

$$(\varphi^* g^Y)_{\mu\nu}(x) = \Omega(x) g^X_{\mu\nu}(x), \quad g^Y \circ \varphi = g^X \quad \text{(active)}, \tag{1.1a}$$

$$(g^Y_{\mu\nu} \circ \varphi)(x) = \Lambda(x) g^X_{\mu\nu}(x), \quad \varphi^* g^Y = g^X \quad \text{(passive)}. \tag{1.1b}$$

The conformal Killing equation

Infinitesimally, a conformal transformation of flat space $(\mathbb{R}^d, \eta)$ is a vector field ϵ^μ whose flow changes the metric only by a local rescaling. Since $\delta g_{\mu\nu} = -(\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu)$ to first order, the

condition is that this be proportional to $\eta_{\mu\nu}$; taking the trace fixes the factor and yields the conformal Killing equation

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{d}\,(\partial\cdot\epsilon)\,\eta_{\mu\nu}. \tag{1.2}$$

The dimension d enters only through the factor $2/d$, but the consequences are drastically different for $d = 2$ and $d \geq 3$.

Lemma 1.1.1. *Any solution of (1.2) satisfies*

$$(d-2)\,\partial_\mu\partial_\nu(\partial\cdot\epsilon) + \eta_{\mu\nu}\,\Box(\partial\cdot\epsilon) = 0. \tag{1.3}$$

Proof. Differentiating (1.2) and taking the combination $\partial_\mu(\cdot)_{\nu\rho}+\partial_\nu(\cdot)_{\mu\rho}-\partial_\rho(\cdot)_{\mu\nu}$ of the resulting identities gives

$$2\,\partial_\mu\partial_\nu\epsilon_\rho = \frac{2}{d}\big(-\eta_{\mu\nu}\partial_\rho + \eta_{\nu\rho}\partial_\mu + \eta_{\rho\mu}\partial_\nu\big)(\partial\cdot\epsilon). \tag{1.4}$$

Applying ∂^ρ and rearranging yields (1.3). □

Proposition 1.1.2. *For $d \geq 3$ the solutions of (1.2) form a Lie algebra of dimension $\frac{1}{2}(d+1)(d+2)$, isomorphic to $\mathfrak{so}(d+1,1)$ and spanned by*

$$\epsilon^\mu = a^\mu \qquad (\textit{translations},\ d), \tag{1.5}$$

$$\epsilon^\mu = \omega^\mu{}_\nu x^\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu} \qquad (\textit{rotations},\ \tfrac{1}{2}d(d-1)), \tag{1.6}$$

$$\epsilon^\mu = \lambda x^\mu \qquad (\textit{dilatation},\ 1), \tag{1.7}$$

$$\epsilon^\mu = 2(b\cdot x)x^\mu - b^\mu x^2 \qquad (\textit{special conformal},\ d). \tag{1.8}$$

Proof. Contracting (1.3) with $\eta^{\mu\nu}$ gives $(2d-2)\Box(\partial\cdot\epsilon) = 0$, so for $d \geq 2$ the divergence is harmonic; (1.3) then reduces to $(d-2)\partial_\mu\partial_\nu(\partial\cdot\epsilon) = 0$. For $d \geq 3$ this forces $\partial\cdot\epsilon$ to be at most linear in x, hence by (1.2) the field ϵ^μ is at most quadratic. Inserting the general quadratic ansatz into (1.2) order by order leaves exactly the four families listed, and the count is $d + \frac{1}{2}d(d-1) + 1 + d = \frac{1}{2}(d+1)(d+2)$. □

Theorem 1.1.3. *For $d = 2$ with Euclidean signature, (1.2) is equivalent to the Cauchy–Riemann equations for $\epsilon := \epsilon_1 + i\epsilon_2$ as a function of $z := x^1 + ix^2$. The conformal algebra of the Euclidean plane is therefore infinite dimensional.*

Proof. With $\eta = \delta$ and $d = 2$, the diagonal components of (1.2) both read $\partial_1\epsilon_1 = \partial_2\epsilon_2$, and the off-diagonal component reads $\partial_1\epsilon_2 + \partial_2\epsilon_1 = 0$. These are precisely the Cauchy–Riemann equations for $\epsilon_1 + i\epsilon_2$. □

Slogan 1.1.4. *Two dimensions are special because (1.3) degenerates: the coefficient $(d-2)$ that forces ϵ to be polynomial vanishes exactly at $d = 2$, and what remains is a first-order elliptic system with an infinite-dimensional solution space.*

Remark 1.1.5. *Only a finite-dimensional part of this algebra integrates to globally defined transformations of the compactified plane: a holomorphic $\epsilon(z)$ generates a global automorphism of $\mathbb{S}^2$ only if it is at most quadratic, which is the $d = 2$ case of Proposition 1.1.2. The distinction between the infinite-dimensional local algebra and the finite-dimensional global group is taken up in the next section.*

SCF and stereographic projection

The inversion with respect to the unit circle, which is part of the special conformal transformation on $\mathbb{R}^2$, is itself a conformal transformation. In fact, inversion is equivalent, through stereographic projection, to the mirror reflection $z \mapsto -z$ of Riemann sphere:

$$\left|\frac{1}{1-z}(x,y)\right|^2 \cdot \left|\frac{1}{1+z}(x,y)\right|^2 = \frac{1-z^2}{(1-z)^2} \cdot \frac{1-z^2}{(1+z)^2} = 1. \tag{1.9}$$

1.2 2D CFT

1.2.1 Conformal invariance

Möbius group as conformal group

The conformal completion of $\mathbb{R}^{2,0}$ is the Riemann sphere,

$$\mathbb{R}^{\bar{2},0} = (\mathbb{S}^2 \times \mathbb{S}^0)/\mathbb{Z}_2 \simeq \mathbb{S}^2. \tag{1.10}$$

By Proposition 1.1.2 the globally defined conformal transformations form a group of dimension $\frac{1}{2}(d+1)(d+2)$, isomorphic to $\mathrm{SO}(d+1,1)$; at $d=2$ this is six-dimensional. Accordingly the group of orientation-preserving global conformal transformations (also called the Möbius group) is

$$\mathrm{Conf}^+(\mathbb{R}^{2,0}) = \mathrm{SO}^+(3,1) \simeq \mathrm{SL}(2,\mathbb{C})/\{\pm 1\} = \mathrm{PSL}(2,\mathbb{C}). \tag{1.11}$$

The action of $\mathrm{Conf}^+(\mathbb{R}^{2,0})$ on $\mathbb{S}^2 = \mathbb{C}P^1 = \mathbb{C} \cup \{\infty\}$ is induced from the action

$$A \cdot z = \frac{az+b}{cz+d}, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{C}). \tag{1.12}$$

For later use (verifying Möbius covariance of correlation functions), we record that

$$A \cdot z_1 - A \cdot z_2 = (z_1 - z_2)(cz_1+d)^{-1}(cz_2+d)^{-1}, \tag{1.13a}$$

$$\frac{d}{dz} A \cdot z = (cz+d)^{-2}. \tag{1.13b}$$

Without orientation-preserving condition, the global conformal automorphisms of $\mathbb{R}^{2,0}$ is

$$\mathrm{Conf}(\mathbb{R}^{2,0}) = \mathrm{Conf}^+(\mathbb{R}^{2,0}) \times \mathrm{Conf}^-(\mathbb{R}^{2,0}) \tag{1.14}$$

where $\mathrm{Conf}^-(\mathbb{R}^{2,0}) = \mathrm{Conf}^+(\mathbb{R}^{2,0}) \cdot \mathcal{K}$ with complex conjugation $\mathcal{K}$.

(Anti-)holomorphic functions as conformal algebra

Theorem 1.1.3 identifies the solutions of the conformal Killing equation with holomorphic functions. Restoring the anti-holomorphic half, the conformal algebra of the Euclidean plane is the $\mathbb{Z}_2$-graded algebra

$$\mathfrak{conf}(\mathbb{R}^{2,0}) = \mathfrak{hol}_0 \oplus \bar{\mathfrak{hol}}_1 \tag{1.15}$$

where $\mathfrak{hol}$ ($\bar{\mathfrak{hol}}$) denotes the $\mathbb{C}-$algebra of holomorphic (anti-holomorphic) maps and the subscripts stand for the grading.

Notably, $\mathfrak{conf}(\mathbb{R}^{2,0}) \gneq \text{Lie Conf}(\mathbb{R}^{2,0})$ since the former is infinite dimensional while the latter is finite (in fact, 12 over $\mathbb{R}$) dimensional. This is precisely the phenomenon of Remark 1.1.5: the generators L_n with $|n| \geq 2$ are vector fields with a singularity at 0 or at ∞, and integrate only to locally defined maps. In terms of the usual Virasoro generators,

$$\text{Lie Conf}^+(\mathbb{R}^{2,0}) = \mathbb{C}L_1 \oplus \mathbb{C}L_0 \oplus \mathbb{C}L_{-1}, \quad L_n := z^{1-n}\frac{\partial}{\partial z}, \tag{1.16a}$$

$$\text{Lie Conf}^-(\mathbb{R}^{2,0}) = \mathbb{C}\bar{L}_1 \oplus \mathbb{C}\bar{L}_0 \oplus \mathbb{C}\bar{L}_{-1}, \quad \bar{L}_n := \bar{z}^{1-n}\frac{\partial}{\partial \bar{z}}, \tag{1.16b}$$

$$\mathfrak{conf}(\mathbb{R}^{2,0}) = \bigoplus_{n\in\mathbb{Z}} \left(\mathbb{C}L_n \oplus \mathbb{C}\bar{L}_n\right). \tag{1.16c}$$

The Witt algebra

The generators L_n of the conformal algebra has a Lie algebra structure characterized by the following, which is called the *Witt algebra*.

$$[L_m, L_n] = (m-n)L_{m+n} \tag{1.17}$$

1.2.2 Primary and secondary fields

Definition 1.2.1. *A primary field of conformal dimensions $h, \bar{h}$ is a field $\phi(z, \bar{z})$ that transforms under the local conformal transformation $z \to w(z)$ as*

$$\phi'(w(z), \bar{w}(\bar{z})) = \left(\frac{dw}{dz}\right)^{-h} \left(\frac{d\bar{w}}{d\bar{z}}\right)^{-\bar{h}} \phi(z, \bar{z}). \tag{1.18}$$

That is, they are sections of the line bundle $\Omega_X^{\otimes h} \otimes \bar{\Omega}_X^{\otimes \bar{h}}$ over the conformal spacetime X.

In the axiomatic treatment of §2.1, only the Möbius part of (1.18) is imposed directly, as (G4) for quasi-primaries. The full local law is replaced by the operator product expansion with T (Definition 2.2.6).

If $w = z + \xi(z)$ is an infinitesimal transformation, Eq. (1.18) is equivalent to the *Lie derivative*

$$\delta_\xi \phi(z) = -(h\partial\xi + \xi\partial)\phi. \tag{1.19}$$

Remark 1.2.2 (The spin is an integer)**.** *For real $h, \bar{h}$ the two factors on the right of (1.18) are not separately defined: each needs a branch of* $\log(dw/dz)$*. Their product is nevertheless well defined whenever the* spin $s := h - \bar{h}$ *is an integer, if we set, for $\lambda \in \mathbb{C}^\times$,*

$$\lambda^h \bar{\lambda}^{\bar{h}} := \lambda^s |\lambda|^{2\bar{h}}, \qquad s = h - \bar{h} \in \mathbb{Z}, \tag{1.20}$$

which involves only an integer power and a positive real power. Correspondingly the bundle in the definition is $K^s \otimes |K|^{2\bar{h}}$, where $K = \Omega_X$ is the canonical bundle and $|K|^\alpha$ the real line bundle of α-densities; the expression $\Omega^{\otimes h} \otimes \bar{\Omega}^{\otimes \bar{h}}$ has no meaning for $s \notin \mathbb{Z}$.

Convention. From now on an expression $z^a\bar{z}^{\bar{a}}$ with $a-\bar{a}\in\mathbb{Z}$ always means $z^{a-\bar{a}}|z|^{2\bar{a}}$. It is a single-valued real-analytic function on $\mathbb{C}^\times$, and it is multiplicative: $(zw)^a\overline{(zw)}^{\bar{a}}=z^a\bar{z}^{\bar{a}}\,w^a\bar{w}^{\bar{a}}$.

Two consequences are worth recording. (i) A chiral field ($\bar{h}=0$) must have $h\in\mathbb{Z}$. Chiral objects with $h\notin\mathbb{Z}$, such as the conformal blocks of §2.5, are therefore multivalued functions rather than correlators, and statements about them are statements about the full correlators from which they are extracted. (ii) Fermions have $s\in\frac{1}{2}+\mathbb{Z}$; then $\lambda^{1/2}$ requires a spin structure, a choice of square root of K, as in Remark 3.4.4. Below we assume $s\in\mathbb{Z}$; the fermionic case changes signs but not arguments.

Lemma 1.2.3 (cocycle)**.** *For a holomorphic map f with nowhere vanishing derivative put $j_f(z;h,\bar{h}):=f'(z)^h\,\overline{f'(z)}^{\bar{h}}$ in the sense of (1.20). Then*

$$j_{f\circ g}(z;h,\bar{h})=j_f\big(g(z);h,\bar{h}\big)\,j_g(z;h,\bar{h}). \tag{1.21}$$

Proof. $(f\circ g)'(z)=f'(g(z))\,g'(z)$, and both $\lambda\mapsto\lambda^s$ ($s\in\mathbb{Z}$) and $\lambda\mapsto|\lambda|^{2\bar{h}}$ are multiplicative. For $s\notin\mathbb{Z}$ any fixed branch of $\lambda\mapsto\lambda^h$ is multiplicative only up to factors $e^{2\pi ihk}$, $k\in\mathbb{Z}$; this is the source of the stray phases met below. □

Example 1.2.4 (Identity field)**.** *The identity field $\mathbb{I}(z)\equiv 1$ is primary and has conformal dimension $h=\bar{h}=0$.*

1.2.3 The stress tensor

Conformal symmetry, like any continuous symmetry, is carried by a conserved current. In a conformal field theory all of these currents are built from a single object.

Translation invariance supplies a conserved stress tensor $T^{\mu\nu}$ with $\partial_\mu T^{\mu\nu}=0$, and rotation invariance allows it to be chosen symmetric. To a conformal Killing vector ϵ^μ one associates the current

$$j^\mu:=T^{\mu\nu}\epsilon_\nu,\qquad \partial_\mu j^\mu=T^{\mu\nu}\,\partial_\mu\epsilon_\nu=\frac{1}{d}\,T^\mu{}_\mu\,(\partial\cdot\epsilon), \tag{1.22}$$

where the last equality uses the symmetry of T together with (1.2). Taking ϵ to be a translation or a rotation gives a conserved current for any T; but for a dilatation $\partial\cdot\epsilon\neq 0$, and conservation requires the trace to vanish.

Proposition 1.2.5. *A theory with a symmetric conserved stress tensor is conformally invariant if and only if $T^\mu{}_\mu=0$.*

Remark 1.2.6. *The statement is exact only up to improvement terms: the stress tensor is defined by translation invariance only up to the addition of an identically conserved piece, and scale invariance guarantees tracelessness only after a suitable choice. We take the traceless representative throughout.*

In two dimensions tracelessness is unusually powerful, because it leaves only two independent components and forces each to depend on a single variable. Pass to complex coordinates $z=x^1+ix^2$, in which

$$g_{z\bar{z}}=g_{\bar{z}z}=\tfrac{1}{2},\qquad g_{zz}=g_{\bar{z}\bar{z}}=0,\qquad g^{z\bar{z}}=g^{\bar{z}z}=2. \tag{1.23}$$

Theorem 1.2.7. *In two dimensions the components T_{zz} and $T_{\bar{z}\bar{z}}$ of a traceless conserved symmetric stress tensor satisfy*

$$\partial_{\bar{z}} T_{zz} = 0, \qquad \partial_z T_{\bar{z}\bar{z}} = 0. \tag{1.24}$$

Proof. The trace is $T^{\mu}{}_{\mu} = g^{\mu\nu} T_{\mu\nu} = 4T_{z\bar{z}}$, so tracelessness is the statement $T_{z\bar{z}} = T_{\bar{z}z} = 0$. Conservation $g^{\mu\rho}\partial_\rho T_{\mu\nu} = 0$ with $\nu = z$ has only two surviving terms,

$$g^{z\bar{z}} \partial_{\bar{z}} T_{zz} + g^{\bar{z}z} \partial_z T_{\bar{z}z} = 0, \tag{1.25}$$

and the second vanishes. The case $\nu = \bar{z}$ is conjugate. □

Accordingly we write

$$T(z) := T_{zz}(z), \qquad \bar{T}(\bar{z}) := T_{\bar{z}\bar{z}}(\bar{z}), \tag{1.26}$$

up to a normalization constant, fixed once and for all by the convention for the operator product expansion. This is the object whose Laurent modes will generate the Virasoro algebra. In (1.26) the argument records which coordinate the field depends on, so that T is holomorphic and $\bar{T}$ antiholomorphic as fields on the worldsheet; the bar is a label, not complex conjugation. The distinction is invisible here and becomes sharp only when $\bar{T}$ is continued off the surface, as in Theorem 4.1.11.

Slogan 1.2.8. *The holomorphic factorization of two-dimensional conformal field theory is already visible at the classical level: tracelessness plus conservation splits the stress tensor into a holomorphic and an antiholomorphic half, which thereafter never mix.*

1.2.4 The Virasoro algebra

The Virasoro algebra is a central extension of the Witt algebra. That the extension is by a single parameter, and that the cocycle can always be brought to the form below, is a statement about $H^2(\mathfrak{Witt}; \mathbb{C})$; we take it for granted here. The constant c is not visible classically — it measures the failure of T to transform as a tensor, and will be produced by the operator product expansion of T with itself in §2.2.2.

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0} \tag{1.27}$$

Looking ahead

Everything so far has been local. Two things change once the plane is compactified to the Riemann sphere $S^2 = \mathbb{C} \cup \{\infty\}$. First, the global symmetry group shrinks: only the Möbius transformations (1.12) act by genuine automorphisms, which is why the algebra $\mathfrak{conf}$ is strictly larger than Lie Conf. Second — and this is the substantial point — the two distinguished points 0 and ∞ turn the radial foliation of the plane into a genuine time evolution, so that operators may be traded for states. Chapter 2 develops this, and the representation theory of the Virasoro algebra that it makes possible.

Chapter 2

The sphere

Compactifying the plane to the sphere does two things. The global symmetry shrinks to the Möbius group, and the points 0 and ∞ turn the radial direction into time, so that fields become states. Section 2.1 states the axioms for correlation functions on the sphere, from which the rest of the note is derived; Möbius covariance alone fixes the two- and three-point functions. Section 2.2 derives the operator product expansion and the Ward identities from these axioms and computes three free theories. Section 2.3 introduces the Hermitian structure and the Verma modules, whose null vectors are located by the Kac determinant. Section 2.4 turns null vectors into differential equations for conformal blocks; this produces the minimal models and their fusion rules. Section 2.5 imposes the consistency condition proper to genus zero, crossing symmetry.

Positivity enters only in Definition 2.3.2. Everything before it holds in non-unitary theories as well.

2.1 Correlation functions on the sphere

The arguments of this chapter and the next treat correlation functions as functions of the insertion points that are covariant under Möbius maps and that behave in a definite way at ∞. We make these properties explicit as axioms; everything else in this section is then a theorem. They are stated only at the level needed here: no topology is put on the space of fields, and positivity is not assumed until §2.3.

2.1.1 Axioms

Write $\mathcal{X}_n := \{\, (z_1, \dots, z_n) \in \mathbb{C}^n \mid z_i \neq z_j \text{ for } i \neq j \,\}$ for the configuration space of n distinct points of the plane.

Definition 2.1.1 (genus-zero correlation system)**.** *A genus-zero correlation system consists of a complex vector space of* fields

$$F = \bigoplus_{(h,\bar{h})} F_{(h,\bar{h})}, \qquad h, \bar{h} \in \mathbb{R}, \quad h - \bar{h} \in \mathbb{Z}, \tag{2.1}$$

and, for each $n \geq 0$, multilinear maps assigning to $\phi_1, \dots, \phi_n \in F$ a real-analytic function $\langle \phi_1(z_1) \cdots \phi_n(z_n) \rangle$ on $\mathcal{X}_n$, subject to the following.

(G1) Locality. *The function is invariant under simultaneous permutations of the pairs (ϕ_i, z_i).*

(G2) Identity. *There is $\mathbb{I} \in F_{(0,0)}$ with $\langle \mathbb{I}(z_1)\phi_2(z_2) \cdots \phi_n(z_n) \rangle = \langle \phi_2(z_2) \cdots \phi_n(z_n) \rangle$, and the empty correlator is* 1.

(G3) Derivatives. *There are operators $\partial, \bar{\partial}$ on F of degrees $(1,0)$ and $(0,1)$ such that*

$$\langle \cdots (\partial\phi)(z_i) \cdots \rangle = \partial_{z_i} \langle \cdots \phi(z_i) \cdots \rangle,$$

and likewise for $\bar{\partial}$.

(G4) Möbius covariance. *There is a graded subspace $F^{\mathrm{qp}} \subset F$ of quasi-primary fields such that F is spanned by the $\partial^m \bar{\partial}^n \phi$ with $\phi \in F^{\mathrm{qp}}$, and such that for $\phi_i \in F^{\mathrm{qp}}_{(h_i, \bar{h}_i)}$ and every $M \in \mathrm{PSL}(2, \mathbb{C})$ with $M(z_i) \neq \infty$ for all i,*

$$\langle \phi_1(Mz_1) \cdots \phi_n(Mz_n) \rangle \prod_{i=1}^{n} j_M(z_i; h_i, \bar{h}_i) = \langle \phi_1(z_1) \cdots \phi_n(z_n) \rangle. \tag{2.2}$$

(G5) Operator product expansion. *For $\phi, \chi \in F$ there are a countable set $S \subset \mathbb{R}^2$ of pairs $(a, \bar{a})$ with $a - \bar{a} \in \mathbb{Z}$, both entries bounded below and only finitely many pairs with $a + \bar{a} \leq \Lambda$ for each Λ, and fields $\psi_{(a,\bar{a})} \in F$, such that*

$$\langle \phi(z)\chi(w)\phi_3(z_3) \cdots \phi_n(z_n) \rangle = \sum_{(a,\bar{a}) \in S} (z-w)^a (\bar{z}-\bar{w})^{\bar{a}} \, \langle \psi_{(a,\bar{a})}(w)\phi_3(z_3) \cdots \phi_n(z_n) \rangle \tag{2.3}$$

for all $\phi_3, \dots, \phi_n$, the series converging absolutely whenever $0 < |z-w| < \min_{j \geq 3} |w - z_j|$. The data $S, \psi_{(a,\bar{a})}$ depend on ϕ, χ only.

Remark 2.1.2 (on the axioms)**.** *(i) Covariance is imposed only on quasi-primaries, since a general field transforms inhomogeneously; the law for $\partial^m \bar{\partial}^n \phi$ follows by differentiating (2.2). That quasi-primaries and their derivatives span F is part of the assumption. Lemma 1.2.3 makes (2.2) consistent under composition of Möbius maps, which would fail by phases if the weights were not spin-adapted.*

(ii) (G1) is stated for bosonic fields; fermionic fields acquire signs.

(iii) (G5) is an axiom here, and nothing in this note proves it in general. Comparing scaling weights in (2.3) under (G4) gives $\psi_{(a,\bar{a})} \in F_{(h_\phi + h_\chi + a,\, \bar{h}_\phi + \bar{h}_\chi + \bar{a})}$ when ϕ, χ are homogeneous.

(iv) Not assumed: positivity, i.e. a Hilbert space structure, and local conformal (Virasoro) symmetry. The latter enters with the stress tensor in Chapter 2.

Remark 2.1.3 (fermionic fields)**.** *A fermionic theory has $F = F_{\bar{0}} \oplus F_{\bar{1}}$, graded by parity $|\phi|$. The axioms change in three places.*

(i) In (G1) a permutation acts with the Koszul sign: one factor (-1) for each transposition of two odd fields.

(ii) Spins lie in $\frac{1}{2}\mathbb{Z}$, with $2s \equiv |\phi| \pmod 2$.

(iii) For $s \in \frac{1}{2}+\mathbb{Z}$, the factor j_M in (G4) is built from $(cz+d)^{-2s}|cz+d|^{-4\bar{h}}$ with $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{C})$. The sign ambiguity of the representative cancels in $\prod_i j_M(z_i)$, because a non-vanishing correlator has even total parity.

On the plane this describes the Neveu–Schwarz sector of Definition 3.4.1: fields single-valued around 0, antiperiodic on the cylinder (Remark 3.1.4). The Ramond sector is not part of F. A field creating a Ramond state, such as the spin field σ of §3.4, has a square-root branch point in its expansion with ψ, so (G1) and (G5) fail for the pair (ψ,σ). The bosonic Ising model, with local fields $\{\mathbb{I},\sigma,\varepsilon\}$, and the fermionic theory, with $\{\mathbb{I},\psi,\bar{\psi},\varepsilon\}$, are therefore two different correlation systems sharing $\mathbb{I}$ and ε; the identification at the end of §3.4 relates them.

2.1.2 Behaviour at infinity

Correlators are defined for points of $\mathbb{C}$. A point at ∞ is reached through the chart $u=-1/\zeta$, which is the image of the standard chart at 0 under the Möbius map $S(z)=-1/z$.

Lemma 2.1.4. *Let $\phi \in F^{\mathrm{qp}}_{(h,\bar{h})}$, let $\phi_2,\dots,\phi_n \in F$, and fix distinct $z_2,\dots,z_n \in \mathbb{C}$. Then $\zeta^{2h}\bar{\zeta}^{2\bar{h}}\langle\phi(\zeta)\phi_2(z_2)\cdots\phi_n(z_n)\rangle$, as a function of $u=-1/\zeta$, extends real-analytically to $u=0$. In particular the limit*

$$\langle\phi(\infty)\phi_2(z_2)\cdots\phi_n(z_n)\rangle := \lim_{\zeta\to\infty} \zeta^{2h}\bar{\zeta}^{2\bar{h}}\,\langle\phi(\zeta)\phi_2(z_2)\cdots\phi_n(z_n)\rangle \tag{2.4}$$

exists, and $\langle\phi(\zeta)\phi_2(z_2)\cdots\rangle = O(|\zeta|^{-2(h+\bar{h})})$ as $\zeta\to\infty$.

Proof. Suppose first that all ϕ_j are quasi-primary and all $z_j \neq 0$. Here $S'(z)=z^{-2}$, so $j_S(z;h,\bar{h}) = z^{-2h}\bar{z}^{-2\bar{h}}$. Applying (2.2) with $M=S$ to the points $(\zeta,z_2,\dots,z_n)$, $\zeta\neq 0$, gives

$$\zeta^{2h}\bar{\zeta}^{2\bar{h}}\langle\phi(\zeta)\phi_2(z_2)\cdots\phi_n(z_n)\rangle = \prod_{j\geq 2} j_S(z_j;h_j,\bar{h}_j)\,\big\langle\phi(u)\,\phi_2(-1/z_2)\cdots\phi_n(-1/z_n)\big\rangle. \tag{2.5}$$

The right side is real-analytic in u near $u=0$, because $(0,-1/z_2,\dots,-1/z_n)\in\mathcal{X}_n$. If some $z_j=0$, translate all points first; translations have $j=1$. Finally, a general ϕ_j is a combination of $\partial^m\bar{\partial}^n$ of quasi-primaries, and derivatives in z_j of a function real-analytic in $(u,z_2,\dots,z_n)$ are again real-analytic. The decay follows from $|\zeta^{2h}\bar{\zeta}^{2\bar{h}}| = |\zeta|^{2(h+\bar{h})}$. □

Remark 2.1.5. *The field at ∞ must be quasi-primary: for a derivative $\partial\phi$ the corresponding limit diverges in general unless $h=0$, since derivatives transform inhomogeneously. The limit (2.4) is the correlator form of the bra (2.80b). Had we used the chart $u=1/\zeta$, the limit would change by $j=(-1)^h(-1)^{\bar{h}}=(-1)^s$. This is an honest sign, because $s\in\mathbb{Z}$, but it is a convention. It is the factor $(-1)^{s_3}$ that the proof of Theorem 2.1.8 produces with the chart w'/w; in the earlier, chiral version it appeared as $(-1)^{h_3}$, and as $(-1)^{h_4}$ in Corollary 2.1.12. Throughout this note the chart at ∞ is $u=-1/\zeta$. The one exception is Lemma 2.3.7, where $1/z$ is used so that the mode relation carries no sign.*

2.1.3 Two- and three-point functions

Theorem 2.1.6 (two-point function)**.** *Let* $\phi_i \in F^{\mathrm{qp}}_{(h_i,\bar{h}_i)}$, $i = 1, 2$. *Then* $\langle \phi_1(z_1)\phi_2(z_2)\rangle = 0$ *unless* $(h_1, \bar{h}_1) = (h_2, \bar{h}_2) =: (h, \bar{h})$, *in which case*

$$\langle \phi_1(z_1)\phi_2(z_2)\rangle = \frac{C_{12}}{z_{12}^{2h}\,\bar{z}_{12}^{2\bar{h}}}, \qquad C_{12} = \langle \phi_1(\infty)\phi_2(0)\rangle = C_{21}. \tag{2.6}$$

Proof. Considering the global conformal transformation $z \to z + z_2$, we notice that the correlation function is invariant under translation:

$$\langle \phi_1(z_1)\phi_2(z_2)\rangle = \langle \phi_1(z_1 - z_2)\phi_2(0)\rangle. \tag{2.7}$$

Let $C_{12} = \langle \phi_1(1)\phi_2(0)\rangle$. now using another global transformation, $z \mapsto (z_1 - z_2)z$, we get

$$\langle \phi_1(z_1)\phi_2(z_2)\rangle = z_{12}^{-(h_1+h_2)}\,\bar{z}_{12}^{-(\bar{h}_1+\bar{h}_2)}\langle \phi_1(1)\phi_2(0)\rangle = \frac{C_{12}}{z_{12}^{h_1+h_2}\,\bar{z}_{12}^{\bar{h}_1+\bar{h}_2}}. \tag{2.8}$$

Under $z \mapsto \lambda z$ both fields pick up a factor, $\lambda^{h_1}\bar{\lambda}^{\bar{h}_1}$ and $\lambda^{h_2}\bar{\lambda}^{\bar{h}_2}$ (here $\lambda = z_{12}$), so the exponent is $h_1 + h_2$. The powers are spin-adapted, since $(h_1 + h_2) - (\bar{h}_1 + \bar{h}_2) = s_1 + s_2 \in \mathbb{Z}$, and the antiholomorphic part is carried along from the start.

Translations and dilations alone cannot relate h_1 to h_2. That needs the special conformal transformation, here in the form $S(z) = -1/z$ applied at $z_1, z_2 \in \mathbb{C}^\times$. Since $Sz_1 - Sz_2 = z_{12}/(z_1 z_2)$ and $j_S(z; h, \bar{h}) = z^{-2h}\bar{z}^{-2\bar{h}}$, (2.2) and (2.8) give

$$\frac{C_{12}}{z_{12}^{h_1+h_2}\,\bar{z}_{12}^{\bar{h}_1+\bar{h}_2}} = \frac{C_{12}}{z_{12}^{h_1+h_2}\,\bar{z}_{12}^{\bar{h}_1+\bar{h}_2}} \cdot w^{\,h_2-h_1}\,\bar{w}^{\,\bar{h}_2-\bar{h}_1}, \qquad w := z_1/z_2. \tag{2.9}$$

As w ranges over $\mathbb{C}^\times \setminus \{1\}$, either $C_{12} = 0$ or $w^{s_2-s_1}|w|^{2(\bar{h}_2-\bar{h}_1)} \equiv 1$. The latter forces $s_1 = s_2$ and $\bar{h}_1 = \bar{h}_2$, which gives (2.6). By (2.4), $\langle \phi_1(\infty)\phi_2(0)\rangle = \lim \zeta^{2h}\bar{\zeta}^{2\bar{h}}\, C_{12}\, \zeta^{-2h}\bar{\zeta}^{-2\bar{h}} = C_{12}$, so the two descriptions of C_{12} agree. Finally, (G1) and $z_{21}^{-2h}\bar{z}_{21}^{-2\bar{h}} = (-1)^{2s} z_{12}^{-2h}\bar{z}_{12}^{-2\bar{h}}$ give $C_{21} = C_{12}$. □

Remark 2.1.7. *In the chiral case the theorem reads* $\langle \phi_1(z_1)\phi_2(z_2)\rangle = C\,\delta_{h_1 h_2}\, z_{12}^{-2h}$ *with* $h \in \mathbb{Z}$ *(Remark 1.2.2).*

Theorem 2.1.8 (three-point function)**.** *Let* $\phi_i \in F^{\mathrm{qp}}_{(h_i,\bar{h}_i)}$, $i = 1, 2, 3$, *and for* $\{i, j, k\} = \{1, 2, 3\}$ *put* $h_{ij} := h_i + h_j - h_k$, $\bar{h}_{ij} := \bar{h}_i + \bar{h}_j - \bar{h}_k$. *Then*

$$\langle \phi_1(z_1)\phi_2(z_2)\phi_3(z_3)\rangle = \frac{C_{123}}{z_{32}^{h_{23}}\bar{z}_{32}^{\bar{h}_{23}}\; z_{21}^{h_{12}}\bar{z}_{21}^{\bar{h}_{12}}\; z_{31}^{h_{13}}\bar{z}_{31}^{\bar{h}_{13}}}, \qquad C_{123} = \langle \phi_3(\infty)\phi_2(1)\phi_1(0)\rangle. \tag{2.10}$$

Each factor is single-valued, since $h_{ij} - \bar{h}_{ij} = s_i + s_j - s_k \in \mathbb{Z}$.

Proof. Let $C_{123} = \langle \phi_3(\infty)\phi_2(1)\phi_1(0)\rangle$. We will exploit the fact that there is a unique Möbius transformation that maps the triple $(\infty, 1, 0)$ to another arbitrary triple (z_3, z_2, z_1). We start by finding this transformation explicitly.

Since the desired transformation maps $[1 : 0]$ to $[z_3 : 1]$ and $[0 : 1]$ to $[z_1 : 1]$, it is represented by a matrix of the form $A = \begin{pmatrix} z_3 & \lambda z_1 \\ 1 & \lambda \end{pmatrix}$. For the image of $[1 : 1]$ to be $[z_2 : 1]$, there must be

some $\mu \in \mathbb{C}$ such that

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} z_3 + \lambda z_1 \\ 1 + \lambda \end{pmatrix} = \mu \begin{pmatrix} z_2 \\ 1 \end{pmatrix}. \tag{2.11}$$

Solving Eq. (2.11) produces $\lambda = (z_1 - z_2)/(z_2 - z_3)$. Rescaling the matrix A, we find the Möbius transform

$$\begin{bmatrix} z \\ z' \end{bmatrix} = \begin{bmatrix} z_3(z_2 - z_1) & z_1(z_3 - z_2) \\ z_2 - z_1 & z_3 - z_2 \end{bmatrix} \begin{bmatrix} w \\ w' \end{bmatrix}. \tag{2.12}$$

Call this map M. Apply (2.2) to M at the points $(\zeta, 1, 0)$ and let $\zeta \to \infty$. In the chart $u = -w'/w$ at ∞, which is the chart of (2.4), we have, with $z_3(u) := M(-1/u)$,

$$j_M(\zeta; h_3, \bar{h}_3) = \left(\frac{dz_3}{du}\right)^{h_3} \left(\overline{\frac{dz_3}{du}}\right)^{\bar{h}_3} \zeta^{-2h_3} \bar{\zeta}^{-2\bar{h}_3}.$$

Multiplying by $\zeta^{2h_3}\bar{\zeta}^{2\bar{h}_3}$ and using Lemma 2.1.4 as $u \to 0$ leads to the following, where $(\lambda)^{-h_i}$ abbreviates the spin-adapted $\lambda^{-h_i}\bar{\lambda}^{-\bar{h}_i}$:

$$\begin{aligned} \langle \phi_3(z_3)\phi_2(z_2)\phi_1(z_1)\rangle &= C_{123} \left(\frac{d(z/z')}{d(-w'/w)}\bigg|_{-\frac{w'}{w}=0}\right)^{-h_3} \left(\frac{d(z/z')}{d(w/w')}\bigg|_{\frac{w}{w'}=1}\right)^{-h_2} \left(\frac{d(z/z')}{d(w/w')}\bigg|_{\frac{w}{w'}=0}\right)^{-h_1} \\ &= C_{123} \left(\frac{z_{32}z_{31}}{z_{21}}\right)^{-h_3} \left(\frac{z_{21}z_{32}}{z_{31}}\right)^{-h_2} \left(\frac{z_{21}z_{31}}{z_{32}}\right)^{-h_1} \\ &= C_{123}(z_3 - z_2)^{h_1 - h_2 - h_3}(z_2 - z_1)^{h_3 - h_1 - h_2}(z_3 - z_1)^{h_2 - h_1 - h_3}. \end{aligned} \tag{2.13}$$

Be cautious that the first derivative in the first line of Eq. (2.13) is against $-w'/w$, not w/w'. This is because the former is a well-defined coordinate chart around the point $[w : w'] = [1 : 0]$ while the latter is not. The second line uses the three derivatives of M; in the last line the spin-adapted powers are split multiplicatively and the exponents collected for each z_{ij}, each together with its antiholomorphic partner as in (2.10). With the chart w'/w instead of $-w'/w$, the first derivative changes sign and a factor $(-1)^{s_3}$ appears. It is well defined because $s_3 \in \mathbb{Z}$, and it is fixed by the choice of chart, which is the one that defines the bra (2.80b) (Remark 2.1.5). □

Remark 2.1.9 (covariance of the standard form)**.** *The right side of* (2.10) *can also be checked to satisfy* (2.2) *directly. Denote its denominator by P, represent M by $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2, \mathbb{C})$ and put $\gamma_i := cz_i + d \neq 0$. Then $Mz_i - Mz_j = z_{ij}/(\gamma_i\gamma_j)$ and $j_M(z_i; h_i, \bar{h}_i) = \gamma_i^{-2h_i}\bar{\gamma}_i^{-2\bar{h}_i}$. By multiplicativity, $P(Mz) = P(z)\prod_i \gamma_i^{-(h_{ij}+h_{ik})}\bar{\gamma}_i^{-(\bar{h}_{ij}+\bar{h}_{ik})}$, and $h_{ij} + h_{ik} = 2h_i$. Hence $P(Mz)^{-1}\prod_i j_M(z_i) = P(z)^{-1}$. All exponents of γ_i have even integer spin part, so the sign of the representative is irrelevant. Since $\mathrm{PSL}(2, \mathbb{C})$ acts simply transitively on ordered triples of distinct points, the ratio of any covariant three-point function to P^{-1} is constant, which gives a second proof. The same computation is used for Corollary 2.1.12.*

Remark 2.1.10 (the sign)**.** *Reversing a difference changes a factor by an integer sign, $z_{ji}^a \bar{z}_{ji}^{\bar{a}} = (-1)^{a-\bar{a}} z_{ij}^a \bar{z}_{ij}^{\bar{a}}$. With (G1) this gives, for instance,*

$$\langle \phi_3(\infty)\phi_1(1)\phi_2(0)\rangle = (-1)^{s_1+s_2-s_3}\, C_{123}. \tag{2.14}$$

This sign, like the chart sign $(-1)^{s_3}$ noted in the proof of Theorem 2.1.8, is an integer sign: it depends only on the order of the reference points (and the chart at ∞, Remark 2.1.5), and it carries no branch ambiguity. In purely chiral notation one is led to $(-1)^h$, which is undefined for $h \notin \mathbb{Z}$; this is another way of saying that chiral three-point "functions" with non-integer weights are not functions.

For later use, we record the following result.

Corollary 2.1.11 (cross ratio)**.** *Suppose the Möbius transformation M maps the tuple $(\infty, 1, x, 0)$ to (z_4, z_3, z_2, z_1). Then*

$$x = \frac{z_{12}z_{34}}{z_{13}z_{24}} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)}. \tag{2.15}$$

Corollary 2.1.12 (four-point function)**.** *For $\phi_i \in F^{\mathrm{qp}}_{(h_i,\bar{h}_i)}$, $i = 1, \dots, 4$,*

$$\langle \phi_4(z_4)\phi_3(z_3)\phi_2(z_2)\phi_1(z_1)\rangle = Q(z_1, \dots, z_4)\, G(x, \bar{x}), \qquad G(x, \bar{x}) := \langle \phi_4(\infty)\phi_3(1)\phi_2(x)\phi_1(0)\rangle, \tag{2.16}$$

with x the cross ratio (2.15) and

$$Q = z_{43}^{h_1+h_2-h_3-h_4}\, z_{31}^{h_4-h_1-h_2-h_3}\, z_{41}^{h_2+h_3-h_1-h_4}\, z_{42}^{-2h_2} \times (\text{the same with } z \to \bar{z},\ h \to \bar{h}), \tag{2.17}$$

each factor being spin-adapted. G is a single-valued real-analytic function on $\mathbb{C} \setminus \{0, 1\}$.

Proof. The exponents of Q add up to $-2h_i$ at each z_i, so the computation of Remark 2.1.9 shows that Q is covariant. Hence $I := \langle \phi_4\phi_3\phi_2\phi_1\rangle / Q$ is Möbius invariant on $\mathcal{X}_4$. If $z, w \in \mathcal{X}_4$ have the same cross ratio, the Möbius map sending (z_1, z_3, z_4) to (w_1, w_3, w_4) sends z_2 to w_2 and keeps all points finite, so I is a function of x alone. At $(z_1, z_2, z_3, z_4) = (0, x, 1, \zeta)$ the cross ratio tends to x and $\zeta^{2h_4}\bar{\zeta}^{2\bar{h}_4} Q \to 1$ as $\zeta \to \infty$, so $I(x) = G(x, \bar{x})$ by (2.4). □

Remark 2.1.13. *With the chart $u = -1/\zeta$ at ∞ no sign appears in Corollary 2.1.12; the factor $(-1)^{h_4}$ of the earlier version is the chart sign of Remark 2.1.5. The constraints of §2.5 are constraints on the function G, and its single-valuedness is the condition imposed there.*

2.2 Radial quantization

2.2.1 Radial ordering

In §2.1 correlators were functions on $\mathcal{X}_n$ subject to Definition 2.1.1. The operator language of this chapter, with states, mode operators and commutators written as differences of contour integrals, is reconstructed from them here. Only (G1)–(G5) are used, and the space of states is F itself. As throughout, fields with identical correlators are identified.

Fields as operators

Let $\bar{F} := \prod_{(h,\bar{h})} F_{(h,\bar{h})}$ be the space of formal sums of homogeneous fields. For $\phi \in F$ and $z \in \mathbb{C}^\times$ define $\phi(z)\colon F \to \bar{F}$ by

$$\phi(z)\chi := \sum_{(a,\bar{a})\in S} z^a \bar{z}^{\bar{a}}\, \psi_{(a,\bar{a})}, \tag{2.18}$$

where S and $\psi_{(a,\bar{a})}$ are the OPE data of (ϕ,χ) in (G5), expanded at $w = 0$. By Remark 2.1.2(iii) each weight space receives finitely many terms, so the right side is a well-defined element of $\bar{F}$. Put $|0\rangle := \mathbb{I}$ and define $\langle 0|\colon \bar{F} \to \mathbb{C}$ by $\langle 0|\psi := \langle \psi(0)\rangle$. This makes sense on $\bar{F}$: one-point functions are constant by translation invariance, a quasi-primary one-point function obeys $\langle\psi\rangle = \lambda^h\bar{\lambda}^{\bar{h}}\langle\psi\rangle$ for all $\lambda \in \mathbb{C}^\times$ and so vanishes unless $(h,\bar{h}) = (0,0)$, and the one-point function of a derivative is the derivative of a constant. Hence $\langle 0|$ sees only the component in $F_{(0,0)}$.

Theorem 2.2.1 (radial ordering)**.** *Let $\phi_1,\dots,\phi_n,\chi \in F$ and $|z_1| > |z_2| > \cdots > |z_n| > 0$. Then*

$$\langle \phi_1(z_1)\cdots\phi_n(z_n)\chi(0)\rangle = \langle 0|\phi_1(z_1)\,\phi_2(z_2)\cdots\phi_n(z_n)\,\chi, \tag{2.19}$$

where the right side is evaluated from the right: $\phi_n(z_n)\chi$ is formed first, $\phi_{n-1}(z_{n-1})$ is applied term by term, and so on. Each of the resulting iterated series converges absolutely.

Proof. Induction on n; for $n = 0$, (2.19) is the definition of $\langle 0|$. For $n \geq 1$, use (G1) to bring the pair $(\phi_n(z_n),\chi(0))$ to the front and apply (G5) to it, with $z_1,\dots,z_{n-1}$ as the remaining points. This is admissible because $|z_n - 0| < \min_{j<n}|z_j - 0|$ by hypothesis, and it gives the absolutely convergent expansion

$$\langle \phi_1(z_1)\cdots\phi_n(z_n)\chi(0)\rangle = \sum_{(a,\bar{a})} z_n^a \bar{z}_n^{\bar{a}}\, \langle \phi_1(z_1)\cdots\phi_{n-1}(z_{n-1})\psi_{(a,\bar{a})}(0)\rangle. \tag{2.20}$$

The points $z_1,\dots,z_{n-1}$ are still radially ordered, so the induction hypothesis applies to each term. □

Remark 2.2.2. *The order of the operators on the right of (2.19) is forced by the order of the $|z_i|$, while by (G1) the left side does not depend on any order. The operator products belonging to different radial orderings are therefore expansions of a single real-analytic function in different regions of $\mathcal{X}_n$. This is the precise meaning of the radially ordered product,*

$$\mathcal{R}\big[\phi_1(z_1)\phi_2(z_2)\big] := \begin{cases} \phi_1(z_1)\phi_2(z_2) & |z_1| > |z_2|, \\ \phi_2(z_2)\phi_1(z_1) & |z_2| > |z_1|, \end{cases} \tag{2.21}$$

and the time ordering on the cylinder $w = \log z$ is the same statement read through $z = e^w$. Nothing here is an assumption about operators; the operators (2.18) are defined by the correlators.

Corollary 2.2.3 (contour deformation)**.** *Let ξ be holomorphic on a neighbourhood of the closed annulus $r_- \leq |z| \leq r_+$, and suppose that the only insertion point with modulus in $[r_-,r_+]$ is w,*

with $r_- < |w| < r_+$. If $A(z)$ is a field whose correlators are holomorphic in z away from the other insertion points, then for radially ordered matrix elements

$$\langle 0|\cdots\Big(\oint_{|z|=r_+} - \oint_{|z|=r_-}\Big)\frac{dz}{2\pi i}\,\xi(z)\,\mathcal{R}\big[A(z)\phi(w)\big]\cdots\chi = \oint_{C_w}\frac{dz}{2\pi i}\,\xi(z)\,\langle\cdots A(z)\phi(w)\cdots\chi(0)\rangle. \tag{2.22}$$

Proof. By Theorem 2.2.1 both integrands are the same holomorphic function of z on the annulus minus w; Cauchy's theorem combines the two circles into a small circle around w. □

This corollary is what justifies the second equality in the derivation of (2.25) below, where the commutator $[Q_\xi, \phi(w)]$ is written as a difference of two contour integrals and then as a single one.

Remark 2.2.4 (states)**.** *By (G2), $\langle\phi(z)\mathbb{I}(0)\cdots\rangle = \langle\phi(z)\cdots\rangle$ is real-analytic near $z = 0$, so by (G3) and uniqueness of the expansion (2.3),*

$$\phi(z)|0\rangle = \sum_{m,n\geq 0}\frac{z^m\bar{z}^n}{m!\,n!}\,\partial^m\bar{\partial}^n\phi, \qquad \lim_{z\to 0}\phi(z)|0\rangle = \phi. \tag{2.23}$$

In this framework the state–operator correspondence of §2.3.1 is the identity map of F. What that section adds is a Hermitian form, which requires positivity (Definition 2.3.2).

2.2.2 Operator product expansion

OPE of $T(z)\phi(w)$

Motivation. This subsubsection derives (2.25) from the classical transformation law (1.19). In the axiomatic treatment adopted from §2.1 on, the logic runs the other way: (2.25) is the definition of a primary field (Definition 2.2.6).

Again, let ϕ be a chiral primary field of conformal dimension h. Its Lie derivative is represented by the charge $Q_\xi \equiv \oint \frac{dz}{2\pi i}\xi(z)T(z)$, hence

$$\begin{aligned}\delta_\xi\phi(w) &= -[Q_\xi, \phi(w)]\\ &= -\oint_{|z|>|w|}\frac{dz}{2\pi i}\xi(z)T(z)\phi(w) + \oint_{|z|<|w|}\frac{dz}{2\pi i}\xi(z)T(z)\phi(w)\\ &= -\oint_{C_w}\frac{dz}{2\pi i}\xi(z)T(z)\phi(w).\end{aligned} \tag{2.24}$$

Comparing to Eq. (1.19), this implies the OPE

$$T(z)\phi(w) \sim \frac{h\phi(w)}{(z-w)^2} + \frac{\partial\phi(w)}{z-w} \tag{2.25}$$

where we write $A \sim B$ if and only if $A - B$ is a regular function.

Remark 2.2.5. *The second equality is Corollary 2.2.3: by Theorem 2.2.1 the two radially ordered products are expansions of one function, holomorphic in z away from the insertions, so the two circles may be joined into C_w. In the axiomatic treatment the logic runs the other way: (2.25) is taken as the definition of a primary field (Definition 2.2.6), and this computation is its motivation.*

OPE of $T(z)T(w)$ and Schwarzian derivative

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{2.26a}$$

$$\Leftrightarrow \qquad \delta_\xi T(z) = -2\partial\xi(z)T(z) - \xi(z)\partial T(z) - \frac{c}{12}\partial^3\xi(z) \tag{2.26b}$$

We want to exponentiate Eq. (2.26b). The first two terms there are identical to Eq. (1.19). Since the third term does not depend on $T(z)$, let us define $\{z,w\}$ by

$$T'(w) = \left(\frac{dw}{dz}\right)^{-2} T(z) + \frac{c}{12}\{z,w\}. \tag{2.27}$$

It must satisfy the following properties.

$$(i) \quad \{w,z\} = -\left(\frac{dw}{dz}\right)^2 \{z,w\} \qquad \text{(inverse)} \tag{2.28a}$$

$$(ii) \quad \{z+\epsilon,z\} = \epsilon'''(z) + O(\epsilon^2) \qquad \text{(infinitesimal limit)} \tag{2.28b}$$

$$(iii) \quad \{u,z\} = \left(\frac{dw}{dz}\right)^2 \{u,w\} + \{w,z\} \qquad \text{(composition)} \tag{2.28c}$$

$$(iv) \quad \{z,w\} = 0 \quad \Leftrightarrow \quad w = \frac{az+b}{cz+d}, \quad ad-bc \neq 0 \qquad \text{(anomalous)} \tag{2.28d}$$

Taking $u = w + \delta w$, we find

$$\delta\{w,z\} \equiv \{w+\delta w,z\} - \{w,z\} \quad = (w')^2\partial_w^3\delta w \tag{2.29a}$$

$$= \delta\left\{\frac{w'''}{w'} - \frac{3}{2}\left(\frac{w''}{w'}\right)^2\right\}. \tag{2.29b}$$

Invoking the second condition in Eq. (2.28), this is integrated to what is called the *Schwarzian derivative*

$$\{w,z\} = \frac{w'''}{w'} - \frac{3}{2}\left(\frac{w''}{w'}\right)^2. \tag{2.30}$$

Free-field realizations of (2.26a), with $c = 1$, $\frac{1}{2}$ and $-2(6\lambda^2 - 6\lambda + 1)$, are computed in §2.2.4.

2.2.3 Ward identities

The stress tensor as an axiom

In the framework of §2.1 the stress tensor is an element of F, and primarity is a property of an operator product expansion.

(T) There is $T \in F^{\mathrm{qp}}_{(2,0)}$ with $\bar\partial T = 0$.

By (G3), $\langle T(z)\cdots\rangle$ is holomorphic in z away from the other insertions, so only pairs $(a,0)$ with $a \in \mathbb{Z}$ occur in its expansions (2.3). In terms of (2.18), $T(z)\chi = \sum_{n\in\mathbb{Z}} z^{-n-2}L_n\chi$ defines operators $L_n\colon F \to F$. We require that these modes implement the global transformations already present in (G3)–(G4): $L_{-1} = \partial$ on F, and L_0 acts on $F_{(h,\bar h)}$ by h, and likewise for $\bar L_{-1}, \bar L_0$ when $\bar T$ is present.

Definition 2.2.6. *$\phi \in F_{(h,\bar{h})}$ is primary if the singular part of the expansion of $\langle T(z)\phi(w)\cdots\rangle$ at $z = w$ is that of (2.25), i.e. $L_n\phi = 0$ for $n > 0$, $L_0\phi = h\phi$ and $L_{-1}\phi = \partial\phi$; if the theory also contains $\bar{T}$, the conjugate conditions are imposed as well.*

The derivation of (2.25) from the Lie derivative (1.19) motivates this definition, but in the present framework the definition comes first. With it, Lemma 2.2.7 is immediate and its $\sim$ acquires a precise meaning: near z_j the j-th summand of the right side is the singular part of the left side, and the other summands are holomorphic at z_j. Hence the difference of the two sides extends holomorphically to every z_j.

Correlation function of primaries

Just as gauge redundancy and general covariance induces gauge and gravitational Ward identities that constrain the correlation functions, conformal symmetry produces its own set of Ward identities. To establish the conformal Ward identities, we need the following lemma.

Lemma 2.2.7.

$$\langle T(z)\phi_1(z_1)\cdots\phi_N(z_N)\rangle \sim \sum_{j=1}^{N}\left[\frac{h_j}{(z-z_j)^2} + \frac{1}{z-z_j}\frac{\partial}{\partial z_j}\right]\langle\phi_1(z_1)\cdots\phi_N(z_N)\rangle. \tag{2.31}$$

Proof. We first observe that

$$\begin{aligned}\langle T(z)\phi_1(z_1)\cdots\phi_N(z_N)\rangle &= \oint_z \frac{dw}{2\pi i}\frac{1}{w-z}\langle T(w)\phi_1(z_1)\cdots\phi_N(z_N)\rangle \\ &\sim -\sum_{j=1}^{N}\oint_{z_j}\frac{dw}{2\pi i}\frac{1}{w-z}\langle T(w)\phi_1(z_1)\cdots\phi_N(z_N)\rangle.\end{aligned} \tag{2.32}$$

Eq. (2.32) is justified by the following. For each i, $T(w)\phi(z_i)$ develops a singularity at z_i as in Eq. (2.25). It follows that the sum of the contour integrals around z and all z_i equals the single contribution from the point at infinity. Using $\zeta = 1/w$, this contribution is

$$\begin{aligned}&\frac{1}{2\pi i}\oint_0 \frac{d\zeta}{\zeta^2}\frac{\zeta}{1-z\zeta}\langle T(\zeta^{-1})\phi_1(z_1)\cdots\phi_N(z_N)\rangle \\ &= \frac{1}{2\pi i}\sum_{n=0}^{\infty}\oint_0\frac{d\zeta}{\zeta}(z\zeta)^n\langle T'(\zeta)\zeta^4\phi_1(z_1)\cdots\phi_N(z_N)\rangle \\ &= \frac{1}{2\pi i}\sum_{n=0}^{\infty}\oint_0\frac{d\zeta}{\zeta}(z\zeta)^n\langle\sum_{k\in\mathbb{Z}}L'_k\zeta^{2-k}\phi_1(z_1)\cdots\phi_N(z_N)\rangle \\ &= \frac{1}{2\pi i}\sum_{n=0}^{\infty}z^n\langle L'_{n+2}\phi_1(z_1)\cdots\phi_N(z_N)\rangle,\end{aligned} \tag{2.33}$$

where T' is the stress-energy tensor in the coordinate chart ζ. Its component L'_{n+2} is inserted at the point of infinity ($\zeta = 0$). Eq. (2.33) is a regular function of z, which is hidden in Eq. (2.32).

We turn to evaluating Eq. (2.32). Using the formula

$$\begin{aligned} -\frac{1}{w-z} &= \frac{1}{[(z-z_i)+(z_i-w)]} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n n!}{(z-z_i)^{n+1}} \frac{(z_i-w)^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(w-z_i)^n}{(z-z_i)^{n+1}}, \end{aligned} \tag{2.34}$$

we find that the j-th summand of Eq. (2.32) is

$$\begin{aligned} &\sum_{n=0}^{\infty} \oint_{z_j} \frac{dw}{2\pi i} \frac{(w-z_j)^n}{(z-z_j)^{n+1}} \left[\frac{h_j}{(w-z_j)^2} + \frac{1}{w-z_j} \frac{\partial}{\partial z_j} + \text{reg.} \right] \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle \\ &\qquad = \left[\frac{h_j}{(z-z_j)^2} + \frac{1}{z-z_j} \frac{\partial}{\partial z_j} \right] \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle. \end{aligned} \tag{2.35}$$

□

Theorem 2.2.8 (conformal Ward identity)**.** *The similarity relation in Lemma 2.2.7 is in fact an honest equality.*

$$\langle T(z) \phi_1(z_1) \cdots \phi_N(z_N) \rangle = \sum_{j=1}^{N} \left[\frac{h_j}{(z-z_j)^2} + \frac{1}{z-z_j} \frac{\partial}{\partial z_j} \right] \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle \tag{2.36}$$

An incorrect attempt. We discuss an incorrect proof of Thm. 2.2.8. Denote the LHS of Eq. (2.36) by $A(z; z_1, \ldots, z_N)$, and the RHS of Eq. (2.36) by $B(z; z_1, \ldots, z_N)$. Thanks to Lemma 2.2.7, we could write the difference $A - B$ as a regular function

$$A - B = \sum_{n=0}^{\infty} c_n(z_1, \ldots, z_N) z^n. \tag{2.37}$$

Now, for arbitrary meromorphic function $\epsilon(z)$, a contour integral encircling all points $z, z_1, \ldots, z_N$ yields

$$\begin{aligned} \oint_{|z| \gg 1} \frac{dz}{2\pi i} \epsilon(z) A(z) &= \delta_\epsilon \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle \\ &= \sum_{j=1}^{N} \langle \delta_\epsilon \phi_j(z_j) \cdot \phi_1(z_1) \cdots \hat{\phi_j} \cdots \phi_N(z_N) \rangle \\ &= \sum_{j=1}^{N} \left[h_j (\partial \epsilon)(z_j) + \epsilon(z_j) \frac{\partial}{\partial z_j} \right] \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle \\ &= \oint_{|z| \gg 1} \frac{dz}{2\pi i} \epsilon(z) B(z) \\ \Rightarrow \quad 0 &= \sum_{n=0}^{\infty} c_n \oint_{|z| \gg 1} \frac{dz}{2\pi i} \epsilon(z) z^n. \end{aligned} \tag{2.38}$$

Taking $\epsilon(z) = z^{-n-1}$, we conclude $c_n = 0$.

This proof attempt fails at the third equality of Eq. (2.38): It expands the meromorphic function $\epsilon(z)$ at $z = z_j$, whose radius of convergence is at most $|z_j|$ due to the pole of $\epsilon(z)$ at 0. If there is another point of insertion z_k with $|z_k| > 2|z_j|$, then the contour must run outside of the domain of convergence (as the radius of the contour exceeds $|z_k|$). This impedes the Laurent expansion used in the third line of Eq. (2.38).

Proof of Thm. 2.2.8. Let $A(z)$ and $B(z)$ be the two sides of (2.36), as functions of $z \in \mathbb{C} \setminus \{z_1, \dots, z_N\}$, with the ϕ_j primary. By (T), A is holomorphic there, and by Definition 2.2.6 the difference $A - B$ has a removable singularity at every z_j. So $A - B$ is entire. As $z \to \infty$, $A(z) = O(|z|^{-4})$ by Lemma 2.1.4 applied to the quasi-primary field T of weight $(2, 0)$, whereas $B(z) = O(|z|^{-1})$. Thus $A - B$ is a bounded entire function tending to 0 at infinity, and it vanishes by Liouville's theorem. □

The proof uses three inputs: holomorphy of T (local conformal symmetry), the singular part of the $T\phi$ expansion (primarity), and Möbius covariance of T itself, which is what produces the decay at ∞. The incorrect attempt above never uses the third input. Without it nothing controls A at ∞: Lemma 2.2.7 alone only says that $A - B$ is entire.

Remark 2.2.9 (Operator-formalism reading of the proof)**.** *In operator language the vanishing of the coefficients in Eq.* (2.33) *is*

$$\langle L'_n \phi_1(x_1) \cdots \phi_N(x_N) \rangle = 0 \quad \forall n \geq 2. \tag{2.39}$$

By Theorem 2.2.1 the left side is a vacuum matrix element with T at the largest radius, $\langle 0 | L'_n \cdots | 0 \rangle$. The bra $\langle 0|$ is the state of $\mathbb{I}$ at ∞ (for $\mathbb{I}$ the chart is immaterial), and the modes L'_m of T in the chart $\zeta = 1/z$ (Remark 2.3.8) annihilate it for $m \geq -1$ for the same reason that L_m annihilates $|0\rangle$ (Remark 2.3.1). With $L'_m = L_{-m}$ (Lemma 2.3.7),

$$\langle 0 | L'_m = \langle 0 | L_{-m} = 0 \quad \forall m \geq -1. \tag{2.40}$$

This is the decay $A = O(z^{-4})$ written in modes; the cases $m = -1, 0, 1$ give Corollary 2.2.10. No adjoint is involved. In a unitary theory (2.40) *is also the adjoint of $L_m|0\rangle = 0$ (Theorem 2.3.9), but the Ward identity holds in non-unitary theories as well.*

It is not true that $\delta_\epsilon \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle = 0$ for arbitrary meromorphic function $\epsilon(z)$, which would imply

$$\langle T(z) \phi_1(z_1) \cdots \phi_N(z_N) \rangle = 0 \tag{2.41}$$

in Thm. 2.2.8. It is because the infinitesimal transformation δ_ϵ cannot be globalized to a true symmetry (element of the global conformal group $\mathrm{PSL}(2, \mathbb{C})$) in general. However, if δ_ϵ is restricted to be the generators of global conformal symmetry, i.e. $\epsilon(z) = c_0 + c_1 z + c_2 z^2$, the correlation functions are invariant under δ_ϵ. This proves the following corollary of Thm. 2.2.8.

Corollary 2.2.10 (Ward identities for global conformal symmetries)**.**

$$\sum_{j=1}^{N} \frac{\partial}{\partial z_j} \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle = 0,$$

$$\sum_{j=1}^{N} \left[h_j + z_j \frac{\partial}{\partial z_j} \right] \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle = 0,$$

$$\sum_{j=1}^{N} \left[2h_j z_j + z_j^2 \frac{\partial}{\partial z_j} \right] \langle \phi_1(z_1) \cdots \phi_N(z_N) \rangle = 0. \tag{2.42}$$

Remark 2.2.11. *Corollary 2.2.10 also follows from the proof of Theorem 2.2.8 without any appeal to δ_ϵ. Expanding $(z - z_j)^{-1} = \sum_{k \geq 0} z_j^k z^{-k-1}$ and $(z - z_j)^{-2} = \sum_{k \geq 0} (k+1) z_j^k z^{-k-2}$, the coefficients of z^{-1}, z^{-2}, z^{-3} in $B(z)$ are exactly the left sides of the three identities. Since $B = A = O(z^{-4})$, they vanish. Conversely, the three identities are the infinitesimal form of (G4) for the ϕ_j.*

2.2.4 Free fields

Normal ordering and Wick's theorem

Definition 2.2.12 (normal ordering)**.** *The normal ordering of a pair of operators A and B at a point w is the constant part of the OPE of $A(z)B(w)$ as $z \to w$. More formally,*

$$:A(w)B(w): \; = \oint_{C_w} \frac{dz}{2\pi i} \frac{1}{z - w} A(z)B(w). \tag{2.43}$$

In the framework of §2.1 the definition reads as follows. $:AB:(w)$ is the field $\psi_{(0,0)}$ in the expansion (2.3) of $A(z)B(w)$, so it is an element of F. When A is holomorphic only $a \in \mathbb{Z}$, $\bar{a} = 0$ occur, and the contour integral above extracts that term. For free fields the definition becomes the familiar subtraction of contractions.

Definition 2.2.13 (free fields)**.** *A family of fields, each even or odd, is free if every correlator of its members and their derivatives, at distinct points, is the sum over complete pairings of products of two-point functions $\langle XY \rangle$, the contractions. Each pairing carries the Koszul sign of the permutation that brings paired odd fields next to each other.*

Lemma 2.2.14. *For free fields, a correlator containing $:X_1 \cdots X_k:(w)$ is given by the same sum over pairings, with all pairings among $X_1, \ldots, X_k$ omitted.*

Proof. For $k = 2$: by Definition 2.2.13, $X_1(z)X_2(w) = \langle X_1(z)X_2(w) \rangle + R(z,w)$, where the correlators of R are the pairing sums without the pair (X_1, X_2). These are regular at $z = w$, so the constant term of the expansion is $R(w,w)$. Induction on k gives the general case. □

Proposition 2.2.15 (Wick's theorem)**.** *Let* $A = :A_1 \cdots A_k:(z)$ *and* $B = :B_1 \cdots B_l:(w)$ *be normal-ordered products of free fields. Then*

$$A\,B = \sum_{\mu} \epsilon_\mu \prod_{(i,j)\in\mu} \langle A_i(z) B_j(w)\rangle \; :\prod_{i\notin\mu} A_i(z) \prod_{j\notin\mu} B_j(w):\,, \tag{2.44}$$

where μ *runs over partial matchings between* $\{A_i\}$ *and* $\{B_j\}$ *(the empty matching included),* ϵ_μ *is the Koszul sign, and the normal ordering on the right omits every pairing among the unmatched fields.*

Proof. Insert both sides into a correlator with further free fields Y. By Lemma 2.2.14, the left side is the sum over pairings of $\{A_i\} \cup \{B_j\} \cup \{Y\}$ with no pair inside $\{A_i\}$ and none inside $\{B_j\}$. Group these pairings by the set μ of pairs between $\{A_i\}$ and $\{B_j\}$. For fixed μ, the remaining pairings match the unmatched A_i, B_j only with Y's, which by Lemma 2.2.14 is the correlator of the normal-ordered product on the right. □

The singular part of (2.44) at $z = w$ comes from $\mu \neq \emptyset$, after the unmatched $A_i(z)$ are Taylor expanded about w. Free fields are all that this note needs.

The free boson

Example 2.2.16 (Free boson)**.**

$$S = \frac{g}{2} \int \mathrm{d}^2x\, \partial_\mu\varphi\partial^\mu\varphi \tag{2.45}$$

Denoting the functional trace $\int d^2x$ *by* $\star$*,*

$$\begin{aligned}
\mathcal{Z}[J] &= \int D\varphi \exp\left[-S\right] \\
&= \int D\varphi \exp\left[-\frac{1}{2}\varphi\star(-g\partial^2)\varphi + J\star\varphi\right] \\
&= \mathcal{Z}[0] \exp\left[\frac{1}{2}J(x)\star G(x-y)\star J(y)\right]
\end{aligned} \tag{2.46}$$

where $G(x-y) = (-g\Delta)^{-1} = (-4g\partial\bar\partial)^{-1}$*. Using Corollary A.0.2,*

$$\langle\varphi(x)\varphi(y)\rangle = -\frac{1}{4\pi g}\left[\log(z-w) + \log(\bar z - \bar w)\right] + \text{const.} \tag{2.47}$$

To find the energy-momentum tensor, we make the metric tensor in (2.45) and perform local transformation $x_\mu \to x_\mu + \epsilon_\mu(x)$. Under this transformation, the variation of action is given by

$$\begin{aligned}
T_{\mu\nu} &= \frac{\partial\mathcal{L}}{\partial(\partial^\mu\varphi)}\partial_\nu\varphi - \eta_{\mu\nu}\mathcal{L} \\
&= g\left[\partial_\mu\varphi\partial_\nu\varphi - \frac{1}{2}\eta_{\mu\nu}(\partial^\lambda\varphi\partial_\lambda\varphi)\right]
\end{aligned} \tag{2.48}$$

In particular, $T_{00} = (g/2)(\varphi_t^2 + \varphi_x^2)$ matches the Hamiltonian density that would have been obtained if we directly Legendre-transformed (2.45). After Wick rotation and in the complex coordinates, we have the holomorphic part of the energy-momentum tensor $T_{zz} = g\partial\varphi\partial\varphi$. However, it is more convenient to use

$$T(z) = -2\pi T_{zz} = -2\pi g\partial\varphi(z)\partial\varphi(z). \tag{2.49}$$

The free fermion

The Euclidean action of a Majorana fermion is

$$S = \frac{g}{2}\int \mathrm{d}^2x\, \big(\psi\,\bar\partial\psi + \bar\psi\,\partial\bar\psi\big), \tag{2.50}$$

with a coupling g playing the role of the constant in (2.45). Varying and integrating by parts — the two terms of $\delta(\psi\bar\partial\psi)$ coincide, since interchanging two Grassmann fields costs a sign that the integration by parts restores — gives $\delta S = g\int \mathrm{d}^2x\,\delta\psi\,\bar\partial\psi$ and hence

$$\bar\partial\psi = 0, \qquad \partial\bar\psi = 0. \tag{2.51}$$

The field ψ is holomorphic and $\bar\psi$ antiholomorphic; the two decouple and we suppress $\bar\psi$ except where stated.

Since (2.50) is Gaussian with the antisymmetric kernel $g\bar\partial$, the propagator is its inverse, $g\,\bar\partial_z\langle\psi(z)\psi(w)\rangle = \delta(z-w)$, and Corollary A.0.2 solves this at once:

$$\big\langle\psi(z)\psi(w)\big\rangle = \frac{1}{\pi g}\,\frac{1}{z-w}. \tag{2.52}$$

Remark 2.2.17. *Unlike the boson, the fermion has no modulus: the rescaling $\psi \mapsto (\pi g)^{-1/2}\psi$ removes g from (2.52) without changing anything else. We therefore set $g = 1/\pi$ once and for all, so that*

$$\big\langle\psi(z)\psi(w)\big\rangle = \frac{1}{z-w}. \tag{2.53}$$

Every quantity computed below is independent of this choice.

We proceed exactly as for the boson. The canonical stress tensor of (2.50) is

$$T_{\mu\nu} = \frac{\partial^R\mathcal{L}}{\partial(\partial^\mu\psi)}\,\partial_\nu\psi - \eta_{\mu\nu}\mathcal{L}, \tag{2.54}$$

the derivative being taken from the right, as is appropriate for Grassmann variables. In complex coordinates $\eta_{zz} = 0$ and $\partial^z = g^{z\bar z}\partial_{\bar z} = 2\bar\partial$, so that only the first term survives and

$$T_{zz} = \frac{1}{2}\,\frac{\partial^R\mathcal{L}}{\partial(\bar\partial\psi)}\,\partial\psi = \frac{g}{4}\,\psi\,\partial\psi. \tag{2.55}$$

As in (2.49) it is more convenient to use

$$T(z) = -2\pi T_{zz} = -\frac{\pi g}{2}\,{:}\psi(z)\,\partial\psi(z){:} \overset{g=1/\pi}{=} -\frac{1}{2}\,{:}\psi(z)\,\partial\psi(z){:}\,, \tag{2.56}$$

the normal ordering being the same regularisation applied to (2.49).

Proposition 2.2.18. *ψ is primary of weight $h = \frac{1}{2}$:*

$$T(z)\psi(w) \sim \frac{\frac{1}{2}\,\psi(w)}{(z-w)^2} + \frac{\partial\psi(w)}{z-w}. \tag{2.57}$$

Proof. From (2.53), $\partial\psi(z)\psi(w) \sim -(z-w)^{-2}$. Contracting the external $\psi(w)$ with each factor of $:\psi\partial\psi:(z)$, with a sign when it is moved past $\partial\psi(z)$,

$$:\psi\,\partial\psi:(z)\,\psi(w) \sim -\frac{\psi(z)}{(z-w)^2} - \frac{\partial\psi(z)}{z-w} = -\frac{\psi(w)}{(z-w)^2} - \frac{2\,\partial\psi(w)}{z-w}, \tag{2.58}$$

where the second equality expands $\psi(z)$ about w. Multiplying by $-\frac{1}{2}$ gives the claim. Note that a single overall constant reproduces both poles with the ratio demanded by (2.25); this is the content of the statement that ψ is primary rather than a mere check of normalization. □

The ghost system and the central charges

Example 2.2.19 (The ghost system)**.** *Let b, c be Grassmann fields of weights $(\lambda, 0)$ and $(1-\lambda, 0)$, with action*

$$S = \frac{1}{\pi}\int \mathrm{d}^2x\; b\,\bar\partial c. \tag{2.59}$$

The integrand $b\,\bar\partial c$ is a section of $K \otimes \bar K$ for every λ, so S is conformally invariant. The computation that gave (2.53) gives

$$b(z)c(w) \sim \frac{1}{z-w} \sim c(z)b(w), \qquad b(z)b(w) \sim 0 \sim c(z)c(w). \tag{2.60}$$

The factor $\frac{1}{2}$ of the Majorana action is absent because b and c are independent. With

$$T = -\lambda\,:b\,\partial c: + (1-\lambda)\,:(\partial b)\,c:, \tag{2.61}$$

the fields b and c are primary of weights λ and $1-\lambda$, and

$$c_{bc} = -2\,(6\lambda^2 - 6\lambda + 1) \tag{2.62}$$

(Example 2.2.20). For $\lambda = \frac{1}{2}$ this gives $c = 1$, the value for two Majorana fermions, as it should. For $\lambda = 2$ it gives $c = -26$. Because $c < 0$, Corollary 2.3.5 shows that this system admits no unitary structure: $\|T\|^2 = c/2 < 0$.

Example 2.2.20 (central charges)**.** Free boson. *$T = -2\pi g :\partial\varphi\partial\varphi:$ and $\partial\varphi(z)\partial\varphi(w) \sim -\frac{1}{4\pi g}(z-w)^{-2}$ by (2.47). The two complete matchings give $(2\pi g)^2 \cdot 2 \cdot (4\pi g)^{-2}(z-w)^{-4} = \frac{1}{2}(z-w)^{-4}$, so $c = 1$. The four single matchings give $-4\pi g :\partial\varphi(z)\partial\varphi(w):(z-w)^{-2} = 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1} + \cdots$, the remaining poles of (2.26a).*

Free fermion. *With $\langle\psi\psi\rangle = 1/(z-w)$, the two complete matchings of $:\psi\partial\psi:(z):\psi\partial\psi:(w)$ give*

$$-\langle\psi(z)\psi(w)\rangle\langle\partial\psi(z)\partial\psi(w)\rangle + \langle\psi(z)\partial\psi(w)\rangle\langle\partial\psi(z)\psi(w)\rangle = \frac{2}{(z-w)^4} - \frac{1}{(z-w)^4}, \tag{2.63}$$

the minus sign being the Koszul sign. Multiplying by $(-\frac{1}{2})^2$ gives $\frac{1}{4}(z-w)^{-4}$, so $c = \frac{1}{2}$.

Ghosts. *The same bookkeeping, applied to (2.61), gives the fourth-order pole $\frac{1}{2}c_{bc}(z-w)^{-4}$ with c_{bc} as in (2.62). The single matchings of $T(z)$ with $b(w)$ and $c(w)$ give the primary expansions of weights λ and $1-\lambda$.*

2.2.5 Conformal families

Descendant fields

If ϕ is primary, the OPE of $T(z)\phi(w)$ implies

$$T(z)|\phi\rangle = \lim_{w\to 0} T(z)\phi(w)|0\rangle = \left[\frac{h\phi(0)}{z^2} + \frac{\partial\phi(0)}{z} + \text{reg.}\right]|0\rangle \tag{2.64a}$$

$$\Rightarrow \quad L_n|\phi\rangle = \oint_{C_0} \frac{dz}{2\pi i} z^{n+1} T(z)|\phi\rangle = \begin{cases} 0 & (n>0) \\ h|\phi\rangle & (n=0) \\ \text{another state} & (n<0) \end{cases} \tag{2.64b}$$

The states of the form $L_n|\phi\rangle$ where $n < 0$ and ϕ : primary are called descendant states. Formally,

$$T(z)\phi(w) = \sum_{k=0}^{\infty} \frac{\phi^{(k)}(w)}{(z-w)^{2-k}}, \tag{2.65a}$$

$$\phi^{(k)}(w) \equiv (\widehat{L}_{-k}\phi)(w) = \oint_w \frac{dz}{2\pi i}(z-w)^{1-k} T(z)\phi(w). \tag{2.65b}$$

Defining the operators $L_m^{(w)} \equiv \frac{1}{2\pi i}\oint_w dz\,(z-w)^{1+m}T(z)$, the field $(\widehat{L}_{-k}\phi)(w)$ as a function of w is obtained by the point-wise application of these operators, $w \mapsto L_{-k}^{(w)}\phi(w)$.

Proposition 2.2.21. *The operators $L_m^{(w)}$ are related to $L_m^{(0)} = L_m$ by*

$$L_m^{(w)} = e^{wL_{-1}} L_m e^{-wL_{-1}}. \tag{2.66}$$

Proof. Eq. (2.66) is obviously true for $w = 0$. It suffices to check the equality of derivatives.

$$\begin{aligned} \frac{d}{dw}(\text{LHS}) &= -(m+1)\oint_w \frac{dz}{2\pi i}\,(z-w)^m T(z) \\ &= -(m+1)L_{m-1} \\ &= [L_{-1}, L_m] = \frac{d}{dw}(\text{RHS}). \end{aligned} \tag{2.67}$$

□

Corollary 2.2.22. *The operators $L_m^{(w)}$ for fixed w form a Virasoro algebra.*

We may take the descendants $\phi^{(k_1,k_2)}$ of a descendant $\phi^{(k_1)}$ and so on, which we collectively call the descendants of ϕ. Together with $\phi^{(0)} = h\phi$, all the descendants $\phi^{\{k\}}$ form the conformal family $[\phi]$.

$$[\phi] := \left\{ \phi^{\{k\}} \middle| \{k\} \in \coprod_{N=1}^{\infty} \left(\mathbb{Z}_{\geq 0}\right)^N \right\} \tag{2.68}$$

We already saw that $T(z)$ is a secondary(=not primary) field. In fact, it belongs to the family $[\mathbb{I}]$ of descendants of the identity operator (constant field 1).

$$\mathbb{I}^{(2)}(w) = \widehat{L}_{-2}\mathbb{I}(w) = \int_{C_w} \frac{dz}{2\pi i}\frac{T(z)}{z-w}\mathbb{I}(w) = T(w) \tag{2.69}$$

Proposition 2.2.23. *The OPE of $T(z)T(w)$ is given by*

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

Proof. We can read the scaling dimension of the stress-energy tensor from its classical definition,

$$T_{\mu\nu} = \frac{-2}{\sqrt{g}} \frac{\delta S}{\delta g^{\mu\nu}}, \tag{2.70}$$

which is 2. Equivalently, $T(\lambda z) = \lambda^{-2} T(z)$. It follows that the most general form of the OPE of $T(z)T(w)$ is

$$T(z)T(w) \sim \frac{A}{(z-w)^4} + \frac{A'\,(\widehat{L}_1 T)(w)}{(z-w)^3} + \frac{B\,T(w)}{(z-w)^2} + \frac{C\,\partial T(w)}{z-w}. \tag{2.71}$$

where we used $\widehat{L}_2 T \propto \mathbb{I}$ in the first summand. Since $T = \widehat{L}_{-2}\mathbb{I}$, we have $\widehat{L}_1 T \propto \widehat{L}_{-1}\mathbb{I} \propto \partial \mathbb{I} = 0$. Thus, we neglect A'. The other coefficients are fixed by demanding that the generators L_n form a Virasoro algebra. Using the property of radial ordering,

$$\begin{aligned}
[L_m, L_n] &= \oint_0 \frac{dw}{2\pi i} w^{1+n} \oint_w \frac{dz}{2\pi i} z^{1+m} T(z)T(w) \\
&= \oint_0 \frac{dw}{2\pi i} w^{1+n} \oint_w \frac{dz}{2\pi i} (z-w+w)^{1+m} \left[\frac{A}{(z-w)^4} + \frac{B\,T(w)}{(z-w)^2} + \frac{C\,\partial T(w)}{z-w} \right] \\
&= \oint_0 \frac{dw}{2\pi i} w^{1+n} \left[\frac{A}{6} w^{m-2}(m^3 - m) + B(1+m) w^m T(w) + C w^{m+1} \partial T(w) \right] \\
&= \frac{A}{6}(m^3 - m)\delta_{m+n,0} + B(1+m) L_{m+n} - C(m+n+2) L_{m+n}.
\end{aligned} \tag{2.72}$$

Comparing with Eq. (1.27), we get $A = c/2,\ B = 2,\ C = 1$. □

Correlation function of descendants

Theorem 2.2.24. $\langle \widehat{L}_{-k}\phi(z)\phi_1(z_1)\cdots\phi_N(z_N)\rangle = \mathcal{L}_{-k}\langle \phi(z)\phi_1(z_1)\cdots\phi_N(z_N)\rangle$, *where*

$$\mathcal{L}_{-k} := \sum_{j=1}^{N} \left[\frac{(k-1)h_j}{(z_j - z)^k} - \frac{1}{(z_j - z)^{k-1}} \frac{\partial}{\partial z_j} \right]. \tag{2.73}$$

Proof. Denoting $\phi_1(z_1)\cdots\phi_N(z_N)$ by X,

$$\begin{aligned}
\langle \widehat{L}_{-k}\phi(z)X\rangle &= \oint_z \frac{dw}{2\pi i} (w-z)^{1-k} \langle T(w)\phi(z)X\rangle \\
&\sim -\sum_{j=1}^{N} \oint_{z_j} \frac{dw}{2\pi i} (w-z)^{1-k} \left[\frac{h_j}{(w-z_j)^2} + \frac{1}{(w-z_j)} \frac{\partial}{\partial z_j} \right] \langle \phi(z)X\rangle
\end{aligned} \tag{2.74}$$

Using the expansion $(w-z)^{1-k} = (z_j - z)^{1-k} + (1-k)(z_j - z)^{-k}(w - z_j) + O((w-z_j)^2)$, we have

$$\langle \widehat{L}_{-k}\phi(z)X\rangle \sim \sum_{j=1}^{N} \left[\frac{(k-1)h_j}{(z_j - z)^k} - \frac{1}{(z_j - z)^{k-1}} \frac{\partial}{\partial z_j} \right] \langle \phi(z)X\rangle. \tag{2.75}$$

It remains to show that the hidden regular term vanishes, which is the contribution of point at infinity:

$$\begin{aligned}
&\int_{|w|\gg 1} \frac{dw}{2\pi i}(w-z)^{1-k}\langle T(w)\phi^{-k_2}(z)X\rangle \\
&= \frac{1}{2\pi i}\int_0 \frac{d\zeta}{\zeta^2}\left(\frac{\zeta}{1-z\zeta}\right)^{k-1}\langle T'(\zeta^{-1})\zeta^4\phi^{-k_2}(z)X\rangle \\
&= \frac{1}{2\pi i}\int_0 \frac{d\zeta}{\zeta^2}\zeta^{k-1}\sum_{n=0}^{\infty}\binom{k+n-2}{n}(z\zeta)^n\langle T'(\zeta^{-1})\zeta^4\phi^{-k_2}(z)X\rangle \\
&= \frac{1}{2\pi i}\int_0 \frac{d\zeta}{\zeta}\zeta^{k-2}\sum_{n=0}^{\infty}\binom{k+n-2}{n}(z\zeta)^n\langle\sum_{k\in\mathbb{Z}} L'_k\zeta^{2-k}\phi^{-k_2}(z)X\rangle \\
&= \sum_{n=0}^{\infty}\binom{k+n-2}{n}z^n\langle L'_{k+n}\phi^{-k_2}(z)X\rangle \\
&= 0, \qquad (2.76)
\end{aligned}$$

where we used Eq. (2.40) once again. □

Remark 2.2.25. *The term at infinity vanishes without the operator formalism. Lemma 2.1.4, applied to T, requires only the field at ∞ to be quasi-primary; the other fields may be arbitrary elements of F, descendants included. Hence $(w-z)^{1-k}\langle T(w)\phi(z)X\rangle = O(|w|^{-3-k})$, and the integral over $|w|=R$ tends to 0 as $R\to\infty$ for $k\geq -1$. The integral is independent of R for large R, so it is zero. The same argument covers Corollary 2.2.26.*

Thm. 2.2.24 generalizes Thm. 2.2.10, which can be recovered by setting $\phi = 1, z = 0$, and $k = 1, 0, -1$. One can say even more:

Corollary 2.2.26. *The following holds.*

$$\langle \widehat{L}_{-k_1}\cdots\widehat{L}_{-k_n}\phi(z)\phi_1(z_1)\cdots\phi_N(z_N)\rangle = \mathcal{L}_{-k_1}\cdots\mathcal{L}_{-k_n}\langle\phi(z)\phi_1(z_1)\cdots\phi_N(z_N)\rangle \qquad (2.77)$$

Proof. The proof of (2.2.24) generalizes verbatim, as we never used the fact that $\phi(z)$ is primary. □

In principle, we can iterate the computation involved in Thm. 2.2.24 to express arbitrary correlation functions of secondary fields in terms of the correlation function of their ancestor primaries. To this end, we must know the OPE between the stress-energy tensor and descendant fields. This can be derived by using the general relation (2.65) and the Virasoro algebra (1.27).

We observe that, for $m, k > 0$,

$$\begin{aligned}
\widehat{L}_m\phi^{(k)}(z) = L^{(z)}_m L^{(z)}_{-k}\phi(z) &= [L^{(z)}_m, L^{(z)}_{-k}]\phi(z) + L^{(z)}_{-k}\underbrace{L^{(z)}_m\phi(z)}_{=0} \\
&= (m+k)L^{(z)}_{m-k}\phi(z) + \frac{c}{12}(k^3-k)\delta_{m,k}\phi(z) \\
&= (m+k)\phi^{(k-m)}(z) + \frac{c}{12}(k^3-k)\delta_{m,k}\phi(z). \qquad (2.78)
\end{aligned}$$

Using this, we prove the following.

Proposition 2.2.27 (OPE of $T(z)$ and descendants)**.**

$$T(w)\phi^{(k)}(z) \sim \frac{\partial\phi^{(k)}(z)}{(w-z)} + \frac{(h+k)\phi^{(k)}(z)}{(w-z)^2} + \sum_{r=1}^{k-1} \frac{(k+r)\phi^{(k-r)}(z)}{(w-z)^{r+2}} + \frac{2kh + \frac{c}{12}(k^3-k)}{(w-z)^{k+2}}\phi(z).$$

Proof. Using (2.78), this is just the general relation (2.65):

$$T(w)\Phi(z) = \sum_{r=0}^{\infty} \frac{\widehat{L}_r\Phi(z)}{(w-z)^{2+r}}.$$

□

Example 2.2.28. *We compute the two-point correlation of descendant fields. Deforming the contour and understanding that the contribution of the point at infinity vanishes as in Theorem 2.2.24,*

$$\begin{aligned}
\langle\phi_1^{(k_1)}(z_1)\phi_2^{(k_2)}(z_2)\rangle &= \oint_{z_1} \frac{dw}{2\pi i}(w-z_1)^{1-k_1}\langle T(w)\phi_1(z_1)\phi_2^{(k_2)}(z_2)\rangle \\
&= -\oint_{z_2} \frac{dw}{2\pi i}(w-z_1)^{1-k_1}\langle \phi_1(z_1)T(w)\phi_2^{(k_2)}(z_2)\rangle \\
&= -\sum_{m=0}^{\infty} \binom{1-k_1}{m} (z_2-z_1)^{1-m-k_1}\langle\phi_1(z_1)F(m;w-z_2)\rangle
\end{aligned}$$

where (cf. Proposition 2.2.27)

$$F(m;w-z_2) = \oint_{z_2} \frac{dw}{2\pi i}(w-z_2)^m T(w)\phi_2^{(k_2)}(z_2) = \begin{cases} \partial\phi_2^{(k_2)}(z_2) & (m=0) \\ (h+k_2)\phi_2^{(k_2)}(z_2) & (m=1) \\ (k_2+m-1)\phi_2^{(k_2-m+1)}(z_2) & (2\le m\le k) \\ [2k_2h + \frac{c}{12}(k_2^3-k_2)]\phi_2(z_2) & (m=k+1) \\ 0 & (\textit{otherwise}). \end{cases}$$

Applying Theorem 2.2.24, the 2-point function of descendant fields completely reduces to that of primary fields.

2.3 States and Virasoro representations

2.3.1 State-operator correspondence

The vacuum

In CFT, the existence of a vacuum is assumed such that the vacuum is invariant under the global conformal transformations. Furthermore, one cannot "lower" the energy level of the vacuum. Hence,

$$L_n|0\rangle = 0 \qquad \forall n \ge -1. \tag{2.79}$$

Remark 2.3.1 (the vacuum is not an extra assumption)**.** *In the framework of Definition 2.1.1 the vacuum is* $|0\rangle = \mathbb{I}$*, and the property just stated is a theorem. By Remark 2.2.4,* $T(z)\mathbb{I} = \sum_{k\geq 0} z^k \partial^k T/k!$ *contains no negative powers of* z*; comparing with* $T(z)\mathbb{I} = \sum_n z^{-n-2} L_n \mathbb{I}$ *gives* $L_n|0\rangle = 0$ *for* $n \geq -1$ *and* $L_{-2}|0\rangle = T$*. Dually,* $\langle 0|T(z)\chi = \langle T(z)\chi(0)\rangle$ *is* $O(z^{-4})$ *by Lemma 2.1.4, so* $\langle 0|L_n = 0$ *for* $n \leq 1$*.*

Kets and bras

By Remark 2.2.4 the in-state of a field is the field itself,

$$|\phi\rangle := \lim_{z\to 0} \phi(z)|0\rangle = \phi \in F. \tag{2.80a}$$

For quasi-primary ϕ of weight $(h, \bar{h})$ the out-state is the linear functional

$$\langle\phi| := \lim_{\zeta\to\infty} \zeta^{2h}\bar{\zeta}^{2\bar{h}}\langle 0|\phi(\zeta, \bar{\zeta}), \qquad \langle\phi|\chi\rangle = \langle\phi(\infty)\chi(0)\rangle, \tag{2.80b}$$

which exists by Lemma 2.1.4. By Theorem 2.1.6, $\langle\phi_1|\phi_2\rangle = C_{12}$ for quasi-primaries of equal weight and vanishes otherwise. This pairing is bilinear (the BPZ pairing). It involves no complex conjugation and exists in every theory, unitary or not. A Hermitian inner product needs an additional structure, described next.

Conjugation is a reflection. On the cylinder $w = \tau - ix$, $z = e^w$, translation in τ is generated by the dilatation operator $D = L_0 + \bar{L}_0$, and a Hermitian Euclidean field satisfies $\phi(\tau, x)^\dagger = \phi(-\tau, x)$, because $\phi(\tau, x) = e^{D\tau}\phi(0, x)e^{-D\tau}$ with D Hermitian. (Up to normalization and a constant, D is the Hamiltonian of §3.1.1.) The map $\tau \mapsto -\tau$ is $w \mapsto -\bar{w}$. On the plane $z = e^w$ it becomes

$$\theta(z) = 1/\bar{z}, \tag{2.81}$$

the reflection in the unit circle. It is anticonformal, an element of Conf^-, and sends $(z, \bar{z})$ to $(1/\bar{z}, 1/z)$. Write $\phi(z, \bar{z}) = z^{-h}\bar{z}^{-\bar{h}}\phi_{\mathrm{cyl}}(w, \bar{w})$, use $(z^h\bar{z}^{\bar{h}})^* = \bar{z}^h z^{\bar{h}}$ (spin-adapted, with base $\bar{z}$), and transport back with $z' = e^{-\bar{w}} = 1/\bar{z}$. This gives

$$\phi(z, \bar{z})^\dagger = \bar{z}^{-2h} z^{-2\bar{h}}\, \phi^\dagger(1/\bar{z}, 1/z), \tag{2.82}$$

where $\phi^\dagger$ is the conjugate field ($\phi^\dagger = \phi$ for a Hermitian field). The holomorphic slot of $\phi^\dagger$ receives $1/\bar{z}$. The cylinder argument presupposes a Hilbert space. In the present framework the logic is reversed: (2.82) becomes the axiom, and the Hilbert space is constructed from it.

Definition 2.3.2 (unitary structure)**.** *A unitary structure on a genus-zero correlation system consists of an antilinear involution* $\phi \mapsto \phi^\dagger$ *of* F^{qp} *preserving each* $F^{\mathrm{qp}}_{(h,\bar{h})}$*, with* $\mathbb{I}^\dagger = \mathbb{I}$ *and* $T^\dagger = T$*, and a sesquilinear form* $(\cdot, \cdot)$ *on* F*, antilinear in the first slot, such that:*

(U1) for quasi-primary ϕ, χ *and* $0 < |w|, |v| < 1$*,*

$$\big(\phi(w)|0\rangle,\, \chi(v)|0\rangle\big) = \big\langle \Theta[\phi(w)]\, \chi(v)\big\rangle, \qquad \Theta[\phi(w)] := \bar{w}^{-2h} w^{-2\bar{h}}\, \phi^\dagger(1/\bar{w}, 1/w); \tag{2.83}$$

(U2) for quasi-primary ϕ, $z \in \mathbb{C}^\times$ and $\chi_1, \chi_2 \in F$,

$$\big(\phi(z)\chi_1,\, \chi_2\big) = \big(\chi_1,\, \bar{z}^{-2h} z^{-2\bar{h}} \phi^\dagger(1/\bar{z}, 1/z)\chi_2\big); \tag{2.84}$$

(U3) $(\cdot,\cdot)$ is positive definite.

Remark 2.3.3. *(U1) is a reflection positivity condition, restricted to single insertions in the unit disc; these suffice, since the Taylor coefficients of the states $\phi(w)|0\rangle$ span F. (U2) is* (2.82) *read as an operator statement. By Proposition 2.3.4 below, which uses only (U1), the form is weight-diagonal, so the right side of* (2.84) *is a finite sum. (U1) is the case $\chi_1 = |0\rangle$ of (U2), once $(\mathbb{I}, \cdot) = \langle 0|$ and Theorem 2.2.1 are used. Letting $w \to 0$ in* (2.83) *gives*

$$(\phi, \chi) = \langle \phi^\dagger(\infty)\chi(0)\rangle = \langle \phi^\dagger | \chi \rangle, \tag{2.85}$$

so the Hermitian form is the BPZ pairing composed with $\dagger$. No sign appears.

Proposition 2.3.4 (norms of global descendants)**.** *Assume (U1). Let ϕ, χ be quasi-primary of weights $(h, \bar{h})$ and $(h', \bar{h}')$. Then $(\partial^m \bar{\partial}^n \phi, \partial^k \bar{\partial}^l \chi) = 0$ unless $(h, \bar{h}) = (h', \bar{h}')$, in which case*

$$\big(\partial^m \bar{\partial}^n \phi,\, \partial^k \bar{\partial}^l \chi\big) = \delta_{mk}\, \delta_{nl}\, m!\, (2h)_m\, n!\, (2\bar{h})_n\, (\phi, \chi), \tag{2.86}$$

where $(a)_m = a(a+1)\cdots(a+m-1)$ and $(\phi, \chi) = \langle \phi^\dagger(\infty)\chi(0)\rangle$.

Proof. $\phi^\dagger$ has weight $(h, \bar{h})$, so by Theorem 2.1.6 the right side of (2.83) vanishes unless the weights agree. If they agree, set $C := \langle \phi^\dagger(\infty)\chi(0)\rangle$. With $z_1 = 1/\bar{w}$ and $z_2 = v$ we have $z_{12} = (1 - \bar{w}v)/\bar{w}$ and $\bar{z}_{12} = (1 - w\bar{v})/w$. Multiplicativity of spin-adapted powers then gives

$$\big\langle \Theta[\phi(w)]\, \chi(v) \big\rangle = C\, (1 - \bar{w}v)^{-2h} (1 - w\bar{v})^{-2\bar{h}} = C \sum_{m,n \geq 0} \frac{(2h)_m}{m!} \frac{(2\bar{h})_n}{n!}\, (\bar{w}v)^m (w\bar{v})^n. \tag{2.87}$$

For $|w|, |v| < 1$ the base lies in the disc $|\lambda - 1| < 1$. There the spin-adapted power equals the product of principal branches, so the binomial series applies. On the other side, Remark 2.2.4 and sesquilinearity give

$$\big(\phi(w)|0\rangle, \chi(v)|0\rangle\big) = \sum_{m,n,k,l} \frac{\bar{w}^m w^n v^k \bar{v}^l}{m!\, n!\, k!\, l!} \big(\partial^m \bar{\partial}^n \phi, \partial^k \bar{\partial}^l \chi\big). \tag{2.88}$$

Comparing coefficients of the independent monomials $\bar{w}^m w^n v^k \bar{v}^l$ gives (2.86); the case $m = n = k = l = 0$ identifies $C = (\phi, \chi)$. □

Corollary 2.3.5 (unitarity bounds)**.** *Assume (U1) and (U3). Every nonzero quasi-primary field has $h \geq 0$ and $\bar{h} \geq 0$. If $h = 0$ then $\partial\phi = 0$, and if $\bar{h} = 0$ then $\bar{\partial}\phi = 0$. Moreover $c \geq 0$, and $c = 0$ only if $T = 0$.*

Proof. By (2.86), $\|\partial\phi\|^2 = 2h\|\phi\|^2$ and $\|\bar{\partial}\phi\|^2 = 2\bar{h}\|\phi\|^2$, with $\|\phi\|^2 > 0$ by (U3). For the central charge, $T^\dagger = T$, so $\|T\|^2 = \langle T(\infty)T(0)\rangle = c/2$, read off from the fourth-order pole of the TT expansion (2.26a) and Theorem 2.1.6. □

Remark 2.3.6. *The bound $h \geq 0$ uses only (U1) and the global conformal group, through Proposition 2.3.4; the Virasoro algebra is not needed. Corollary 2.3.5 is also the quickest way to see that the Lee–Yang model of Remark 2.4.13, which has a primary of negative weight, cannot be unitary.*

Lemma 2.3.7 (modes at infinity)**.** *Let L'_m be the modes of T in the chart $\zeta = 1/z$ at ∞, i.e. $T'(\zeta) = \sum_m \zeta^{-m-2} L'_m$. Then $L'_n = L_{-n}$.*

Proof. By definition, we have the mode expansion

$$T(z) = \sum_n z^{-n-2} L_n, \qquad T'(\zeta) = \sum_m \zeta^{-m-2} L'_m \qquad (\zeta = 1/z). \tag{2.89}$$

Applying (2.27) leads to

$$T(z(\zeta)) = T'(\zeta) \left(\frac{d\zeta}{dz}\right)^2 = T'(\zeta)\zeta^4 + \frac{c}{12}\cancel{\{\zeta, z\}}, \tag{2.90}$$

where the Schwarzian derivative vanishes due to (2.28d). Thus,

$$T(z) = \sum_n z^{-n-2} L_n = \sum_n \zeta^{n+2} L_n = \sum_m \zeta^{-m+2} L'_m, \tag{2.91}$$

which implies $L'_n = L_{-n}$. □

This is a statement about a change of chart and involves no adjoint. Combined with the decay $\langle T(z) \cdots \rangle = O(z^{-4})$ of Lemma 2.1.4, it gives (2.40). The adjoint needs (U2).

Remark 2.3.8 (chart)**.** *This lemma is the one place where the chart at ∞ is $\zeta = 1/z$ rather than the $u = -1/z$ of Lemma 2.1.4. With $u = -1/z$ the same computation gives $L'_n = (-1)^n L_{-n}$. The two charts differ by the rotation $u \mapsto -u$, which acts on $F_{(h,\bar{h})}$ by $(-1)^s$. The vanishing statements (2.40) are insensitive to the choice; $1/z$ is used here only so that the mode relation carries no sign.*

Theorem 2.3.9. *In a theory with a unitary structure, $(L_n\chi_1, \chi_2) = (\chi_1, L_{-n}\chi_2)$ for all $\chi_1, \chi_2 \in F$; that is, $L_n^\dagger = L_{-n}$.*

Proof. Apply (2.84) to T, which has $(h, \bar{h}) = (2, 0)$ and $T^\dagger = T$. Since the form is antilinear in the first slot, the left side is

$$\sum_n \bar{z}^{-n-2} (L_n\chi_1, \chi_2).$$

The right side is $\bar{z}^{-4} \sum_m \bar{z}^{m+2} (\chi_1, L_m\chi_2) = \sum_m \bar{z}^{m-2} (\chi_1, L_m\chi_2)$. Both are finite sums by weight-diagonality. Comparing the coefficients of $\bar{z}^{-n-2}$ gives $m = -n$. □

Remark 2.3.10 (on the earlier convention)**.** *The earlier version of this subsection implemented conjugation by the holomorphic inversion $z \mapsto 1/z$. That map is θ composed with the parity $z \mapsto \bar{z}$ ($x \mapsto -x$), i.e. the rotation $w \mapsto -w$ of the cylinder by π. It preserves orientation, whereas the reflection that implements the adjoint reverses it. In the full theory its factor $(-z^{-2})^h (-\bar{z}^{-2})^{\bar{h}} = (-1)^s z^{-2h} \bar{z}^{-2\bar{h}}$ is well defined. In chiral notation it becomes $(-1)^h$, which is undefined for $h \notin \mathbb{Z}$. With θ no sign occurs at all, and (2.80b) needs no absorbed sign.*

Remark 2.3.11. *The operator reading of Eq.* (2.39) *promised in §2.2 is Remark 2.2.9: it rests on Lemma 2.3.7 and* (2.40)*, not on Theorem 2.3.9.*

2.3.2 Verma module

Replacing all fields $\phi^{\{k\}}$ by the corresponding states $|\phi^{\{k\}}\rangle$ in Eq. (2.68) yields the Verma module $V(c,h)$. There is a natural grading of Verma module by the eigenvalues of L_0.

$$V(c,h) = \bigoplus_{n=0}^{\infty} V^{(n)}(c,h) \tag{2.92a}$$

$$V^{(n)}(c,h) := \left\{ |\phi\rangle \,\middle|\, L_0|\phi\rangle = (h+n)|\phi\rangle \right\} \tag{2.92b}$$

Since $[L_0, L_{-k}] = kL_{-k}$ $(k>0)$, we have $L_0|\phi^{\{k\}}\rangle = h + \mathrm{Sum}\{k\}$.

To count the states at each level, consider the following auxiliary grading of $V^{(n)}(c,h)$ by $\mathbb{Z}_{\geq 0}$.

$$V^{(n)}(c,h) = \bigoplus_{d=0}^{\infty} V^{(n;d)}(c,h) \tag{2.93a}$$

$$V^{(n;d)}(c,h) := \left\{ |\phi^{\{k\}}\rangle \in V^{(n)}(c,h) \,\middle|\, \mathrm{Card}\{k\} = d \right\} \tag{2.93b}$$

where Card = cardinality. For instance, $\deg L_{-4}\phi = 1$ and $\deg L_{-1}L_{-3}\phi = 2$ in $V^{(4)}(c,h)$. Using the Virasoro algebra, we notice that the induced action of L_n $(n<0)$ on the quotient module

$$W^{(n;l)}(c,h) := \bigoplus_{d=0}^{l} V^{(n;d)}(c,h) \Big/ \bigoplus_{d=0}^{l-1} V^{(n;d)}(c,h) \simeq V^{(n;l)}(c,h) \tag{2.94}$$

is abelian; $[L_{-m}, L_{-n}] = (n-m)L_{-m-n} = 0$ $(m,n>0)$ as a map $W^{(k;d)} \to W^{(k+m+n;d+2)}$. This implies that $\dim V^{(n)} = P(n)$ is given by the number theory partition function, which counts the number of ways to partition n into natural numbers, identifying the permuted partitions.

2.3.3 Null states and irreducible representations

However, for certain values of c and h, $V(c,h)$ contains *null states*, which are descendant states but are annihilated by all L_n $(n>0)$. They render the Verma module reducible, which is remedied by quotienting the null states out. Therefore, an irreducible module has at most $P(n)$ states at level n, for general irreducible modules of the Virasoro algebra.

Consider the Gram matrix of all states at level N of the Verma module $V_{c,h}$.

$$[G^{(N)}_{c,h}]_{\boldsymbol{mn}} = \langle h, \boldsymbol{m}|h, \boldsymbol{n}\rangle \qquad (|\boldsymbol{m}| = |\boldsymbol{n}| = N) \tag{2.95}$$

Theorem 2.3.12 (Kac). *A null state of level l appears if and only if (c,h) is a root of the Kac determinant,*

$$\det G^{(N)}_{c,h} = \alpha_l \prod_{\substack{r,s\geq 1\\ rs\leq N}} [h - h_{r,s}(c)]^{P(N-rs)}, \tag{2.96}$$

where P is the partition function. The functions $h_{r,s}(c)$ has several convenient representations.

$$\begin{aligned} h_{r,s}(c) &= h_0 + \frac{1}{4}(r\alpha_+ + s\alpha_-)^2 \\ h_0 &= \frac{c-1}{24} \\ \alpha_\pm &= \frac{\sqrt{1-c} \pm \sqrt{25-c}}{\sqrt{24}} \end{aligned} \tag{2.97}$$

It is useful to have other representations of the Kac formula (2.97).

$$c = 13 - 6\left(t + \frac{1}{t}\right), \qquad h_{r,s}(t) = \frac{1}{4}(r^2-1)t + \frac{1}{4}(s^2-1)\frac{1}{t} - \frac{1}{2}(rs-1) \tag{2.98a}$$

$$c = 1 - \frac{6}{m(m+1)}, \qquad h_{r,s}(m) = \frac{[(m+1)r - ms]^2 - 1}{4m(m+1)} \tag{2.98b}$$

The relation between the representations (2.97), (2.98a), and (2.98b) is given by

$$t = 1 + \frac{1}{12}\left[1 - c \pm \sqrt{(1-c)(25-c)}\right], \qquad m = -\frac{1}{2} \pm \frac{1}{2}\sqrt{\frac{25-c}{1-c}} \tag{2.99a}$$

$$t = \frac{m}{m+1} \quad \text{or} \quad t = \frac{m+1}{m} \tag{2.99b}$$

Theorem 2.3.12 asserts the existence of null states but does not exhibit any. It is worth constructing the simplest one by hand, both as a check on the Kac formula and because this particular vector is the input to the differential equations of Belavin, Polyakov and Zamolodchikov.

Proposition 2.3.13 (level-two singular vector). *Let $h \neq -\frac{1}{2}$. The Verma module $V(c,h)$ contains a singular vector at level 2 if and only if*

$$c = \frac{2h(5-8h)}{2h+1}, \tag{2.100}$$

in which case it is unique up to scale and given by

$$|\chi\rangle = \left(L_{-2} - \frac{3}{2(2h+1)}L_{-1}^2\right)|h\rangle. \tag{2.101}$$

Proof. A level-two vector is $|\chi\rangle = (\alpha L_{-2} + \beta L_{-1}^2)|h\rangle$, and being singular means $L_1|\chi\rangle = L_2|\chi\rangle = 0$; the remaining conditions $L_n|\chi\rangle = 0$ for $n \geq 3$ follow since L_1, L_2 generate. From the Virasoro algebra (1.27),

$$\begin{aligned} L_1 L_{-2}|h\rangle &= [L_1, L_{-2}]|h\rangle = 3L_{-1}|h\rangle, \\ L_1 L_{-1}^2|h\rangle &= \left([L_1, L_{-1}] + L_{-1}L_1\right) L_{-1}|h\rangle = 2L_0 L_{-1}|h\rangle + L_{-1}[L_1, L_{-1}]|h\rangle \\ &= 2(h+1)L_{-1}|h\rangle + 2hL_{-1}|h\rangle = 2(2h+1)L_{-1}|h\rangle, \end{aligned} \tag{2.102}$$

so that

$$L_1|\chi\rangle = \big[3\alpha + 2\beta(2h+1)\big] L_{-1}|h\rangle. \tag{2.103}$$

Similarly, using $[L_2, L_{-2}] = 4L_0 + \frac{c}{2}$ and $[L_2, L_{-1}] = 3L_1$,

$$\begin{aligned} L_2 L_{-2}|h\rangle &= \left(4h + \frac{c}{2}\right)|h\rangle, \\ L_2 L_{-1}^2|h\rangle &= 3L_1 L_{-1}|h\rangle + L_{-1}\cdot 3L_1|h\rangle = 6h|h\rangle, \end{aligned} \tag{2.104}$$

so that

$$L_2|\chi\rangle = \left[\alpha\left(4h + \frac{c}{2}\right) + 6\beta h\right]|h\rangle. \tag{2.105}$$

Suppose first $\alpha = 0$. Then (2.103) forces $\beta(2h+1) = 0$, hence $\beta = 0$ by hypothesis, and there is no singular vector. We may therefore normalize $\alpha = 1$. Now (2.103) gives $\beta = -\frac{3}{2(2h+1)}$, which is (2.101), and substituting into (2.105),

$$4h + \frac{c}{2} = \frac{9h}{2h+1} \quad \Longleftrightarrow \quad \frac{c}{2} = \frac{9h - 4h(2h+1)}{2h+1} = \frac{h(5-8h)}{2h+1}, \tag{2.106}$$

which is (2.100). Uniqueness up to scale is clear since α determines β. □

Remark 2.3.14. *The excluded value $h = -\frac{1}{2}$ carries no level-two singular vector at all: (2.103) forces $\alpha = 0$, and then (2.105) reads $6\beta h = -3\beta$, forcing $\beta = 0$ as well.*

This is a nontrivial check on Theorem 2.3.12: the curve (2.100) in the (c, h) plane must be exactly the locus where the Kac determinant at level 2 vanishes.

Corollary 2.3.15. *Equation (2.100) is precisely the pair of Kac curves $h = h_{1,2}(c)$ and $h = h_{2,1}(c)$.*

Proof. In the parametrization (2.98a) one has $h_{1,2}(t) = \frac{3}{4t} - \frac{1}{2} = \frac{3-2t}{4t}$, whence

$$2h + 1 = \frac{3}{2t}, \qquad 5 - 8h = 5 - \frac{6-4t}{t} = \frac{3(3t-2)}{t}, \tag{2.107}$$

and therefore

$$\frac{2h(5-8h)}{2h+1} = \frac{3-2t}{2t}\cdot\frac{3(3t-2)}{t}\cdot\frac{2t}{3} = \frac{(3-2t)(3t-2)}{t} = 13 - 6\left(t + \frac{1}{t}\right), \tag{2.108}$$

which is c by (2.98a). Since $h_{r,s}(t) = h_{s,r}(1/t)$ while c is invariant under $t \mapsto 1/t$, the same holds for $h_{2,1}$. Conversely, (2.100) rearranges to the quadratic

$$16h^2 + (2c - 10)h + c = 0, \tag{2.109}$$

which has exactly two roots for given c; they must therefore be $h_{1,2}(c)$ and $h_{2,1}(c)$. □

Remark 2.3.16. *The Kac determinant at level 2 also carries a factor $h - h_{1,1} = h$, since $rs = 1$ contributes with exponent $P(2-1) = 1$. That root is invisible in Proposition 2.3.13 because the corresponding null state is $L_{-1}^2|0\rangle$, a descendant of the level-one singular vector $L_{-1}|0\rangle$, which has no L_{-2} component and so is excluded by the normalization $\alpha = 1$.*

Example 2.3.17 ($c=\frac{1}{2}$)**.** *Solving* (2.109) *at* $c=\frac{1}{2}$ *gives* $32h^2-18h+1=0$*, hence*

$$h=\frac{18\pm 14}{64}=\frac{1}{2}\quad \text{or}\quad \frac{1}{16}. \tag{2.110}$$

These are $h_{2,1}$ *and* $h_{1,2}$ *of* $\mathcal{M}(4,3)$*, that is the fields later called* ε *and* σ*; recall* $(2,1)\sim(1,3)$ *under the reflection of the Kac table. Conversely, imposing that a level-two singular vector exist at* $h=\frac{1}{16}$ *returns* $c=\frac{1}{2}$ *from* (2.100) *directly, without any reference to the Kac table. This is the germ of the Belavin–Polyakov–Zamolodchikov argument, in which the existence of degenerate fields and the closure of their operator algebra fix the admissible values of* c*.*

The same check can be pushed one level higher. At level 3 Theorem 2.3.12 predicts the factors $h^{P(2)}=h^2$, $(h-h_{1,2})(h-h_{2,1})$ and $(h-h_{1,3})(h-h_{3,1})$, each of the last four with exponent $P(1)=P(0)=1$.

Proposition 2.3.18 (level three)**.** *In the basis* $L_{-3}|h\rangle$, $L_{-1}L_{-2}|h\rangle$, $L_{-1}^3|h\rangle$ *of* $V^{(3)}(c,h)$ *the Gram matrix is*

$$G^{(3)}_{c,h}=\begin{pmatrix} 2c+6h & 2c+16h & 24h \\ 2c+16h & ch+2c+8h^2+34h & 36h(h+1) \\ 24h & 36h(h+1) & 24h(h+1)(2h+1)\end{pmatrix}, \tag{2.111}$$

and

$$\det G^{(3)}_{c,h}=48\,h^2\,\big(16h^2+(2c-10)h+c\big)\big(3h^2+(c-7)h+c+2\big). \tag{2.112}$$

If $h\neq -1,-2$*, the module* $V(c,h)$ *has a singular vector at level* 3 *with an* L_{-3} *component if and only if* $3h^2+(c-7)h+c+2=0$*, and it is then*

$$|\chi_3\rangle=\Big(L_{-3}-\frac{2}{h+2}L_{-1}L_{-2}+\frac{1}{(h+1)(h+2)}L_{-1}^3\Big)|h\rangle. \tag{2.113}$$

Proof. The entries of $G^{(3)}$ follow from (1.27) as in Proposition 2.3.13; for instance $\langle h|L_3L_{-3}|h\rangle=6h+2c$, and the last diagonal entry is $3!\,(2h)_3$, as in Proposition 2.3.4. Expanding the determinant gives (2.112).

For the singular vector, write $|\chi\rangle=(L_{-3}+\beta L_{-1}L_{-2}+\gamma L_{-1}^3)|h\rangle$. Using $[L_1,L_{-3}]=4L_{-2}$, $[L_1,L_{-1}]=2L_0$ and $L_1L_{-1}^n|h\rangle=n(2h+n-1)L_{-1}^{n-1}|h\rangle$,

$$L_1|\chi\rangle=\big[4+2\beta(h+2)\big]L_{-2}|h\rangle+\big[3\beta+6\gamma(h+1)\big]L_{-1}^2|h\rangle. \tag{2.114}$$

This vanishes if and only if $\beta=-2/(h+2)$ and $\gamma=1/((h+1)(h+2))$, which are the coefficients in (2.113). Using $[L_2,L_{-3}]=5L_{-1}$ and $[L_2,L_{-1}]=3L_1$,

$$\begin{aligned} L_2L_{-3}|h\rangle=5L_{-1}|h\rangle,\qquad L_2L_{-1}L_{-2}|h\rangle=\big(4h+9+\tfrac{c}{2}\big)L_{-1}|h\rangle,\\ L_2L_{-1}^3|h\rangle=6(3h+1)L_{-1}|h\rangle, \end{aligned} \tag{2.115}$$

so that with these β,γ

$$L_2|\chi\rangle=-\frac{3h^2+(c-7)h+c+2}{(h+1)(h+2)}\,L_{-1}|h\rangle. \tag{2.116}$$

Since L_1 and L_2 generate all L_n with $n\geq 1$, $|\chi\rangle$ is singular exactly when the numerator vanishes. □

Corollary 2.3.19. (2.112) *agrees with Theorem 2.3.12 at* $N = 3$*:*

$$3h^2 + (c-7)h + c + 2 = 3\,(h - h_{1,3})(h - h_{3,1}), \tag{2.117}$$

the factor $16h^2 + (2c-10)h + c = 16(h - h_{1,2})(h - h_{2,1})$ *is* (2.109)*, and* h^2 *is* $(h - h_{1,1})^{P(2)}$*.*

Proof. In the parametrization (2.98a), $h_{1,3} = \frac{2}{t} - 1$ and $h_{3,1} = 2t - 1$, so $h_{1,3} + h_{3,1} = 2(t + \frac{1}{t}) - 2 = \frac{7-c}{3}$ and $h_{1,3}h_{3,1} = 5 - 2(t + \frac{1}{t}) = \frac{c+2}{3}$. The second statement is Corollary 2.3.15. □

Example 2.3.20 ($c = \frac{1}{2}$ again)**.** *At* $c = \frac{1}{2}$ *the new quadratic is* $3h^2 - \frac{13}{2}h + \frac{5}{2} = 0$*, with roots* $h = \frac{1}{2}$ *and* $h = \frac{5}{3}$*. Thus* ε*, which has a level-two singular vector by Example 2.3.17, has a second one at level* 3*. It is not* L_{-1} *applied to the first, since only* $|\chi_3\rangle$ *has an* L_{-3} *component. This is the second singular vector expected from* $(2,1) \sim (1,3)$*: the labels give levels* $2 \cdot 1 = 2$ *and* $1 \cdot 3 = 3$*.*

2.4 Conformal blocks and minimal models

2.4.1 Basis of OPE

OPE as eigenbasis expansion

Let ϕ_i, ϕ_j be homogeneous of weights $(h_i, \bar{h}_i)$ and $(h_j, \bar{h}_j)$. By Remark 2.1.2(iii), the field $\psi_{(a,\bar{a})}$ in (2.18) has weight $(h_i + h_j + a,\ \bar{h}_i + \bar{h}_j + \bar{a})$. Each weight occurs for exactly one pair $(a, \bar{a})$, so grouping (2.18) by weight gives

$$\phi_i(z)|\phi_j\rangle = \sum_{(h,\bar{h})} z^{h-h_i-h_j}\,\bar{z}^{\bar{h}-\bar{h}_i-\bar{h}_j}\ P_{(h,\bar{h})}\big[\phi_i(1)|\phi_j\rangle\big], \tag{2.118}$$

where $P_{(h,\bar{h})}$ is the projection of $\bar{F}$ onto $F_{(h,\bar{h})}$. By the compatibility condition in (T), $F_{(h,\bar{h})}$ is the $(L_0, \bar{L}_0)$-eigenspace with eigenvalue $(h, \bar{h})$. The operator product expansion is therefore the expansion of the state $\phi_i(z)|\phi_j\rangle$ in the eigenbasis of $(L_0, \bar{L}_0)$. The coefficient states do not depend on z, and all of the z-dependence sits in powers whose exponents are differences of weights. The leading singularity is set by the lowest weight that occurs. By Theorem 2.2.1, inside correlators the series converges for $|z|$ smaller than the distance from 0 to the other insertions.

OPE as inserting the identity

Denoting by Π_{h_p} the projection onto the Verma module $V(c, h_p)$, the identity operator can be expressed as a sum of projections.

$$\mathbf{1} = \sum_{p\text{: primary}} \Pi_{h_p} \otimes \overline{\Pi}_{\bar{h}_p} \tag{2.119}$$

Suppressing the anti-holomorphic part,

$$\mathbf{1} = \sum_{p:\text{ primary}} \Pi_{h_p} = \sum_{p:\text{ primary}} \sum_{N=0}^{\infty} \sum_{\substack{\boldsymbol{m},\boldsymbol{n} \\ |\boldsymbol{m}|=|\boldsymbol{n}|=N}} |h_p, \boldsymbol{n}\rangle [G^{(N)}_{c,h_p}]^{-1}_{\boldsymbol{n},\boldsymbol{m}} \langle h_p, \boldsymbol{m}|. \tag{2.120}$$

Now the OPE of two primary ϕ_i, ϕ_j is obtained by "inserting the identity operator",

$$\begin{aligned} \phi_i(z_i)\phi_j(z_j)|0\rangle &= \sum_p \Pi_{h_p} \phi_i(z_i)\phi_j(z_j)|0\rangle \\ &= \sum_{p:\text{ primary}} \sum_{\boldsymbol{n}} a_{p,\boldsymbol{n}} |h_p, \boldsymbol{n}\rangle \\ &= \sum_{p:\text{ primary}} \sum_{\boldsymbol{n}} a_{p,\boldsymbol{n}} L_{-\boldsymbol{n}} \phi_p(0)|0\rangle, \end{aligned} \tag{2.121}$$

where the coefficients $a_{p,\boldsymbol{n}}$ are given by

$$a_{p,\boldsymbol{n}} = \sum_{\substack{\boldsymbol{m} \\ |\boldsymbol{m}|=|\boldsymbol{n}|}} [G^{(N)}_{c,h_p}]^{-1}_{\boldsymbol{n},\boldsymbol{m}} \langle h_p, \boldsymbol{m}|\phi_i(z_i)\phi_j(z_j)|0\rangle \tag{2.122}$$

Role of primaries in OPE

The coefficients $a_{p,\boldsymbol{n}}$ above look theory-dependent: they are matrix elements of $\phi_i\phi_j$ between descendant states. They are not. Only one number per primary is theory-dependent, the structure constant, and everything else is fixed by the Virasoro algebra. This is what makes the conformal blocks of (2.129) universal and defines the coefficients $\beta(h_i, h_j, h_p; \boldsymbol{n})$ used there. As in (2.129), write the expansion as $\phi_j(z)|\phi_i\rangle$ and suppress the antiholomorphic factor.

Proposition 2.4.1 (descendant coefficients)**.** *Let ϕ_i, ϕ_j, ϕ_p be primary, and for a sequence $\boldsymbol{m} = (m_1, \dots, m_k)$ of positive integers put $|\boldsymbol{m}| = \sum_l m_l$. Then*

$$\langle \phi_p | L_{m_1} \cdots L_{m_k}\, \phi_j(z)|\phi_i\rangle = \gamma(\boldsymbol{m})\; z^{h_p+|\boldsymbol{m}|-h_i-h_j}\; C_{ijp}, \tag{2.123}$$

$$\gamma(\boldsymbol{m}) = \prod_{l=1}^{k} \Big(h_p + \sum_{l'<l} m_{l'} + m_l h_j - h_i \Big), \tag{2.124}$$

with $C_{ijp} = \langle \phi_p|\phi_j(1)|\phi_i\rangle$. Consequently the level-N part of $\Pi_{h_p}\phi_j(z)|\phi_i\rangle$ is

$$C_{ijp}\, z^{h_p+N-h_i-h_j} \sum_{|\boldsymbol{n}|=N} \beta(h_i, h_j, h_p; \boldsymbol{n})\, L_{-\boldsymbol{n}}|\phi_p\rangle, \tag{2.125}$$

$$\beta(\cdot; \boldsymbol{n}) = \sum_{|\boldsymbol{m}|=N} \big[G^{(N)}_{c,h_p}\big]^{-1}_{\boldsymbol{n},\boldsymbol{m}}\, \gamma(\boldsymbol{m}), \tag{2.126}$$

where the Gram matrix is taken with respect to the same pairing of $\langle \phi_p|L_{m_1}\cdots L_{m_k}$ with $L_{-\boldsymbol{n}}|\phi_p\rangle$. The coefficients β depend only on (c, h_i, h_j, h_p).

Proof. Let $\langle\psi| = \langle\phi_p|L_{m_1}\cdots L_{m_{k-1}}$. It has L_0-weight $h_\psi = h_p + \sum_{l<k} m_l$, so $\langle\psi|\phi_j(z)|\phi_i\rangle = K\, z^{h_\psi - h_j - h_i}$ for a constant K. Since $L_{m_k}|\phi_i\rangle = 0$, Corollary 2.2.3 and primarity of ϕ_j give

$$\begin{aligned} \langle\psi|L_{m_k}\phi_j(z)|\phi_i\rangle &= \big(z^{m_k+1}\partial_z + (m_k+1)h_j z^{m_k}\big) K z^{h_\psi-h_j-h_i} \\ &= \big(h_\psi + m_k h_j - h_i\big)\, z^{m_k}\, K z^{h_\psi-h_j-h_i}. \end{aligned} \tag{2.127}$$

Induction on k gives (2.124). Inserting (2.124) into the coefficients $a_{p,\boldsymbol{n}}$ of the previous subsubsection gives (2.126). □

Example 2.4.2. *At level one, $G^{(1)} = 2h_p$ and $\gamma(1) = h_p + h_j - h_i$, so*

$$\phi_j(z)\phi_i(0) = C_{ijp}\, z^{h_p-h_i-h_j}\Big(\phi_p(0) + \frac{h_p+h_j-h_i}{2h_p}\, z\,\partial\phi_p(0) + O(z^2)\Big) + \cdots . \tag{2.128}$$

For the free boson with $\phi_i = \mathcal{V}_\alpha$, $\phi_j = \mathcal{V}_\beta$, $\phi_p = \mathcal{V}_{\alpha+\beta}$ this coefficient is $\beta/(\alpha+\beta)$, in agreement with the explicit oscillator form of $\mathcal{V}_\beta(z)|\alpha\rangle$ (§3.3). The level-two coefficients agree in the same way.

Remark 2.4.3. *The data of a conformal field theory on the sphere are therefore the spectrum of primaries $(h_p, \bar{h}_p)$ and the structure constants C_{ijp}; everything else in an operator product expansion is kinematics. Where the Gram matrix is singular, i.e. at a zero of the Kac determinant, (2.126) breaks down. The way out is the passage to the irreducible module of §2.3.3, and the consistency of that passage is what produces the fusion rules of §2.4.5.*

2.4.2 Virasoro conformal blocks

4-point functions

Definition 2.4.4. *We saw in Corollary 2.1.12 that, for some function f,*

$$\langle\phi_4(z_4,\bar{z}_4)\phi_3(z_3,\bar{z}_3)\phi_2(z_2,\bar{z}_2)\phi_1(z_1,\bar{z}_1)\rangle = f(\{z_{ij}\})G_{34}^{21}(z,\bar{z}), \tag{2.129a}$$

$$G_{kl}^{ji}(z,\bar{z}) \equiv \langle\phi_l|\phi_k(1,1)\phi_j(z,\bar{z})|\phi_i\rangle = \sum_p C_{ijp}C_{klp}\mathcal{F}_{kl}^{ji}(h_p|z)\overline{\mathcal{F}}_{kl}^{ji}(\bar{h}_p|\bar{z}). \tag{2.129b}$$

The last equality defines the Virasoro conformal blocks, given by

$$\mathcal{F}_{kl}^{ji}(h_p|z) \equiv z^{h_p-h_i-h_j}\sum_{\boldsymbol{n}}\beta(h_i,h_j,h_p;\boldsymbol{n})z^{|\boldsymbol{n}|}\frac{\langle\phi_l|\phi_k(1)L_{-\boldsymbol{n}}|\phi_p\rangle}{\langle\phi_l|\phi_k(1)|\phi_p\rangle}. \tag{2.130}$$

Manifestly, the Virasoro blocks do not depend on a specific theory but are universally determined from the representations of Virasoro algebra. The consequences of comparing this expansion with the one adapted to $z \to 1$ are taken up in §2.5. Applying the "inserting identity" viewpoint to the 4-point functions,

$$\langle\phi_4|\phi_3(1,1)\phi_2(z,\bar{z})|\phi_1\rangle = \langle\phi_4|\phi_3(1,1)\Big(\sum_p \Pi_{h_p}\otimes\overline{\Pi}_{\bar{h}_p}\Big)\phi_2(z,\bar{z})|\phi_1\rangle, \tag{2.131a}$$

$$\mathcal{F}_{34}^{21}(h_p|z)\overline{\mathcal{F}}_{34}^{21}(\bar{h}_p|\bar{z}) = \frac{\langle\phi_4|\phi_3(1,1)\Big(\Pi_{h_p}\otimes\overline{\Pi}_{\bar{h}_p}\Big)\phi_2(z,\bar{z})|\phi_1\rangle}{C_{12p}C_{34p}}. \tag{2.131b}$$

The definition in (2.129) amounts to normalizing the Virasoro blocks by requiring

$$\lim_{z\to 0}\frac{\mathcal{F}_{34}^{21}(h_p|z)}{z^{h_p-h_1-h_2}} = 1. \tag{2.132}$$

2.4.3 Degenerate fields and the BPZ equations

The expansion (2.129) is a sum over all primaries, and nothing so far restricts which of them contribute. There is one situation in which the sum is guaranteed to be finite: when one of the external fields carries a null vector. The mechanism is that a null vector, being of zero norm, decouples from every correlation function, and by Corollary 2.2.26 this decoupling is a linear differential equation.

Definition 2.4.5. *A primary field ϕ of weight h is degenerate at level N if $V(c,h)$ contains a singular vector at level N,*

$$|\chi\rangle = \sum_{|\boldsymbol{k}|=N} a_{\boldsymbol{k}}\, L_{-\boldsymbol{k}}|h\rangle, \qquad L_n|\chi\rangle = 0 \quad (n>0), \tag{2.133}$$

normalized so that the coefficient of L_{-1}^N is $a_{(1^N)} = 1$.

By Theorem 2.3.12 this happens exactly when $h = h_{r,s}(c)$ for some r, s with $rs = N$; the simplest instance is Proposition 2.3.13. Before stating the consequence we record a simplification of $\mathcal{L}_{-1}$.

Lemma 2.4.6. *Acting on $\langle \phi(z)\phi_1(z_1)\cdots\phi_N(z_N)\rangle$ one has $\mathcal{L}_{-1} = \partial_z$.*

Proof. Setting $k=1$ in Theorem 2.2.24 gives $\mathcal{L}_{-1} = -\sum_j \partial_j$, the term proportional to $(k-1)h_j$ being absent. The first identity of Corollary 2.2.10, applied to the full collection of fields $\phi(z), \phi_1(z_1), \dots, \phi_N(z_N)$, reads $\big(\partial_z + \sum_j \partial_j\big)\langle\cdots\rangle = 0$. □

Theorem 2.4.7 (BPZ). *Let ϕ be degenerate at level N with singular vector $|\chi\rangle$ as in Definition 2.4.5. Then for any primaries $\phi_1, \dots, \phi_M$,*

$$\left[\partial_z^N + \sum_{\substack{|\boldsymbol{k}|=N \\ \boldsymbol{k}\neq(1^N)}} a_{\boldsymbol{k}}\, \mathcal{L}_{-\boldsymbol{k}}\right] \langle \phi(z)\phi_1(z_1)\cdots\phi_M(z_M)\rangle = 0, \tag{2.134}$$

where $\mathcal{L}_{-\boldsymbol{k}} = \mathcal{L}_{-k_1}\cdots\mathcal{L}_{-k_n}$ and the $\mathcal{L}_{-k}$ are the differential operators of Theorem 2.2.24. In particular the equation is linear and of order N in z.

Proof. A singular vector is orthogonal to the whole Verma module, in particular to itself, so $|\chi\rangle$ has zero norm and the corresponding field $\chi(z)$ decouples: $\langle \chi(z)\phi_1\cdots\phi_M\rangle = 0$. By Corollary 2.2.26 each $L_{-\boldsymbol{k}}$ acting on ϕ is represented by $\mathcal{L}_{-\boldsymbol{k}}$ acting on the correlator, and by Lemma 2.4.6 the term $\boldsymbol{k} = (1^N)$ contributes ∂_z^N with coefficient 1. □

Caution 2.4.8. *For $k \geq 2$ the operator $\mathcal{L}_{-k}$ contains derivatives ∂_j with respect to the* other *insertion points, so (2.134) is in general a partial differential equation. It becomes an ordinary differential equation in z for the four-point function, where the ∂_j can be eliminated using the remaining two identities of Corollary 2.2.10.*

Example 2.4.9 (level two). *For $|\chi\rangle$ of Proposition 2.3.13, dividing by the coefficient of L_{-1}^2 and inserting $\mathcal{L}_{-2} = \sum_j \left[h_j(z_j - z)^{-2} - (z_j - z)^{-1}\partial_j\right]$ gives*

$$\left[\partial_z^2 - \frac{2(2h+1)}{3}\sum_j\left(\frac{h_j}{(z_j-z)^2} - \frac{1}{z_j - z}\,\partial_j\right)\right]\langle\phi(z)\phi_1\cdots\phi_M\rangle = 0. \tag{2.135}$$

No new input has been used: the coefficient comes from (2.101), the operator $\mathcal{L}_{-2}$ from Theorem 2.2.24, and the nesting from Corollary 2.2.26.

The importance of (2.134) is less that it can be solved — though for four points it can — than that its mere existence bounds the number of channels.

Corollary 2.4.10. *Let ϕ be degenerate at level N. Then any four-point function containing ϕ satisfies a linear ordinary differential equation of order N, so at most N conformal blocks contribute to it.*

The admissible h_p can moreover be read off without solving the equation. Writing ϕ for the degenerate field of weight h and ϕ_1 for the primary it is fused with, a conformal block behaves as $\mathcal{F} \sim z^{h_p - h - h_1}$ as $z \to 0$; substituting this into (2.134) and keeping the leading power produces an indicial equation of degree N in h_p, whose roots are exactly the weights allowed in the channel. Fusion rules are therefore, from this point of view, the characteristic exponents of a differential equation — to be contrasted with the modular derivation of §3.6.2.

Proposition 2.4.11. *Let ϕ be degenerate at level N and let ϕ_1, ϕ_p be primary. If $h_p - h - h_1$ is not a root of the indicial equation of Theorem 2.4.7 at $z = 0$, then $C_{\phi\phi_1 p} = 0$.*

Proof. Placing ϕ_1 at 0 and ϕ_p at infinity, $\langle\phi(z)\phi_1(0)\phi_p(\infty)\rangle = C_{\phi\phi_1 p}\, z^{h_p - h - h_1}$ exactly. Substituting into (2.134) leaves a single power of z, whose coefficient is $C_{\phi\phi_1 p}$ times the indicial polynomial. □

Caution 2.4.12. *Proposition 2.4.11 is a vanishing statement: it bounds the fusion rules from above but does not show that the surviving channels actually occur. That $N_{ij}^k = 1$ rather than 0 requires computing the structure constants, or the independent input of the Verlinde formula.*

2.4.4 The structure of minimal models

An irreducible module is obtained from $V(c,h)$ by quotienting out the maximal proper submodule.

$$M(c,h) = V(c,h)\Big/\sum_i U(\mathfrak{Vir})|\chi_i\rangle, \tag{2.136}$$

where $|\chi_i\rangle$ are the singular vectors of Theorem 2.3.12. For minimal models there are two of them and the sum is not direct; this is worked out in §3.5.2, see (3.110), where the resolution is used to compute characters.

The fusion rules generated by these singular vectors are derived in §2.4.5, by the analytic route of Theorem 2.4.7; they are recovered independently from modular data in §3.6.2.

Eq. (2.97) implies that when there are two coprime integers p, p' such that

$$p\alpha_- + p'\alpha_+ = 0, \tag{2.137}$$

then the conformal weights $h_{r,s}$ enjoy the periodicity property. In terms of the pair of integers (p, p'), the central charge and the Kac formula become

$$\begin{aligned} c &= 1 - 6\frac{(p-p')^2}{pp'} \\ h_{r,s} &= \frac{(pr-p's)^2 - (p-p')^2}{4pp'} \end{aligned} \tag{2.138}$$

From Eq. (2.138) can be read are additional symmetries of the conformal dimensions, which are the basis on which one can construct a Hilbert space that contains finitely many irreducible representations of the Virasoro algebra.

$$h_{r,s} = h_{r+p',s+p} \tag{2.139a}$$

$$h_{r,s} = h_{p'-r,p-s} \tag{2.139b}$$

$$h_{r,s} + rs = h_{p'+r,p-s} = h_{p'-r,p+s} \tag{2.139c}$$

$$h_{r,s} + (p'-r)(p-s) = h_{r,2p-s} \quad = h_{2p'-r,s} \tag{2.139d}$$

If $c \leq 1$, the parameter t of Eq. (2.98a) is real and positive, and equal to $t = -\alpha_+/\alpha_- = p/p'$. The two integers p, p' thus have a same sign, and may be taken as positive using the symmetries of Eq. (2.138). Furthermore, one can possibly exchange $p \leftrightarrow p'$ and $r \leftrightarrow s$ to assume $p > p'$.

As a result of these symmetries, only those Virasoro modules in the range $1 \leq r \leq p'-1$ and $1 \leq s \leq p-1$ remain, the other being decoupled. All four relations follow from a single observation: by (2.138), h depends on (r, s) only through $\lambda_{r,s} := pr - p's$, and only through its square,

$$h_{r,s} = \frac{\lambda_{r,s}^2 - (p-p')^2}{4N}, \qquad N := pp'. \tag{2.140}$$

Writing also $\lambda_{r,-s} := pr + p's$, direct substitution gives

$$\lambda_{r+p',\,s+p} = \lambda_{r,s}, \qquad \lambda_{p'-r,\,p-s} = -\lambda_{r,s}, \tag{2.141a}$$

$$\lambda_{p'+r,\,p-s} = \lambda_{r,-s}, \qquad \lambda_{p'-r,\,p+s} = -\lambda_{r,-s}, \tag{2.141b}$$

$$\lambda_{r,\,2p-s} = \lambda_{r,-s} - 2N, \qquad \lambda_{2p'-r,\,s} = 2N - \lambda_{r,-s}. \tag{2.141c}$$

The first line leaves λ^2 unchanged and hence gives the first two relations. The second and third lines replace s by $-s$, and

$$\frac{\lambda_{r,-s}^2 - \lambda_{r,s}^2}{4N} = rs, \qquad \frac{(\lambda_{r,-s} - 2N)^2 - \lambda_{r,s}^2}{4N} = rs - \lambda_{r,-s} + N = (p'-r)(p-s), \tag{2.142}$$

which are the third and fourth relations.

Only the first two are symmetries of h: they make the rectangle $1 \leq r \leq p'-1$, $1 \leq s \leq p-1$ a fundamental domain for the periodicity, with the reflection identifying its entries in pairs and leaving $\frac{1}{2}(p-1)(p'-1)$ distinct weights. The last two are not symmetries but shift formulas, and the shifts rs and $(p'-r)(p-s)$ are precisely the levels of the two singular vectors identified in Observation 3.5.1 of §3.5.2.

The Kac table

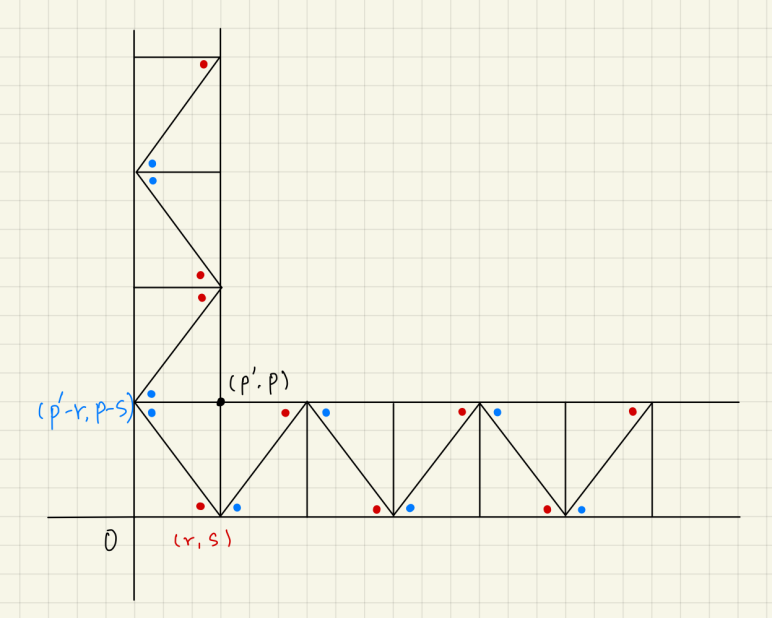


Figure 2.1: Schematic illustration of Kac table

Remark 2.4.13 (unitarity)**.** *The Kac determinant only detects reducibility; it says nothing about the sign of the Gram matrix, and $\mathcal{M}(p,p')$ is defined for any coprime pair. Requiring in addition that the Gram matrix be positive definite — so that $\mathcal{H}$ is a genuine Hilbert space — is a much stronger condition, and by the theorem of Friedan, Qiu and Shenker the unitary minimal models with $c<1$ are exactly those with $p'=p-1$,*

$$c = 1 - \frac{6}{p(p-1)}, \qquad p \geq 3. \tag{2.143}$$

The first few are the Ising model $\mathcal{M}(4,3)$ with $c=\frac{1}{2}$, the tricritical Ising model $\mathcal{M}(5,4)$ with $c=\frac{7}{10}$, and the tetracritical Ising model $\mathcal{M}(6,5)$ with $c=\frac{4}{5}$. Nonunitary minimal models are perfectly consistent as conformal field theories and do occur in statistical mechanics — $\mathcal{M}(5,2)$ with $c=-\frac{22}{5}$ describes the Lee–Yang edge singularity — but they have a negative-norm state and a primary of negative weight.

2.4.5 Fusion rules from null vectors

Theorem 2.4.7 produces a partial differential equation. For the four-point function the extra derivatives can be removed, leaving an ordinary differential equation whose characteristic exponents are the admissible weights in the channel. We carry this out for a field degenerate at level two.

Reduction to an ordinary differential equation

Fix the three spectator fields at $z_1=0$, $z_2=1$, $z_3=\infty$ and write

$$G(z) := \lim_{z_3\to\infty} z_3^{2h_3}\,\langle \phi_3(z_3)\,\phi_2(1)\,\phi(z)\,\phi_1(0)\rangle\,. \tag{2.144}$$

Lemma 2.4.14. *Let*

$$g(z,z_1,z_2) := \lim_{z_3\to\infty} z_3^{2h_3}\langle\phi_3(z_3)\phi_2(z_2)\phi(z)\phi_1(z_1)\rangle,$$

so that $G(z) = g(z,0,1)$. *Then, with* $\Delta := h+h_1+h_2-h_3$,

$$\big[\partial_{z_1} g\big]_{z_1=0,z_2=1} = \big[(z-1)\partial_z + \Delta\big]G, \qquad \big[\partial_{z_2} g\big]_{z_1=0,z_2=1} = -\big[z\partial_z + \Delta\big]G. \tag{2.145}$$

We abbreviate these as ∂_1 *and* ∂_2.

Proof. Since $\phi_3(\infty)$ is well defined, the correlator behaves as $z_3^{-2h_3}\big[G + O(z_3^{-1})\big]$ for large z_3, whence $\partial_3 = -2h_3 z_3^{-2h_3-1}G + O(z_3^{-2h_3-2})$. Multiplying the identities of Corollary 2.2.10 by $z_3^{2h_3}$ and letting $z_3\to\infty$, the third insertion contributes

$$\partial_3 \longrightarrow 0, \qquad z_3\partial_3 + h_3 \longrightarrow -2h_3 + h_3 = -h_3, \tag{2.146}$$

so that the first two identities become, after setting $z_1 = 0$, $z_2 = 1$,

$$\partial_z G + \partial_1 + \partial_2 = 0, \tag{2.147a}$$

$$z\partial_z G + \partial_2 + (h + h_1 + h_2 - h_3)G = 0. \tag{2.147b}$$

The second gives ∂_2, and substituting into the first gives ∂_1. The third Ward identity is not used: in the same limit it produces $(2h_3-1)c_1$, where c_1 is the subleading coefficient in the expansion of the correlator at large z_3, and therefore carries no information about G alone. It is not needed, since ∂_3 has already decoupled and only two unknowns remain. □

Proposition 2.4.15. *Let* ϕ *be degenerate at level two and put* $\kappa := \frac{2}{3}(2h+1)$. *Then*

$$\left\{\partial_z^2 - \kappa\left[\left(\frac{1}{1-z} - \frac{1}{z}\right)\partial_z + \frac{h_1}{z^2} + \frac{h_2}{(1-z)^2} + \frac{\Delta}{z(1-z)}\right]\right\} G(z) = 0. \tag{2.148}$$

Proof. In (2.135) the term $j=3$ drops out in the limit, since $h_3(z_3-z)^{-2}\to 0$ and $(z_3-z)^{-1}\partial_3 \to 0$ like z_3^{-2}. The remaining two terms are

$$\frac{h_1}{z^2} + \frac{1}{z}\partial_1 + \frac{h_2}{(1-z)^2} - \frac{1}{1-z}\partial_2, \tag{2.149}$$

and inserting (2.145) the coefficient of ∂_z becomes

$$\frac{z-1}{z} + \frac{z}{1-z} = \frac{(z-1)^2 - z^2}{z(z-1)} = \frac{1-2z}{z(z-1)} = \frac{1}{1-z} - \frac{1}{z}, \tag{2.150}$$

while the terms proportional to Δ combine into $\Delta\big[z^{-1} + (1-z)^{-1}\big] = \Delta/z(1-z)$. □

The indicial equation and the fusion rules

By (2.129) a conformal block behaves as $z^{h_p-h-h_1}$ as $z\to 0$. Substituting $G\sim z^\alpha$ into (2.148) and keeping the terms of order $z^{\alpha-2}$ gives $\alpha(\alpha-1) = \kappa(h_1-\alpha)$, that is

Proposition 2.4.16. *The exponents $\alpha = h_p - h - h_1$ of (2.148) at $z = 0$ satisfy*

$$\alpha^2 + (\kappa - 1)\alpha - \kappa h_1 = 0. \tag{2.151}$$

In particular exactly two weights h_p occur in the operator product expansion of ϕ with ϕ_1.

Theorem 2.4.17 (fusion rule of $\phi_{1,2}$)**.** *For $h = h_{1,2}$ and $h_1 = h_{r,s}$ the two roots of (2.151) are $h_p = h_{r,s-1}$ and $h_p = h_{r,s+1}$, that is*

$$\phi_{1,2} \times \phi_{r,s} = \phi_{r,s-1} + \phi_{r,s+1}. \tag{2.152}$$

Proof. In the parametrization (2.98a), $h_{1,2} = \frac{3}{4t} - \frac{1}{2}$, so $\kappa = \frac{2}{3}(2h_{1,2} + 1) = 1/t$. Writing $\alpha_\mp := h_{r,s\mp1} - h_{1,2} - h_{r,s}$ one computes

$$\alpha_- = \frac{r+1}{2} - \frac{s+1}{2t}, \qquad \alpha_+ = \frac{1-r}{2} + \frac{s-1}{2t}, \tag{2.153}$$

whence $\alpha_- + \alpha_+ = 1 - \frac{1}{t} = -(\kappa - 1)$ and

$$\alpha_-\alpha_+ = \frac{1-r^2}{4} - \frac{s^2-1}{4t^2} + \frac{rs-1}{2t} = -\frac{1}{t}\, h_{r,s} = -\kappa h_1, \tag{2.154}$$

so $\alpha_\pm$ are the roots of (2.151). □

Remark 2.4.18. *Truncation of the Kac table is automatic: if $s = 1$ then $\phi_{r,0}$ does not exist and only one channel survives, and similarly at the upper edge. The analogous rule for $\phi_{2,1}$ follows by $t \mapsto 1/t$, and the two together generate the fusion rules of the whole table.*

Example 2.4.19 (Ising model)**.** *Take $\phi = \phi_j = \sigma$ in $\mathcal{M}(4,3)$, so $h = h_j = \frac{1}{16}$, $\kappa = \frac{3}{4}$ and $\Delta = \frac{1}{8}$. Equation (2.148) is solved by*

$$\mathcal{F}_\pm(z) = \big[z(1-z)\big]^{-1/8} \sqrt{\frac{1 \pm \sqrt{1-z}}{2}}, \tag{2.155}$$

as one verifies by direct substitution. The indicial equation (2.151) reads $64\alpha^2 - 16\alpha - 3 = 0$ with roots $\alpha = -\frac{1}{8}, \frac{3}{8}$, so that $h_p = \alpha + \frac{1}{8}$ takes the values

$$h_p = 0 \quad \text{and} \quad h_p = \tfrac{1}{2}, \qquad \text{i.e.} \qquad \sigma \times \sigma = \mathbb{1} + \varepsilon, \tag{2.156}$$

in agreement with $\mathcal{F}_+ \sim z^{-1/8}$ and $\mathcal{F}_- \sim \frac{1}{2} z^{3/8}$.

2.5 Crossing symmetry

All of the preceding is chiral: Verma modules, fusion rules and conformal blocks are determined by the Virasoro algebra alone and know nothing about how left- and right-moving sectors are paired. A physical correlation function is not a block but a particular combination of blocks with their antiholomorphic partners, and it must be single-valued. This is the consistency condition available at genus zero, and we now extract from it the one piece of data the chiral analysis cannot supply: the structure constants.

2.5.1 The crossing equation

By (2.129) the four-point function of primaries decomposes in the channel $z \to 0$ as

$$G(z,\bar{z}) = \sum_p C_{12p} C_{34p} \left|\mathcal{F}^{21}_{34}(h_p|z)\right|^2, \tag{2.157}$$

where $|\mathcal{F}|^2$ abbreviates the product of a block with its antiholomorphic partner. The same function admits a second expansion, adapted to $z \to 1$, obtained by exchanging the roles of ϕ_1 and ϕ_3.

The crossing matrix

The blocks (2.155) are adapted to the channel $z \to 0$. The crossing equation compares them with the blocks adapted to $z \to 1$, which are $\mathcal{F}_\pm(1-z)$; since both are bases of the same two-dimensional solution space, they are related by an invertible matrix.

Proposition 2.5.1. *With $\mathcal{F}_+, \mathcal{F}_-$ labeled by $h_p = 0, \frac{1}{2}$,*

$$\mathcal{F}_\pm(1-z) = \sum_q F_{\pm q}\, \mathcal{F}_q(z), \qquad F = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{2.158}$$

Proof. Write $u := \sqrt{\frac{1+\sqrt{1-z}}{2}}$ and $v := \sqrt{\frac{1-\sqrt{1-z}}{2}}$, so that $\mathcal{F}_+ = [z(1-z)]^{-1/8} u$ and $\mathcal{F}_- = [z(1-z)]^{-1/8} v$. Then $u^2 + v^2 = 1$ and $u^2 v^2 = z/4$, whence

$$(u \pm v)^2 = 1 \pm 2uv = 1 \pm \sqrt{z}, \qquad \text{so} \qquad \sqrt{\frac{1 \pm \sqrt{z}}{2}} = \frac{u \pm v}{\sqrt{2}}. \tag{2.159}$$

Since $[z(1-z)]^{-1/8}$ is invariant under $z \mapsto 1-z$, replacing z by $1-z$ in (2.155) gives $\mathcal{F}_\pm(1-z) = [z(1-z)]^{-1/8}(u \pm v)/\sqrt{2}$, which is (2.158). □

Remark 2.5.2. *One has $F^2 = \mathbb{1}$ and $\det F = -1$, and the vacuum entry is $F_{\mathbb{1}\mathbb{1}} = 1/\sqrt{2} = 1/d_\sigma$, the inverse quantum dimension of Example 3.6.15. This matrix is the simplest instance of an F-symbol: where the modular S-matrix encodes the behavior of characters on the torus, F encodes the change of basis between operator product channels on the sphere.*

What crossing determines

The crossing matrix is a statement about blocks, hence about chiral data. To obtain a constraint on a theory we must impose that the physical correlator, which combines the two chiralities, be a single-valued function of z. For the Ising four-spin function this suffices to fix the remaining structure constant.

Proposition 2.5.3 (integrality of the spin)**.** *Expand the four-point function in the channel $z \to 0$ as*

$$G(z,\bar{z}) = \sum_{p,\bar{q}} X_{p\bar{q}}\, \mathcal{F}_p(z)\, \overline{\mathcal{F}_{\bar{q}}(z)}, \tag{2.160}$$

where p and $\bar{q}$ run independently over the channels of the holomorphic and antiholomorphic factors. Single-valuedness of G requires

$$X_{p\bar{q}} \neq 0 \qquad \Longrightarrow \qquad h_p - \bar{h}_{\bar{q}} \in \mathbb{Z}. \tag{2.161}$$

Proof. By the normalization in (2.129), $\mathcal{F}_p \sim z^{h_p - h_1 - h_2}$ and $\overline{\mathcal{F}_{\bar{q}}} \sim \bar{z}^{\bar{h}_{\bar{q}} - \bar{h}_1 - \bar{h}_2}$ as $z \to 0$. Taking the external fields to have integer spin, the continuation $z \mapsto e^{2\pi i} z$, $\bar{z} \mapsto e^{-2\pi i}\bar{z}$ multiplies the $(p, \bar{q})$ term by

$$\exp\bigl[2\pi i\,(h_p - \bar{h}_{\bar{q}})\bigr]. \tag{2.162}$$

Grouping the terms of (2.160) according to the value of $h_p - \bar{h}_{\bar{q}}$ modulo 1, each group carries its own phase and the groups have distinct leading powers of $|z|$, so no cancellation between them is possible. Single-valuedness of the sum therefore requires every nonvanishing group to carry phase 1. □

Remark 2.5.4. *This is the statement that a local field has integer spin. It concerns which left and right modules may be combined, not which channels occur; the latter is fixed by the fusion rules of §2.4.5. The same condition will reappear on the torus as Proposition 3.7.2, there constraining the whole spectrum rather than one correlator.*

Remark 2.5.5 (two directions)**.** *In the framework of §2.1 the condition $h - \bar{h} \in \mathbb{Z}$ is part of the definition of F (Remark 1.2.2). For a genus-zero correlation system Proposition 2.5.3 is therefore automatic: the intermediate states of the s-channel expansion are elements of F (Theorem 2.2.1). The proposition is the converse direction. Blocks know nothing about F; when correlators are assembled from them, as in this section, single-valuedness forces left and right modules to be paired only with integer spin. The axiom is thus a property that any such construction must produce.*

Corollary 2.5.6. *For $\langle\sigma\sigma\sigma\sigma\rangle$ in $\mathcal{M}(4,3)$ the expansion (2.160) is diagonal:*

$$G = A\,\bigl|\mathcal{F}_+\bigr|^2 + B\,\bigl|\mathcal{F}_-\bigr|^2, \qquad A = \bigl|C_{\sigma\sigma\mathbb{1}}\bigr|^2, \qquad B = 4\,\bigl|C_{\sigma\sigma\varepsilon}\bigr|^2. \tag{2.163}$$

Proof. By Example 2.4.19 the channels are $h_p = 0, \frac{1}{2}$, whose difference is $\frac{1}{2} \notin \mathbb{Z}$; Proposition 2.5.3 then forces $X_{p\bar{q}} = 0$ for $p \neq \bar{q}$. The solutions (2.155) behave as $\mathcal{F}_+ \sim z^{-1/8}$ and $\mathcal{F}_- \sim \frac{1}{2} z^{3/8}$, so the blocks normalized as in (2.129) are $\mathcal{F}_{\mathbb{1}} = \mathcal{F}_+$ and $\mathcal{F}_\varepsilon = 2\mathcal{F}_-$; inserting these into (2.157) gives (2.163). □

Theorem 2.5.7. *Normalize the primaries so that their two-point functions are unit normalized, whence $C_{\sigma\sigma\mathbb{1}} = 1$. Then*

$$C_{\sigma\sigma\varepsilon} = \tfrac{1}{2}. \tag{2.164}$$

Proof. The four insertions are the same field σ, of weight $h = \bar{h} = \frac{1}{16}$ and hence of integer spin, so the correlator is unchanged when the insertions at 0 and 1 are exchanged. In the coordinates of §2.4.5 that exchange is $z \mapsto 1 - z$, so

$$G(z, \bar{z}) = G(1 - z, \overline{1 - z}). \tag{2.165}$$

Expanding the right-hand side with Proposition 2.5.1,

$$\begin{aligned} A\left|\mathcal{F}_+(1-z)\right|^2 + B\left|\mathcal{F}_-(1-z)\right|^2 &= \frac{A}{2}\left|\mathcal{F}_+ + \mathcal{F}_-\right|^2 + \frac{B}{2}\left|\mathcal{F}_+ - \mathcal{F}_-\right|^2 \\ &= \frac{A+B}{2}\left(\left|\mathcal{F}_+\right|^2 + \left|\mathcal{F}_-\right|^2\right) + (A-B)\,\mathrm{Re}\left(\mathcal{F}_+\overline{\mathcal{F}_-}\right), \end{aligned} \tag{2.166}$$

all blocks on the right now evaluated at z. Subtracting (2.163),

$$\frac{B-A}{2}\left|\mathcal{F}_+\right|^2 + \frac{A-B}{2}\left|\mathcal{F}_-\right|^2 + (A-B)\,\mathrm{Re}\left(\mathcal{F}_+\overline{\mathcal{F}_-}\right) = 0. \tag{2.167}$$

The first two terms are single-valued at $z = 0$, whereas the third behaves as $z^{-1/8}\bar{z}^{3/8}$ and is multiplied by $e^{-2\pi i(1/8+3/8)} = -1$ under $z \mapsto e^{2\pi i}z$; the third can therefore not cancel against the first two, and each must vanish separately. Both conditions give $A = B$, whence $4\left|C_{\sigma\sigma\varepsilon}\right|^2 = 1$. □

Slogan 2.5.8. *Genus zero and genus one each contribute one piece of the data of a conformal field theory: crossing symmetry on the sphere fixes the structure constants C_{ijk}, and modular invariance on the torus fixes the pairing M_{ij} of left- and right-movers. Neither is visible in the chiral representation theory alone.*

The sphere has now been exhausted: the Virasoro algebra has produced its representations, its null vectors have produced differential equations, and those in turn have produced fusion rules, the crossing matrix and the structure constants. What is still missing is the pairing of the two chiralities, and no computation on the sphere can supply it. That question is settled on the torus, which is the subject of Chapter 3.

2.6 Notes

These notes collect pointers to more advanced frameworks and to the literature. They are outside the logical line of the chapter: nothing in the main text depends on them. Each item records where it was moved from.

- *Wick's theorem for interacting fields* (moved from after Proposition 2.2.15). When contractions are no longer numbers, (2.44) is replaced by the generalized Wick theorem, which contains an additional term in which a contraction is itself expanded against the remaining field [DMS97].
- *Ghosts* (moved from Example 2.2.19). The case $\lambda = 2$, $c = -26$, is the system of reparametrization ghosts of the bosonic string.

- *Reflection positivity* (moved from Remark 2.3.3). (U1) is reflection positivity in the sense of Osterwalder and Schrader [OS73]. For unitary full vertex operator algebras the corresponding Osterwalder–Schrader axioms are established in [AMT24]; see also [Sch08].

- *The kernel in Proposition 2.3.4* (moved from the remark after Corollary 2.3.5). $(1-\bar{w}v)^{-2h}$ is the reproducing kernel of the holomorphic discrete series of $\mathrm{SU}(1,1)$ (for $2h > 0$), and it is positive definite exactly when $h \geq 0$.

- *BPZ conjugation* (moved from Remark 2.3.8). The relation $L'_n = (-1)^n L_{-n}$ in the chart $-1/z$ is the BPZ conjugation used in string field theory.

- *The holomorphic inversion* (moved from Remark 2.3.10). The map $z \mapsto 1/z$, i.e. the rotation of the cylinder by π, gives the Euclidean form of the PCT operation.

- *Fusing matrix and modular data* (author's sentence, moved from the remark on the fusing matrix F in §2.5). Together they are the genus-one and genus-zero data of the Moore–Seiberg formalism.

Chapter 3

The cylinder and the torus

The map from the plane to the cylinder turns $L_0 + \bar{L}_0$ into a Hamiltonian, and gluing the two ends of the cylinder gives a torus. The partition function must then be invariant under the modular group $\mathrm{PSL}(2,\mathbb{Z})$. Of its two generators, T only re-imposes the integrality of the spin, already met on the sphere. S exchanges space and time and relates the multiplicities of different modules; it is what distinguishes the theories built on the same chiral data.

Section 3.1 sets up the cylinder, the partition function and the modular group, and §3.2 the theta functions on which every S-transformation rests. The free boson (§3.3) and the free fermion with the Ising model (§3.4) are computed directly. Sections 3.5 and 3.6 treat minimal models in general: characters, the S-matrix and the Verlinde formula. Section 3.7 asks which combinations of characters are modular invariant.

3.1 From the cylinder to the torus

3.1.1 The cylinder

Let the cylinder of circumference L have coordinate $w = \tau - ix$, $x \sim x + L$; this convention is used throughout the note. For $\tau = it$ this is $w = i(t-x)$, so holomorphic fields are right-movers, and map it to the plane by $z = e^{2\pi w/L}$. Circles $\tau = \text{const}$ go to circles $|z| = e^{2\pi\tau/L}$, so radial ordering is τ-ordering (Remark 2.2.2). A field of weight $(h,\bar{h})$ on the cylinder is

$$\phi_{\mathrm{cyl}}(w,\bar{w}) = \Big(\frac{2\pi}{L}\Big)^{h+\bar{h}} z^h \bar{z}^{\bar{h}}\, \phi(z,\bar{z}), \tag{3.1}$$

with the spin-adapted power of Remark 1.2.2; for quasi-primaries this is (1.18) applied to $w \mapsto z$.

The stress tensor on the cylinder. T is not primary, so (3.1) does not apply to it; its transformation law is (2.27). For $z = e^{2\pi w/L}$ the Schwarzian (2.30) is $\{z,w\} = (2\pi/L)^2\big(1-\frac{3}{2}\big) = -\frac{1}{2}(2\pi/L)^2$, so

$$T_{\mathrm{cyl}}(w) = \Big(\frac{2\pi}{L}\Big)^2\Big[z^2T(z) - \frac{c}{24}\Big] = \Big(\frac{2\pi}{L}\Big)^2\Big[\sum_{n\in\mathbb{Z}} L_n\, e^{-2\pi n w/L} - \frac{c}{24}\Big]. \tag{3.2}$$

The constant is the whole content of the Schwarzian here: a Möbius map would give none (property (iv) of (2.28)), and $z = e^{2\pi w/L}$ is not Möbius.

Definition 3.1.1. *The Hamiltonian and momentum on the cylinder are the charges of T_{cyl} on a circle $\tau = const$,*

$$H := \frac{1}{2\pi}\int_0^L dx\,\big[T_{cyl}(w) + \bar{T}_{cyl}(\bar{w})\big], \qquad P := \frac{1}{2\pi}\int_0^L dx\,\big[T_{cyl}(w) - \bar{T}_{cyl}(\bar{w})\big]. \tag{3.3}$$

Proposition 3.1.2. *H and P are independent of τ, and*

$$H = \frac{2\pi}{L}\Big(L_0 + \bar{L}_0 - \frac{c}{12}\Big), \qquad P = \frac{2\pi}{L}\big(L_0 - \bar{L}_0\big). \tag{3.4}$$

For quasi-primary ϕ, $[H, \phi_{cyl}] = \partial_\tau \phi_{cyl}$ and $[P, \phi_{cyl}] = i\partial_x \phi_{cyl}$; the latter is the sign of

$$\phi(t,x) = e^{i(Ht - Px)}\phi(0)e^{-i(Ht-Px)}.$$

Hence

$$e^{aH}\phi_{cyl}(\tau,x)e^{-aH} = \phi_{cyl}(\tau + a, x), \qquad e^{i\theta P}\phi_{cyl}(\tau,x)e^{-i\theta P} = \phi_{cyl}(\tau, x-\theta). \tag{3.5}$$

Proof. In (3.2), $\int_0^L dx\, e^{-2\pi n(\tau - ix)/L} = L\,\delta_{n0}$, so only L_0 survives and the τ-dependence drops out. This gives (3.4). For the commutators, Corollary 2.2.3 and the expansion of $T(w)\phi(z)$ give

$$[L_0, \phi(z,\bar{z})] = \oint_{C_z} \frac{dw}{2\pi i}\, w\, T(w)\phi(z,\bar{z}) = (h + z\partial_z)\phi(z,\bar{z}). \tag{3.6}$$

Here only the double and simple poles contribute: $L_0\phi = h\phi$, $L_{-1}\phi = \partial\phi$, and a pole $(w-z)^{-k-2}$ with $k \geq 1$ multiplied by $w = z + (w-z)$ has no residue. Since $\partial_w = (2\pi/L)z\partial_z$, (3.1) gives $[L_0, \phi_{\rm cyl}] = (L/2\pi)\partial_w\phi_{\rm cyl}$, and likewise $[\bar{L}_0, \phi_{\rm cyl}] = (L/2\pi)\partial_{\bar{w}}\phi_{\rm cyl}$. The constant drops out of commutators, and $\partial_\tau = \partial_w + \partial_{\bar{w}}$, $i\partial_x = \partial_w - \partial_{\bar{w}}$. A holomorphic excitation of weight h thus carries energy and momentum $\frac{2\pi}{L}h$, as a right-mover should. □

Remark 3.1.3 (what fixes the constant)**.** *The commutators determine H only up to an additive constant. The constant $-\pi c/6L$ is fixed by declaring T_{cyl} to be the energy density in the cylinder coordinate, with T normalized on the plane so that $\langle T\rangle = 0$. The ground-state energy*

$$E_0(L) = -\frac{\pi c}{6L} \tag{3.7}$$

is then the finite-size (Casimir) energy relative to the infinite plane.

Remark 3.1.4 (spin and periodicity)**.** *On $F_{(h,\bar{h})}$, P acts by $(2\pi/L)(h-\bar{h}) = (2\pi/L)s$. The translation e^{iLP} around the circle is $e^{2\pi i s} = 1$, as it must be on a periodic cylinder. This is Remark 1.2.2 seen from the cylinder: integer spin on the plane is periodicity on the cylinder. For $s \in \frac{1}{2} + \mathbb{Z}$ one gets antiperiodicity, which is the Neveu–Schwarz boundary condition of* (3.72).

In a unitary theory H is Hermitian and bounded below by $E_0(L)$; see Corollary 3.1.5.

Corollary 3.1.5 (spectrum in a unitary theory)**.** *In a theory with a unitary structure (Definition 2.3.2), H and P are Hermitian and $H \geq -\pi c/6L$. The minimum is attained exactly on $F_{(0,0)}$, whose elements ϕ satisfy $\partial\phi = \bar{\partial}\phi = 0$ and therefore have correlators independent of their position. In particular $|0\rangle = \mathbb{I}$ attains it.*

Proof. Hermiticity is Theorem 2.3.9 with $n = 0$, together with its analogue for $\bar{T}$. By Corollary 2.3.5, every quasi-primary has $h, \bar{h} \geq 0$, so every $\partial^m \bar{\partial}^n \phi$ has $L_0 + \bar{L}_0$-eigenvalue $h + \bar{h} + m + n \geq 0$, with equality only for $m = n = 0$ and $h = \bar{h} = 0$. Such ϕ have $\partial\phi = \bar{\partial}\phi = 0$ by the same corollary, and (G3) gives the position independence. □

Remark 3.1.6. *Uniqueness of the vacuum, $F_{(0,0)} = \mathbb{C}\,\mathbb{I}$, does not follow. It is an additional requirement, and it fails for instance in a direct sum of two theories. Prop. 2.3.4 also says that the global descendants of a quasi-primary of weight $h > 0$ contribute $\sum_{m\geq 0} q^{h+m} = q^h/(1-q)$ to $\operatorname{Tr} q^{L_0}$, while for $h = 0$ only $m = 0$ survives.*

3.1.2 Torus partition function

Since a CFT is invariant under rigid rotations and rescaling, we can fix the first period $\omega_1 = 1$. Then the only remaining structure parameter of the torus is the modular parameter $\tau = \omega_2/\omega_1$. Then the partition function is

$$\mathcal{Z}(\tau) = \int \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{3.8}$$

where the path integral measure is over the fields ϕ such that

$$\phi(0) = \phi(\tau) = \phi(\operatorname{Re}\tau, \operatorname{Im}\tau) = e^{-(\operatorname{Im}\tau)H} e^{i(\operatorname{Re}\tau)P} \phi(0,0) = e^{-(\operatorname{Im}\tau)H} e^{i(\operatorname{Re}\tau)P} \phi(0). \tag{3.9}$$

Thus,

$$\begin{aligned} \mathcal{Z}(\tau) &= \sum_{\phi_0} \langle \phi_0 | e^{-(\operatorname{Im}\tau)H} e^{i(\operatorname{Re}\tau)P} | \phi_0 \rangle \\ &= \operatorname{Tr} \exp\left[-(\operatorname{Im}\tau)H + i(\operatorname{Re}\tau)P\right] \\ &= \operatorname{Tr} \exp 2\pi i \left[\tau\left(L_0 - \frac{c}{24}\right) - \bar{\tau}\left(\bar{L}_0 - \frac{c}{24}\right)\right] \\ &= \operatorname{Tr}\, q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}} \end{aligned} \tag{3.10}$$

The third equality is (3.4) with $L = 1$, Euclidean time $\operatorname{Im}\tau$ and twist $\operatorname{Re}\tau$. The $-c/24$ is the Casimir energy of Remark 3.1.3, split equally between the two chiralities.

3.1.3 The modular group $\mathrm{PSL}(2,\mathbb{Z})$

A torus is a quotient $\mathbb{C}/\Lambda$ by a rank-two lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$, and the complex structure remembers the lattice, not the chosen basis. Two bases related by

$$\begin{pmatrix} \omega_2' \\ \omega_1' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \omega_2 \\ \omega_1 \end{pmatrix}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z}), \tag{3.11}$$

describe the same torus; in terms of $\tau = \omega_2/\omega_1$ this is

$$\tau \longmapsto \frac{a\tau + b}{c\tau + d}, \tag{3.12}$$

which preserves $\operatorname{Im}\tau > 0$ since $\operatorname{Im}\frac{a\tau+b}{c\tau+d} = \frac{\operatorname{Im}\tau}{|c\tau+d|^2}$. The element $-I_2$ acts trivially, so the group acting faithfully is

$$\mathrm{PSL}(2,\mathbb{Z}) = SL(2,\mathbb{Z})/\{\pm I_2\} = \left\langle S, T \,\middle|\, S^2 = (ST)^3 = 1 \right\rangle, \tag{3.13}$$

generated by

$$S : \tau \mapsto -\frac{1}{\tau}, \qquad T : \tau \mapsto \tau + 1. \tag{3.14}$$

Slogan 3.1.7. *The partition function is not a function of τ but a function on the moduli space $\mathbb{H}/\mathrm{PSL}(2,\mathbb{Z})$ of tori:*

$$\mathcal{Z}(-1/\tau) = \mathcal{Z}(\tau), \qquad \mathcal{Z}(\tau+1) = \mathcal{Z}(\tau). \tag{3.15}$$

Of the two generators, T costs nothing: it is already visible in the trace formula above, where $\tau \to \tau + 1$ multiplies each state by $e^{2\pi i(L_0 - \bar{L}_0)}$, so that T-invariance is the statement that the spins $h - \bar{h}$ are integers. All the content of (3.15) therefore lies in S, which is not visible in the trace formula at all — it exchanges the space and time cycles of the torus. Computing the action of S rests on the theta functions of §3.2.

3.2 Theta functions

Every S-transformation in this chapter reduces to that of a theta function, and that in turn to Poisson summation (Theorem A.0.4). This section collects the transformation laws.

3.2.1 Theta functions of level N

For integers $N \geq 1$ and λ, the theta function of level N is

$$\Theta_\lambda(\tau) := \sum_{n\in\mathbb{Z}} q^{\frac{(\lambda+2nN)^2}{4N}} = \sum_{n\in\mathbb{Z}} e^{2\pi i\tau N\left(n+\frac{\lambda}{2N}\right)^2}, \qquad q = e^{2\pi i\tau}. \tag{3.16}$$

Reindexing the sum immediately gives the two elementary properties

$$\Theta_{\lambda+2N} = \Theta_\lambda, \qquad \Theta_{-\lambda} = \Theta_\lambda, \tag{3.17}$$

so that Θ_λ depends only on λ mod $2N$, up to sign of λ.

Since Θ_λ depends on λ only through λ mod $2N$ and through λ^2, the generator $T : \tau \mapsto \tau+1$ acts diagonally.

Proposition 3.2.1.

$$\Theta_\lambda(\tau+1) = e^{2\pi i\frac{\lambda^2}{4N}}\,\Theta_\lambda(\tau). \tag{3.18}$$

Proof. Write (3.16) as $\Theta_\lambda(\tau) = \sum_{n\in\mathbb{Z}} e^{2\pi i\tau E_n}$ with $E_n := \frac{(\lambda+2Nn)^2}{4N}$. Replacing τ by $\tau+1$ multiplies the n-th term by the phase $e^{2\pi i E_n}$. These phases are all equal, because

$$E_n - E_0 = \frac{(\lambda+2Nn)^2-\lambda^2}{4N} = n\lambda + Nn^2 \in \mathbb{Z} \tag{3.19}$$

for $\lambda, n, N \in \mathbb{Z}$; that is, the exponents E_n share a common fractional part $E_0 = \lambda^2/4N$. The common phase $e^{2\pi i E_0}$ therefore factors out of the sum. □

The generator $S : \tau \mapsto -1/\tau$ is the substantial one. It mixes the Θ_λ among themselves by a finite Fourier transform on $\mathbb{Z}/2N\mathbb{Z}$; the mechanism is Poisson summation (Theorem A.0.4), which trades the sum over the lattice $\mathbb{Z}$ in (3.16) for a sum over its dual.

Theorem 3.2.2. *For $\tau \in \mathbb{H}$,*

$$\Theta_\lambda(-1/\tau) = \sqrt{\frac{-i\tau}{2N}} \sum_{\lambda' \bmod 2N} e^{i\pi\lambda\lambda'/N}\, \Theta_{\lambda'}(\tau), \tag{3.20}$$

with the principal branch of Definition A.0.7, so that $\sqrt{-i\tau} = 1$ at $\tau = i$.

Proof. Set

$$a := \frac{2\pi i N}{\tau}, \qquad c := \frac{\lambda}{2N}, \qquad f(x) := e^{-a(x+c)^2}, \tag{3.21}$$

so that $\Theta_\lambda(-1/\tau) = \sum_{n\in\mathbb{Z}} f(n)$ by (3.16). By Corollary A.0.11 we have $a \in H_+$, and c is real, so f is a Schwartz function and Theorem A.0.4 applies.

To compute $\hat f$, substitute $u = x + c$ and apply Theorem A.0.10 with $b = 2\pi i k$:

$$\hat f(k) = e^{2\pi i k c}\int_{\mathbb{R}} e^{-au^2 - 2\pi i k u}\,\mathrm{d}u = e^{i\pi k\lambda/N}\sqrt{\frac{\pi}{a}}\; e^{-\pi^2 k^2/a}. \tag{3.22}$$

Corollary A.0.11 identifies $\sqrt{\pi/a}$ with $\sqrt{-i\tau/2N}$, while

$$-\frac{\pi^2 k^2}{a} = -\frac{\pi^2 k^2 \tau}{2\pi i N} = 2\pi i\tau\,\frac{k^2}{4N}, \qquad \text{so} \qquad e^{-\pi^2 k^2/a} = q^{\frac{k^2}{4N}}. \tag{3.23}$$

Poisson summation therefore gives

$$\Theta_\lambda(-1/\tau) = \sqrt{\frac{-i\tau}{2N}} \sum_{k\in\mathbb{Z}} e^{i\pi k\lambda/N}\, q^{\frac{k^2}{4N}}. \tag{3.24}$$

It remains to reorganise the sum. Split $k = \lambda' + 2Nm$ with λ' running over $\mathbb{Z}/2N\mathbb{Z}$ and $m \in \mathbb{Z}$. The phase factorises as $e^{i\pi\lambda\lambda'/N}\, e^{2\pi i m\lambda}$, and the second factor equals 1 because $\lambda, m \in \mathbb{Z}$; the phase is thus independent of m, and the residual sum over m reassembles $\Theta_{\lambda'}$ by (3.16). □

3.2.2 The Dedekind eta function

The Dedekind eta function is

$$\eta(\tau) := q^{1/24}\varphi(q), \qquad \varphi(q) := \prod_{n\geq 1}(1-q^n). \tag{3.25}$$

Both Θ_λ and η are holomorphic on $\mathbb{H}$. The transformation law of η is usually quoted from the theory of modular forms, but it is in fact a corollary of Theorem 3.2.2: the Dedekind eta function is itself a difference of two theta functions.

Lemma 3.2.3. *At level $N = 6$ one has $\eta = \Theta_1 - \Theta_5$.*

Proof. Euler's pentagonal number theorem gives $\varphi(q) = \sum_{k\in\mathbb{Z}} (-1)^k q^{k(3k-1)/2}$, and $\frac{1}{24} + \frac{k(3k-1)}{2} = \frac{(6k-1)^2}{24}$, so

$$\eta = q^{1/24}\varphi(q) = \sum_{k\in\mathbb{Z}} (-1)^k q^{\frac{(6k-1)^2}{24}}. \tag{3.26}$$

With $N = 6$ we have $2N = 12$ and $4N = 24$. For $k = 2n$ the index is $6k - 1 = 12n - 1$, contributing to $\Theta_{-1} = \Theta_1$ with sign $+$; for $k = 2n + 1$ it is $12n + 5$, contributing to Θ_5 with sign $-$. The pentagonal number theorem is proved in Corollary A.0.15. □

Corollary 3.2.4. *$\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ and $\eta(\tau + 1) = e^{i\pi/12}\eta(\tau)$.*

Proof. The second statement is immediate from $\eta = q^{1/24}\varphi(q)$. For the first, apply (3.20) at $N = 6$ to Lemma 3.2.3. Writing $\zeta := e^{i\pi/6}$, the coefficient of $\Theta_{\lambda'}$ is $c_{\lambda'} = \zeta^{\lambda'} - \zeta^{5\lambda'}$. Pairing λ' with $12 - \lambda'$ and using $\Theta_{-\lambda'} = \Theta_{\lambda'}$,

$$c_{\lambda'} + c_{12-\lambda'} = 2\cos\frac{\pi\lambda'}{6} - 2\cos\frac{5\pi\lambda'}{6}, \tag{3.27}$$

which vanishes for $\lambda' = 0, 2, 3, 4, 6$ and equals $\pm 2\sqrt{3}$ for $\lambda' = 1, 5$ respectively. Hence

$$\eta(-1/\tau) = \sqrt{\frac{-i\tau}{12}} \cdot 2\sqrt{3}\,(\Theta_1 - \Theta_5) = \sqrt{-i\tau}\,\eta(\tau), \tag{3.28}$$

since $2\sqrt{3}/\sqrt{12} = 1$. □

Remark 3.2.5. *The argument is not circular: Theorem 3.2.2 rests on Poisson summation and Lemma 3.2.3 on a combinatorial identity, and neither uses the modularity of η. Note also that the consistency of $S^2 = 1$ with the branch conventions was verified in Example A.0.12 of the appendix. In conformal field theory Lemma 3.2.3 states that the vacuum character of the (trivial) minimal model $\mathcal{M}(3, 2)$, which has $c = 0$, equals 1.*

3.2.3 Jacobi theta functions

The free fermion of §3.4 needs three further theta functions,

$$\theta_2(\tau) := \sum_{n\in\mathbb{Z}} q^{\frac{1}{2}(n+\frac{1}{2})^2}, \qquad \theta_3(\tau) := \sum_{n\in\mathbb{Z}} q^{\frac{n^2}{2}}, \qquad \theta_4(\tau) := \sum_{n\in\mathbb{Z}} (-1)^n q^{\frac{n^2}{2}}. \tag{3.29}$$

Proposition 3.2.6. *The functions (3.29) have the product forms*

$$\theta_3 = \prod_{m\geq 1} (1 - q^m)(1 + q^{m-\frac{1}{2}})^2, \qquad \theta_4 = \prod_{m\geq 1} (1 - q^m)(1 - q^{m-\frac{1}{2}})^2,$$
$$\theta_2 = 2q^{\frac{1}{8}} \prod_{m\geq 1} (1 - q^m)(1 + q^m)^2, \tag{3.30}$$

and transform as

$$\theta_3(-1/\tau) = \sqrt{-i\tau}\,\theta_3(\tau), \qquad \theta_4(-1/\tau) = \sqrt{-i\tau}\,\theta_2(\tau), \qquad \theta_2(-1/\tau) = \sqrt{-i\tau}\,\theta_4(\tau), \tag{3.31}$$
$$\theta_3(\tau + 1) = \theta_4(\tau), \qquad \theta_4(\tau + 1) = \theta_3(\tau), \qquad \theta_2(\tau + 1) = e^{i\pi/4}\theta_2(\tau). \tag{3.32}$$

Proof. Products. Set $z = 1$ and $z = -1$ in Lemma A.0.13. For θ_2 set $z = q^{1/2}$: the right side of (A.19) becomes $\sum_n q^{(n^2+n)/2} = q^{-1/8}\theta_2$, and the left side becomes $\prod_m (1-q^m)(1+q^m)(1+q^{m-1}) = 2\prod_m (1-q^m)(1+q^m)^2$.

S. At level $N = 2$, (3.16) reads $\Theta_\lambda = \sum_n q^{(\lambda+4n)^2/8}$. Sorting the terms of (3.29) by the residue of $2n$ or $2n+1$ modulo 4 gives

$$\theta_3 = \Theta_0 + \Theta_2, \qquad \theta_4 = \Theta_0 - \Theta_2, \qquad \theta_2 = \Theta_1 + \Theta_3. \tag{3.33}$$

By Theorem 3.2.2 with $N = 2$, $\Theta_\lambda(-1/\tau) = \frac{1}{2}\sqrt{-i\tau}\sum_{\lambda' \bmod 4} i^{\lambda\lambda'}\Theta_{\lambda'}(\tau)$. The coefficient of $\Theta_{\lambda'}$ is $\frac{1}{2}(1+(-1)^{\lambda'})$ for θ_3, $\frac{1}{2}(1-(-1)^{\lambda'})$ for θ_4, and $\frac{1}{2}(i^{\lambda'} + i^{3\lambda'}) = \cos(\pi\lambda'/2)$ for θ_2. These give (3.31).

T. Under $\tau \mapsto \tau + 1$ the term $q^{n^2/2}$ acquires $e^{i\pi n^2} = (-1)^n$, and $q^{\frac{1}{2}(n+\frac{1}{2})^2}$ acquires $e^{i\pi(n^2+n)}e^{i\pi/4} = e^{i\pi/4}$. □

3.3 The free boson

3.3.1 Canonical quantization on cylinder

The Lagrangian of the free boson is

$$\mathcal{L} = \frac{g}{2}\int \mathrm{d}x\, \partial_\mu\varphi\partial^\mu\varphi. \tag{3.34}$$

On a cylinder of circumference L, a Bose field may be expanded a la Fourier:

$$\varphi(x,t) = \sum_{n\in\mathbb{Z}} e^{2\pi i n x/L}\varphi_n(t), \tag{3.35a}$$

$$\varphi_n(t) = \frac{1}{L}\int_0^L \mathrm{d}x\, e^{-2\pi i n x/L}\varphi(x,t), \tag{3.35b}$$

$$\mathcal{L} = \frac{g}{2}L\sum_{n\in\mathbb{Z}}\left[\dot\varphi_n\dot\varphi_{-n} - \left(\frac{2\pi n}{L}\right)^2\varphi_n\varphi_{-n}\right]. \tag{3.35c}$$

We apply the canonical quantization to (3.35c). The momentum operator conjugate to φ_n is

$$\pi_n = gL\dot\varphi_{-n}, \qquad [\varphi_n,\pi_m] = i\delta_{nm}, \qquad [\varphi_n,\varphi_m] = [\pi_n,\pi_m] = 0, \tag{3.36}$$

and the Hamiltonian is

$$H = \frac{1}{2gL}\sum_{n\in\mathbb{Z}}\Big[\pi_n\pi_{-n} + (2\pi n g)^2\varphi_n\varphi_{-n}\Big]. \tag{3.37}$$

We note from Eq. (3.35b) that $\varphi_n^\dagger = \varphi_{-n}$ and $\pi_n^\dagger = \pi_{-n}$. For $n > 0$, the summands for $\{n,-n\}$ pair up to form a 2D harmonic oscillator with phase space variables (x,y,p_x,p_y) through

$$\begin{aligned} \varphi_n &= \frac{1}{\sqrt{2}}(x+iy), & \varphi_{-n} &= \frac{1}{\sqrt{2}}(x-iy),\\ \pi_n &= \frac{1}{\sqrt{2}}(p_x - ip_y), & \pi_{-n} &= \frac{1}{\sqrt{2}}(p_x+ip_y), \end{aligned} \tag{3.38}$$

with obvious commutation relations. In terms of these variables, the Hamiltonian is

$$H = \frac{1}{2M}\pi_0^2 + \sum_{n>0}\frac{1}{2M}(p_x^2+p_y^2) + \frac{1}{2}M\Omega_n^2(x^2+y^2), \qquad M = gL, \quad \Omega_n = 2\pi n/L. \tag{3.39}$$

Lesson: For each $n > 0$, $(\varphi_n, \varphi_{-n})$ are just the complexified coordinates for the planar harmonic oscillator (x, y), with conjugate momenta (π_{-n}, π_n). Conventionally, one often proceeds by defining the annihilation operators (where we extend to all $n \in \mathbb{Z}\backslash\{0\}$)

$$a_n^{\text{pre}} = \frac{1}{\sqrt{2M|\Omega_n|}}(i\pi_{-n} + M\Omega_n\varphi_n) = \frac{1}{\sqrt{4\pi g|n|}}(i\pi_{-n} + 2\pi g n\,\varphi_n), \tag{3.40a}$$

$$\bar{a}_n^{\text{pre}} = \frac{1}{\sqrt{2M|\Omega_n|}}(i\pi_n + M\Omega_n\varphi_{-n}) = \frac{1}{\sqrt{4\pi g|n|}}(i\pi_n + 2\pi g n\,\varphi_{-n}), \tag{3.40b}$$

$$a_{-n}^{\text{pre}} = -(a_n^{\text{pre}})^\dagger, \qquad \bar{a}_{-n}^{\text{pre}} = -(\bar{a}_n^{\text{pre}})^\dagger. \tag{3.40c}$$

But here we shall use a different normalization:

$$\begin{aligned} a_n &= -i\sqrt{n}\,a_n^{\text{pre}}, & \bar{a}_n &= -i\sqrt{n}\,\bar{a}_n^{\text{pre}} & (n>0),\\ a_{-n} &= a_n^\dagger, & \bar{a}_{-n} &= \bar{a}_n^\dagger & (n>0). \end{aligned} \tag{3.41}$$

Then the Hamiltonian is

$$H = \frac{1}{2gL}\pi_0^2 + \frac{2\pi}{L}\sum_{n>0}\left(a_{-n}a_n + \bar{a}_{-n}\bar{a}_n\right). \tag{3.42}$$

Due to the normalization, each creation operator a_{-n} increases the energy by $2\pi n/L$:

$$[H, a_{-n}] = \frac{2\pi}{L}n a_{-n} \qquad (n \geq 0). \tag{3.43}$$

Now, we can expand the field $\varphi(x)$ by the canonical operators in Eq. (3.41): From Eq. (3.40), the Fourier modes are

$$\varphi_n = \frac{i}{n\sqrt{4\pi g}}(a_n - \bar{a}_{-n}), \tag{3.44}$$

Eq. (3.35a) says

$$\varphi(x, t=0) = \varphi_0 + \frac{i}{\sqrt{4\pi g}}\sum_{n\neq 0}\frac{1}{n}(a_n - \bar{a}_{-n})e^{2\pi i n x/L}. \tag{3.45}$$

Eq. (3.43) is enough to determine the time evolution:

$$\begin{aligned} \varphi_0(t) &= \varphi_0(0) + \frac{1}{gL}\pi_0 t & a_n(t) &= a_n(0)e^{-2\pi i n t/L}\\ & & \bar{a}_n(t) &= \bar{a}_n(0)e^{-2\pi i n t/L} \end{aligned} \tag{3.46}$$

$$\implies \quad \varphi(x,t) = \varphi_0 + \frac{1}{gL}\pi_0 t + \frac{i}{\sqrt{4\pi g}}\sum_{n\neq 0}\frac{1}{n}\left(a_n e^{2\pi i n(x-t)/L} - \bar{a}_{-n}e^{2\pi i n(x+t)/L}\right) \tag{3.47}$$

Now we Wick-rotate to Euclidean spacetime ($t \mapsto -i\tau$) and use the conformal coordinates

$$z = e^{2\pi(\tau - ix)/L}, \qquad \bar{z} = e^{2\pi(\tau+ix)/L} \tag{3.48}$$

so that

$$\begin{aligned}\varphi(z,\bar{z}) &= \varphi_0 - \frac{i}{4\pi g}\pi_0 \log(z\bar{z}) + \frac{i}{\sqrt{4\pi g}}\sum_{n\neq 0}\frac{1}{n}\left(a_n z^{-n} + \bar{a}_n \bar{z}^{-n}\right) \\ &= \varphi(z) + \bar{\varphi}(\bar{z}).\end{aligned} \tag{3.49}$$

The split fields $\varphi(z), \bar{\varphi}(\bar{z})$ themselves are not primary, but the derivatives $\partial\varphi$ and $\bar{\partial}\bar{\varphi}$ are. Defining

$$a_0 \equiv \bar{a}_0 := \frac{\pi_0}{\sqrt{4\pi g}}, \tag{3.50}$$

the derivatives are

$$i\partial\varphi(z) = \frac{1}{\sqrt{4\pi g}}\sum_{n\in\mathbb{Z}} a_n z^{-n-1}, \qquad i\bar{\partial}\bar{\varphi}(\bar{z}) = \frac{1}{\sqrt{4\pi g}}\sum_{n\in\mathbb{Z}} \bar{a}_n \bar{z}^{-n-1}. \tag{3.51}$$

3.3.2 Vertex operators

Since the boson φ has scaling dimension zero, it is possible to construct local fields out of arbitrary functions of φ. An outstanding example is the *vertex operators*

$$\begin{aligned}\mathcal{V}_\alpha(z,\bar{z}) = :e^{i\alpha\varphi(z,\bar{z})}: \; &= \exp\left[i\alpha\varphi_0 + \frac{\alpha}{\sqrt{4\pi g}}\sum_{n>0}\frac{1}{n}(a_{-n}z^n + \bar{a}_{-n}\bar{z}^n)\right] \\ &\quad \times \exp\left[\frac{\alpha}{4\pi g}\pi_0\log(z\bar{z}) - \frac{\alpha}{\sqrt{4\pi g}}\sum_{n>0}\frac{1}{n}(a_n z^n + \bar{a}_n\bar{z}^n)\right].\end{aligned} \tag{3.52}$$

Using (2.47), we compute the OPE

$$\begin{aligned}\partial\varphi(z)\mathcal{V}_\alpha(w,\bar{w}) &= \sum_{n=0}^{\infty}\frac{(i\alpha)^n}{n!}\partial\varphi(z):\varphi(w,\bar{w})^n: \\ &\sim \frac{-1}{4\pi g}\frac{1}{z-w}\sum_{n=1}^{\infty}\frac{(i\alpha)^n}{(n-1)!}:\varphi(w,\bar{w})^{n-1}: \\ &\sim \frac{-i\alpha}{4\pi g}\frac{\mathcal{V}_\alpha(w,\bar{w})}{(z-w)}.\end{aligned} \tag{3.53}$$

This implies that the OPE of $T(z)$ and $\mathcal{V}_\alpha(w,\bar{w})$ is

$$\begin{aligned}T(z)\mathcal{V}_\alpha(w,\bar{w}) =& -2\pi g\sum_{n=0}^{\infty}\frac{(i\alpha)^n}{n!}:\partial\varphi(z)\partial\varphi(z):\;:\varphi(w,\bar{w})^n: \\ \sim& -\frac{1}{8\pi g}\frac{1}{(z-w)^2}\sum_{n=2}^{\infty}\frac{(i\alpha)^n}{(n-2)!}:\varphi(w,\bar{w})^{n-2}: \\ &+\frac{1}{z-w}\sum_{n=1}^{\infty}\frac{(i\alpha)^n}{n!}n:\partial\varphi(z)\varphi(w,\bar{w})^{n-1}: \\ \sim& \frac{\alpha^2}{8\pi g}\frac{\mathcal{V}_\alpha(w,\bar{w})}{(z-w)^2} + \frac{\partial_w\mathcal{V}_\alpha(w,\bar{w})}{z-w}.\end{aligned} \tag{3.54}$$

(3.54) reveals that $\mathcal{V}_\alpha$ is primary with conformal dimensions $h(\alpha) = \bar{h}(\alpha) = \alpha^2/8\pi g$.

3.3.3 The Fock space

Since the Hamiltonian in Eq. (3.42) is independent of φ_0, and all the creation/annihilation operators commute with π_0, independent Fock spaces can be built upon a one-parameter family of vacua $|\alpha\rangle$ parameterized by the eigenvalue of $\pi_0 = \sqrt{4\pi g}\, a_0$. By definition,

$$a_0|\alpha\rangle = \bar{a}_0|\alpha\rangle = \frac{\alpha}{\sqrt{4\pi g}}|\alpha\rangle, \qquad a_n|\alpha\rangle = \bar{a}_n|\alpha\rangle = 0 \quad (n > 0). \tag{3.55}$$

Now we construct the whole spectrum over $|\alpha\rangle$. In quantum theory, the energy-momentum tensor in Eq. (2.49) must be regularized, which we do by normal ordering:

$$T(z) = -2\pi g : \partial\varphi(z)\partial\varphi(z) : \; = \frac{1}{2}\sum_{n,m\in\mathbb{Z}} z^{-n-m-2} : a_n a_m : \; = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}, \tag{3.56}$$

which implies

$$L_0 = \frac{1}{2}a_0^2 + \sum_{n>0} a_{-n}a_n, \qquad L_n = \frac{1}{2}\sum_{m\in\mathbb{Z}} a_{n-m}a_m \quad (n \neq 0). \tag{3.57}$$

The Hamiltonian is then (to wit: Eq. (3.42))

$$H = \frac{2\pi}{L}(L_0 + \bar{L}_0). \tag{3.58}$$

Remark 3.3.1. *Compared with (3.4) at $c = 1$, the constant $-\frac{2\pi}{L}\cdot\frac{1}{12}$ is missing, because normal ordering discards the zero-point energy $\frac{1}{2}\sum_{n\geq 1}\frac{2\pi n}{L}$ of each chirality. Regularizing $\sum_{n\geq 1} n = \zeta(-1) = -\frac{1}{12}$ restores it: $2\cdot\frac{1}{2}\cdot\frac{2\pi}{L}\cdot(-\frac{1}{12}) = -\frac{2\pi}{L}\frac{c}{12}$. The Schwarzian in (3.2) produces the same constant without any regularization (Remark 3.1.3).*

The other states of the Fock space over $|\alpha\rangle$ are obtained by successive applications of creation operators:

$$\prod_{i>0} a_{-i}^{n_i}\bar{a}_{-i}^{m_i}|\alpha\rangle \qquad (n_i, m_i \geq 0). \tag{3.59}$$

These states are eigenstates of $L_0, \bar{L}_0$ with conformal dimensions

$$h = \frac{\alpha^2}{8\pi g} + \sum_j j n_j, \qquad \bar{h} = \frac{\alpha^2}{8\pi g} + \sum_j j m_j. \tag{3.60}$$

The state $|\alpha\rangle$ is a state of definite zero-mode momentum $\pi_0 = \alpha$. This "1-particle" state can be obtained from the "absolute" vacuum $|0\rangle$ by application of the vertex operator:

Proposition 3.3.2. *The vertex operator $\mathcal{V}_\alpha(z,\bar{z}) = :e^{i\alpha\varphi(z,\bar{z})}:$ generates the state $|\alpha\rangle$ by*

$$|\alpha\rangle = \mathcal{V}_\alpha(0)|0\rangle. \tag{3.61}$$

Proof. When the commutator $[B, A]$ is a constant, we have

$$[B, e^A] = e^A [B, A]. \tag{3.62}$$

Now set $B = \pi_0$ and $A = i\alpha\varphi(z, \bar{z})$ to get (using (3.52))

$$[\pi_0, \mathcal{V}_\alpha] = \alpha \mathcal{V}_\alpha. \tag{3.63}$$

applying Eq. (3.63) at $z = 0$ to the absolute vacuum $|0\rangle$, we get

$$\pi_0 \mathcal{V}_\alpha(0)|0\rangle = \alpha \mathcal{V}_\alpha(0)|0\rangle. \tag{3.64}$$

Setting $B = a_n$, Eq. (3.52) implies

$$[a_n, \mathcal{V}_\alpha(z, \bar{z})] = \frac{\alpha z^n}{\sqrt{4\pi g}} \mathcal{V}_\alpha(z, \bar{z}) \quad \Rightarrow \quad a_n \mathcal{V}_\alpha(0)|0\rangle = 0 \quad (n > 0) \tag{3.65}$$

and similarly for $\bar{a}_n$. Equations (3.64), (3.65) imply that the right-hand side of (3.61) has the desired properties of $|\alpha\rangle$. □

3.3.4 Compactification

So far φ took values in $\mathbb{R}$, and the momenta α formed a continuum. Let it instead take values in a circle,

$$\varphi \sim \varphi + 2\pi R. \tag{3.66}$$

This has two effects. First, the shift $\varphi_0 \mapsto \varphi_0 + 2\pi R$ is generated by $e^{2\pi i R \pi_0}$, which must act trivially, so $\pi_0 = n/R$ with $n \in \mathbb{Z}$. Second, the field may wind: $\varphi(x + L, t) = \varphi(x, t) + 2\pi R w$ with $w \in \mathbb{Z}$. The winding adds $2\pi R w\, x/L$ to the mode expansion, which in the coordinates $z = e^{2\pi(\tau - ix)/L}$ of (3.49) reads $\frac{i}{2} R w \,(\log z - \log \bar{z})$. The zero modes of $i\partial\varphi$ and $i\bar{\partial}\varphi$ are then no longer equal:

$$a_0 = \frac{\pi_0 - 2\pi g R w}{\sqrt{4\pi g}}, \qquad \bar{a}_0 = \frac{\pi_0 + 2\pi g R w}{\sqrt{4\pi g}}. \tag{3.67}$$

Since $L_0 = \frac{1}{2} a_0^2 + \sum_{n>0} a_{-n} a_n$, the primary of momentum n and winding w has

$$h_{n,w} = \frac{1}{8\pi g}\left(\frac{n}{R} - 2\pi g R w\right)^2, \qquad \bar{h}_{n,w} = \frac{1}{8\pi g}\left(\frac{n}{R} + 2\pi g R w\right)^2, \qquad h_{n,w} - \bar{h}_{n,w} = -nw \in \mathbb{Z}. \tag{3.68}$$

The spin is an integer, as Remark 1.2.2 requires; by Remark 3.1.4 the momentum of the state is $-\frac{2\pi}{L} n w$. For $w = 0$ these are the vertex operators $\mathcal{V}_{n/R}$ of (3.54). Fields with $w \neq 0$ are vertex operators of the dual field $\varphi(z) - \bar{\varphi}(\bar{z})$; we need only their weights.

Proposition 3.3.3 (T-duality of the spectrum)**.** *Under the map $R \mapsto R' := 1/(2\pi g R)$ one has $h_{n,w}(R') = h_{w,n}(R)$ and $\bar{h}_{n,w}(R') = \bar{h}_{w,n}(R)$. The spectra at R and R' therefore coincide, with momentum and winding exchanged.*

Proof. $n/R' = 2\pi g R n$ and $2\pi g R' w = w/R$; insert into (3.68). □

3.3.5 The partition function

For the compact boson of §3.3.4, the Fock space over each primary (n, w) contributes $q^{h_{n,w}}\bar{q}^{\bar{h}_{n,w}}$ times the oscillator factor $\prod_{k\geq 1}|1-q^k|^{-2}$, and $c = 1$. Hence

$$\mathcal{Z}(R) = \operatorname{Tr} q^{L_0-\frac{1}{24}}\bar{q}^{\bar{L}_0-\frac{1}{24}} = \frac{1}{|\eta(\tau)|^2}\sum_{n,w\in\mathbb{Z}} q^{h_{n,w}}\bar{q}^{\bar{h}_{n,w}}, \tag{3.69}$$

with η as in (3.25).

Proposition 3.3.4. *Write $\tau = \tau_1 + i\tau_2$. Then*

$$\mathcal{Z}(R) = \frac{R\sqrt{2\pi g}}{\sqrt{\tau_2}\,|\eta(\tau)|^2}\sum_{m,w\in\mathbb{Z}} \exp\Big[-\frac{2\pi^2 g R^2}{\tau_2}\,|m + w\tau|^2\Big]. \tag{3.70}$$

Consequently $\mathcal{Z}(R)$ is invariant under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$, and $\mathcal{Z}(R) = \mathcal{Z}(1/(2\pi g R))$.

Proof. By (3.68), $q^h\bar{q}^{\bar{h}} = \exp[2\pi i\tau_1(h-\bar{h}) - 2\pi\tau_2(h+\bar{h})]$, with $h+\bar{h} = (n^2/R^2 + 4\pi^2 g^2 R^2 w^2)/4\pi g$ and $h-\bar{h} = -nw$. For fixed w, the sum over n is $\sum_n f(n)$ with $f(x) = \exp[-\frac{\tau_2}{2gR^2}x^2 - 2\pi i\tau_1 wx]$, a Gaussian. Theorem A.0.4 gives

$$\sum_n f(n) = R\sqrt{\frac{2\pi g}{\tau_2}}\sum_m \exp\Big[-\frac{2\pi^2 g R^2}{\tau_2}(m + \tau_1 w)^2\Big], \tag{3.71}$$

and combining with the remaining factor $\exp(-2\pi^2 g R^2 \tau_2 w^2)$ gives (3.70).

Invariance: $\sqrt{\tau_2}|\eta|^2$ is invariant under both generators by Corollary 3.2.4, since $\tau_2 \mapsto \tau_2/|\tau|^2$ under S. In the lattice sum, $\tau \mapsto \tau+1$ is undone by $(m, w) \mapsto (m-w, w)$. Under $\tau \mapsto -1/\tau$ we have $|m - w/\tau|^2/(\tau_2/|\tau|^2) = |-w + m\tau|^2/\tau_2$, which is undone by $(m, w) \mapsto (w, -m)$. Finally, Poisson resummation in w instead of n gives (3.70) with R replaced by $1/(2\pi g R)$. Hence $\mathcal{Z}(R) = \mathcal{Z}(1/(2\pi g R))$, which is Proposition 3.3.3 at the level of the partition function. □

Remark 3.3.5. *The two forms of $\mathcal{Z}$ make different symmetries manifest. The Hamiltonian form (3.69) is a trace over states. It shows T-invariance (integer spin, Proposition 3.7.2) and T-duality (momentum and winding are exchanged). The form (3.70) is a sum over classical windings (m, w) around the two cycles of the torus. It is what a path integral gives directly, and it makes S-invariance manifest. Poisson resummation is the dictionary between the two. The compact boson is the non-rational companion of the minimal models: $c = 1$, infinitely many primaries, and a one-parameter family of modular invariants labelled by R, with R and $1/(2\pi g R)$ the same theory.*

3.4 The free fermion and the Ising model

The action, the propagator and the stress tensor of the Majorana fermion are in §2.2.4. This section puts the fermion on the cylinder.

3.4.1 Mode expansion and the two sectors

Quantizing on the cylinder requires a boundary condition around the spatial circle, and here the fermion differs from the boson. Under $z = e^w$ a field of weight $\frac{1}{2}$ transforms with a square root,

$$\psi_{\mathrm{cyl}}(w) = \left(\frac{\mathrm{d}z}{\mathrm{d}w}\right)^{1/2} \psi_{\mathrm{pl}}(z) = z^{1/2}\,\psi_{\mathrm{pl}}(z), \tag{3.72}$$

and the rotation $w \mapsto w + 2\pi i$ fixes z while reversing the sign of $z^{1/2}$. A field single-valued on the plane is therefore antiperiodic on the cylinder, and conversely.

Definition 3.4.1. *The two boundary conditions define the Neveu–Schwarz and Ramond sectors, with plane mode expansions*

$$NS: \quad \psi(z) = \sum_{r\in\mathbb{Z}+\frac{1}{2}} \psi_r\, z^{-r-\frac{1}{2}}, \qquad \psi \text{ single-valued at } 0, \tag{3.73}$$

$$R: \quad \psi(z) = \sum_{n\in\mathbb{Z}} \psi_n\, z^{-n-\frac{1}{2}}, \qquad \psi \text{ has a branch point at } 0. \tag{3.74}$$

The vacuum of each sector is annihilated by all modes of positive index.

Caution 3.4.2. *The names refer to the* cylinder*: the Neveu–Schwarz sector is antiperiodic there, hence single-valued on the plane. Both conventions occur in the literature; we always describe the plane behavior.*

Proposition 3.4.3. *In either sector,* $\{\psi_r, \psi_s\} = \delta_{r+s,0}$.

Proof. Inverting (3.73) and using the standard contour prescription for equal-time anticommutators in radial quantization,

$$\{\psi_r, \psi_s\} = \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{s-\frac{1}{2}} \oint_w \frac{\mathrm{d}z}{2\pi i}\, z^{r-\frac{1}{2}}\, \psi(z)\psi(w). \tag{3.75}$$

By (2.53) the inner integral is $w^{r-\frac{1}{2}}$, and the outer one is $\oint_0 \frac{\mathrm{d}w}{2\pi i} w^{r+s-1} = \delta_{r+s,0}$. The Ramond case is identical, the indices now being integers. □

Remark 3.4.4. *Geometrically, a primary of weight h is a section of $\Omega^{\otimes h}$, so $h = \frac{1}{2}$ requires a line bundle L with $L^{\otimes 2} \cong \Omega$ — a square root of the canonical bundle. Two such square roots differ by a line bundle whose square is trivial, and the Kummer sequence*

$$1 \longrightarrow \underline{\mathbb{Z}_2} \longrightarrow \mathcal{O}^* \xrightarrow{\;2\;} \mathcal{O}^* \longrightarrow 1 \tag{3.76}$$

identifies these with $H^1(\Sigma;\mathbb{Z}_2) \cong \mathrm{Hom}(H_1(\Sigma;\mathbb{Z}),\mathbb{Z}_2)$: a choice of sign for each cycle. On the sphere there is one such choice, which is why Definition 3.4.1 can present the two sectors as boundary conditions at a point rather than as global data. On the torus there are four, and Remark 3.4.24 shows that all of them are needed.

Expanding (2.56) in modes, with $\partial\psi(z) = \sum_s (-s - \frac{1}{2})\psi_s z^{-s-3/2}$,

$$T(z) = \frac{1}{2}\sum_{r,s} \left(s + \tfrac{1}{2}\right) : \psi_r \psi_s : \; z^{-r-s-2}, \tag{3.77}$$

so reading off the coefficient of z^{-n-2} and setting $s = n - r$,

$$L_n = \frac{1}{2}\sum_{r} \left(n - r + \tfrac{1}{2}\right) : \psi_r \psi_{n-r} : . \tag{3.78}$$

Relabelling $r \mapsto n - r$ and using the antisymmetry of $: \cdot \, :$ turns this into $L_n = -\frac{1}{2}\sum_r (r + \frac{1}{2}) : \psi_r \psi_{n-r} :$; averaging the two expressions gives the manifestly antisymmetric form

$$L_n = \frac{1}{4}\sum_{r} (n - 2r) \; : \psi_r \psi_{n-r} : , \qquad \text{in particular} \qquad L_0 = \sum_{r>0} r \, \psi_{-r}\psi_r, \tag{3.79}$$

the expression for L_0 following because the $r > 0$ and $r < 0$ halves of the sum contribute equally after normal ordering.

Proposition 3.4.5. *Under the state–operator correspondence, the Neveu–Schwarz field ψ corresponds to the state $\psi_{-1/2}|0\rangle$, which is primary of weight $h = \frac{1}{2}$.*

Proof. Acting with (3.73) on the vacuum and using $\psi_r|0\rangle = 0$ for $r > 0$,

$$\psi(z)|0\rangle = \sum_{r \le -\frac{1}{2}} \psi_r \, z^{-r-\frac{1}{2}}|0\rangle = \psi_{-1/2}|0\rangle + z\,\psi_{-3/2}|0\rangle + O(z^2), \tag{3.80}$$

which is regular at the origin with $\lim_{z\to 0} \psi(z)|0\rangle = \psi_{-1/2}|0\rangle$. By (3.79) and Proposition 3.4.3,

$$L_0 \, \psi_{-1/2}|0\rangle = \tfrac{1}{2}\, \psi_{-1/2}\psi_{1/2}\psi_{-1/2}|0\rangle = \tfrac{1}{2}\, \psi_{-1/2}|0\rangle. \tag{3.81}$$

Finally $L_0 = \sum_{r>0} r\,\psi_{-r}\psi_r$ is a sum of commuting number operators with non-negative eigenvalues, so no state of the Neveu–Schwarz Fock space has $h < 0$. Since L_n lowers h by n, the state $L_n\psi_{-1/2}|0\rangle$ would have weight $\frac{1}{2} - n < 0$ for $n \ge 1$ and therefore vanishes. Hence $\psi_{-1/2}|0\rangle$ is primary. □

Remark 3.4.6. *The regularity of $\psi(z)|0\rangle$ at the origin is not an extra input: it is equivalent to the condition $\psi_r|0\rangle = 0$ for $r > 0$ of Definition 3.4.1, and this is why the Neveu–Schwarz modes are indexed by half-integers — exactly so that $z^{-r-1/2}$ carries non-negative integer powers on the states that survive.*

Theorem 3.4.7. *The Virasoro algebra realized on the Neveu–Schwarz sector has $c = \frac{1}{2}$.*

Proof. Setting $n = -2$ in (3.79), $L_{-2} = -\frac{1}{2}\sum_r (1 + r) \; : \psi_r \psi_{-2-r} : .$ Acting on the vacuum only $r = -\frac{1}{2}$ and $r = -\frac{3}{2}$ have both indices negative, and the two terms coincide, so

$$L_{-2}|0\rangle = -\tfrac{1}{2}\, \psi_{-1/2}\psi_{-3/2}|0\rangle. \tag{3.82}$$

Its norm squared is $\frac{1}{4}$, since $\langle 0|\psi_{3/2}\psi_{1/2}\psi_{-1/2}\psi_{-3/2}|0\rangle = 1$ by Proposition 3.4.3. On the other hand the Virasoro algebra (1.27) gives $[L_2, L_{-2}] = 4L_0 + \frac{c}{2}$, so $\langle 0|L_2 L_{-2}|0\rangle = \frac{c}{2}$. Hence $c = \frac{1}{2}$. □

Remark 3.4.8. *The same number follows from the operator product expansion: a double Wick contraction of (2.56) with itself, using (2.53), produces the quartic pole of (2.27) with coefficient $\frac{1}{4} = c/2$. The mode computation above is that statement read at level two of the vacuum module.*

3.4.2 The Ramond ground state and the spin field

The Ramond sector differs from the Neveu–Schwarz sector in two ways: the mode indices are integers, and there is a zero mode. Both are visible already in the two-point function, and both are responsible for the shift of the ground state energy computed below.

Proposition 3.4.9. *In the Ramond sector, for $|z| > |w|$,*

$$\big\langle \psi(z)\psi(w) \big\rangle_R = \frac{1}{2}\,\frac{\sqrt{z/w} + \sqrt{w/z}}{z-w}. \tag{3.83}$$

Proof. By Proposition 3.4.3 the only nonvanishing expectation values are $\langle \mathrm{R}|\psi_n\psi_{-n}|\mathrm{R}\rangle = 1$ for $n > 0$ and $\psi_0^2 = \frac{1}{2}$. Inserting (3.74),

$$\begin{aligned} \big\langle \psi(z)\psi(w) \big\rangle_{\mathrm{R}} &= \sum_{n\geq 1} z^{-n-\frac{1}{2}} w^{n-\frac{1}{2}} + \frac{1}{2}\, z^{-\frac{1}{2}} w^{-\frac{1}{2}} \\ &= \frac{1}{\sqrt{zw}}\left[\frac{w/z}{1-w/z} + \frac{1}{2}\right] = \frac{1}{\sqrt{zw}} \cdot \frac{z+w}{2(z-w)}, \end{aligned} \tag{3.84}$$

and $\frac{z+w}{2\sqrt{zw}} = \frac{1}{2}(\sqrt{z/w} + \sqrt{w/z})$. □

Remark 3.4.10. *The equation $\bar\partial\psi = 0$ determines the propagator only up to a term holomorphic in both arguments; what selects a solution is the space in which it is required to live. On $\mathbb{C}$, single-valuedness and decay at infinity kill the holomorphic part and leave (2.53); on the double cover branched at the origin the same equation admits (3.83) instead. Both have the same singularity, as they must, since that is a local property. The difference between them is measured by $\langle T\rangle$, which vanishes in the Neveu–Schwarz vacuum and equals $1/16z^2$ in the Ramond sector, as Proposition 3.4.15 below records.*

We proceed to determine the weight of the Ramond ground state. The key point is that the normal ordering in (3.79) is vacuous away from $n = 0$.

Lemma 3.4.11. *For $n \neq 0$ the modes (3.79) are unambiguous: $:\psi_r\psi_{n-r}:$ and $\psi_r\psi_{n-r}$ differ by $\pm\{\psi_r, \psi_{n-r}\} = \pm\delta_{n,0}$.*

Theorem 3.4.12. *The Ramond ground state has $h = \frac{1}{16}$, and the Ramond sector realizes the Virasoro algebra with $c = \frac{1}{2}$.*

Proof. Throughout we use $\psi_n|\mathrm{R}\rangle = 0$ for $n > 0$, $\psi_0^2 = \frac{1}{2}$ (Proposition 3.4.3), and $\psi_n^\dagger = \psi_{-n}$; by Lemma 3.4.11 the modes $L_{\pm1}, L_{\pm2}$ may be used without normal ordering. Note first that for $k > 0$

$$\big\|\psi_{-k}\psi_0|\mathrm{R}\rangle\big\|^2 = \langle \mathrm{R}|\psi_0\,\psi_k\psi_{-k}\,\psi_0|\mathrm{R}\rangle = \langle \mathrm{R}|\psi_0^2|\mathrm{R}\rangle = \tfrac{1}{2}. \tag{3.85}$$

Weight. In $L_{-1} = \frac{1}{4}\sum_r (-1-2r)\,\psi_r\psi_{-1-r}$ a term survives on $|\mathrm{R}\rangle$ only if $r \leq 0$ and $-1-r \leq 0$, hence $r \in \{-1, 0\}$. The two terms have coefficients $\frac{1}{4}$ and $-\frac{1}{4}$, and $\psi_0\psi_{-1} = -\psi_{-1}\psi_0$, so they add:

$$L_{-1}|\mathrm{R}\rangle = \tfrac{1}{2}\,\psi_{-1}\psi_0|\mathrm{R}\rangle, \qquad \big\|L_{-1}|\mathrm{R}\rangle\big\|^2 = \tfrac{1}{4}\cdot\tfrac{1}{2} = \tfrac{1}{8} \tag{3.86}$$

by (3.85). The same index count gives $L_1|\mathrm{R}\rangle = 0$, so

$$\tfrac{1}{8} = \langle \mathrm{R}|L_1 L_{-1}|\mathrm{R}\rangle = \langle \mathrm{R}|[L_1, L_{-1}]|\mathrm{R}\rangle = 2h, \tag{3.87}$$

whence $h = \frac{1}{16}$.

Central charge. In $L_{-2} = \frac{1}{4}\sum_r (-2-2r)\,\psi_r \psi_{-2-r}$ the surviving indices are $r \in \{-2,-1,0\}$; the term $r=-1$ has coefficient zero, and the other two again add, giving

$$L_{-2}|\mathrm{R}\rangle = \psi_{-2}\psi_0|\mathrm{R}\rangle, \qquad \big\|L_{-2}|\mathrm{R}\rangle\big\|^2 = \tfrac{1}{2}. \tag{3.88}$$

Since $L_2|\mathrm{R}\rangle = 0$, the Virasoro relation $[L_2, L_{-2}] = 4L_0 + \frac{c}{2}$ gives $\frac{1}{2} = 4\cdot\frac{1}{16} + \frac{c}{2}$, so $c = \frac{1}{2}$. □

Remark 3.4.13. *The value of c agrees with Theorem 3.4.7. Nothing had guaranteed this: the two sectors are inequivalent representations built on different index sets, and the agreement traces back to Proposition 3.4.3 taking the same form in both.*

The same weight can be recovered from the propagator, at the cost of relating the two sectors to one another.

Lemma 3.4.14. *If $|h\rangle$ is a highest weight state of weight h, then $\langle h|T(z)|h\rangle = h/z^2$.*

Proof. Insert $T(z) = \sum_n L_n z^{-n-2}$. For $n > 0$ one has $L_n|h\rangle = 0$, and for $n < 0$ the adjoint relation $L_n^\dagger = L_{-n}$ gives $\langle h|L_n = \big(L_{-n}|h\rangle\big)^\dagger = 0$. Only $n = 0$ survives, with $L_0|h\rangle = h|h\rangle$. □

Proposition 3.4.15. $\big\langle T(z)\big\rangle_R = \dfrac{1}{16z^2}$, *in agreement with Theorem 3.4.12.*

Proof. The normal ordering in (2.56) is a point-splitting: one subtracts from $\psi(z)\partial\psi(w)$ the c-number singularity, which is local and hence the same in every sector, namely the Neveu–Schwarz propagator (2.53). Since $\langle\psi(z)\partial\psi(w)\rangle = \partial_w\langle\psi(z)\psi(w)\rangle$,

$$\big\langle T(z)\big\rangle_{\mathrm{R}} = -\frac{1}{2}\lim_{w\to z}\partial_w\left[\big\langle\psi(z)\psi(w)\big\rangle_{\mathrm{R}} - \frac{1}{z-w}\right], \tag{3.89}$$

the limit existing because the two propagators have the same singular part. Put $w = z(1+\varepsilon)$. Then $(1+\varepsilon)^{1/2} + (1+\varepsilon)^{-1/2} = 2 + \frac{1}{4}\varepsilon^2 + O(\varepsilon^3)$ and $z - w = -z\varepsilon$, so (3.83) gives

$$\big\langle\psi(z)\psi(w)\big\rangle_{\mathrm{R}} = -\frac{1}{z\varepsilon}\left(1 + \frac{\varepsilon^2}{8}\right) = \frac{1}{z-w} - \frac{w-z}{8z^2}. \tag{3.90}$$

The bracket is therefore $-\frac{w-z}{8z^2}$, whose derivative in w is $-\frac{1}{8z^2}$, giving $\langle T(z)\rangle_{\mathrm{R}} = \frac{1}{16z^2}$; comparison with Lemma 3.4.14 returns $h = \frac{1}{16}$. □

Remark 3.4.16. *Unlike Theorem 3.4.12, this argument is not internal to the Ramond sector: the subtraction that defines the normal ordering is fixed by the Neveu–Schwarz propagator. It is recorded because it is the only place where (3.83) is used, and because it exhibits what the normal ordering actually does.*

Definition 3.4.17. *The operator creating the Ramond ground state under the state–operator correspondence of §2.3 is the spin field $\sigma(z)$, of weight $\frac{1}{16}$. It is characterized by the branch cut it induces,*

$$\psi(z)\,\sigma(0) \sim z^{-1/2}\,\mu(0) + \cdots , \tag{3.91}$$

where the disorder field μ has the same weight.

Remark 3.4.18. *Unlike the Neveu–Schwarz sector, the Ramond sector has a mode of index zero. By (3.79) it commutes with L_0, so it maps ground states to ground states and the ground state cannot be unique unless it acts as a scalar. In the full theory there are two such modes, one from each chirality, with*

$$\psi_0^2 = \bar{\psi}_0^2 = \tfrac{1}{2}, \qquad \{\psi_0, \bar{\psi}_0\} = 0, \tag{3.92}$$

and the combination $a := \frac{1}{\sqrt{2}}(\psi_0 + i\bar{\psi}_0)$ satisfies $a^2 = 0$ and $\{a, a^\dagger\} = 1$. It is therefore a single fermionic oscillator, acting within the space of Ramond ground states and generating exactly two of them,

$$|\Omega\rangle, \qquad a^\dagger|\Omega\rangle, \tag{3.93}$$

both of weight $h = \bar{h} = \frac{1}{16}$ since a carries no L_0. These are the two fields σ and μ of Definition 3.4.17, which differ by one quantum of a.

3.4.3 Identification with the Ising model

Collecting Proposition 3.4.5 and Theorems 3.4.7 and 3.4.12,

$$c = \tfrac{1}{2}, \qquad \text{NS}: \; h = 0,\ \tfrac{1}{2}, \qquad \text{R}: \; h = \tfrac{1}{16}, \tag{3.94}$$

which is exactly the content of the Kac table of $\mathcal{M}(4,3)$, constructed independently from the Kac determinant in §2.4.4. Nothing in the present section uses that construction: the three weights were obtained from the mode algebra alone, and the identification below agrees with it. The identification is

$$\mathbb{1} \leftrightarrow |0\rangle, \qquad \varepsilon \leftrightarrow \psi = \psi_{-1/2}|0\rangle, \qquad \sigma \leftrightarrow \text{Ramond ground state}. \tag{3.95}$$

Slogan 3.4.19. *The three primaries of the Ising model are not three unrelated fields but the two sectors of a single free fermion: $\mathbb{1}$ and ε span the Neveu–Schwarz sector, and σ generates the Ramond sector. The Kac determinant of §2.3.3 finds them by representation theory; the free fermion finds them by boundary conditions.*

Remark 3.4.20. *The two constructions are not equivalent, and the difference is taken up below, when the traces over the sectors are summed (Remark 3.4.24). A Majorana fermion depends on a choice of square root of Ω, whereas the Ising model does not; passing from one to the other requires summing over that choice, which on the torus is the GSO projection. The fusion rule $\sigma \times \sigma = \mathbb{1} + \varepsilon$, derived independently in Example 2.4.19 below, already reflects it: two branch points brought together leave either no cut, or the field ψ itself.*

3.4.4 Ising characters from the fermion

The traces of $q^{L_0-c/24}$ over the three sectors are the characters of the Ising model. Computing them and their S-transformation needs only the mode algebra and the theta functions of §3.2. It uses neither the Kac determinant nor the resolution, on which the characters of general minimal models in §3.5 rest (§A.2). The result is compared with the general formula in Example 3.6.8 and used in Chapter 5.

Let $\mathcal{H}_{\mathrm{NS}}$ be the span of the monomials $\psi_{-r_1}\cdots\psi_{-r_k}|0\rangle$ with $r_1>\cdots>r_k>0$ in $\mathbb{Z}+\frac12$, and $\mathcal{H}_{\mathrm{R}}$ the span of $\psi_{-n_1}\cdots\psi_{-n_k}|\mathrm{R}\rangle$ with $n_1>\cdots>n_k\geq 1$, over a single Ramond ground state on which ψ_0 acts as a scalar (Remark 3.4.18). The monomials are orthonormal for $\psi_r^\dagger=\psi_{-r}$, and by Proposition 3.4.3 each space is preserved by all modes. Fermion parity $(-1)^F$, the parity of k, splits $\mathcal{H}_{\mathrm{NS}}=\mathcal{H}_{\mathrm{NS}}^+\oplus\mathcal{H}_{\mathrm{NS}}^-$. The modes (3.79) are bilinear in ψ, so they preserve this splitting. Following the identification $\mathbb{1}\leftrightarrow|0\rangle$, $\varepsilon\leftrightarrow\psi_{-1/2}|0\rangle$, $\sigma\leftrightarrow|\mathrm{R}\rangle$, we take the three chiral sectors of the Ising model to be $\mathcal{H}_{\mathrm{NS}}^+$, $\mathcal{H}_{\mathrm{NS}}^-$ and $\mathcal{H}_{\mathrm{R}}$, and write

$$\chi_0:=\operatorname{Tr}_{\mathcal{H}_{\mathrm{NS}}^+}q^{L_0-\frac{1}{48}},\qquad \chi_{\frac12}:=\operatorname{Tr}_{\mathcal{H}_{\mathrm{NS}}^-}q^{L_0-\frac{1}{48}},\qquad \chi_{\frac{1}{16}}:=\operatorname{Tr}_{\mathcal{H}_{\mathrm{R}}}q^{L_0-\frac{1}{48}},\tag{3.96}$$

with $c/24=1/48$ by Theorem 3.4.7.

Lemma 3.4.21 (sector traces)**.**

$$\chi_0\pm\chi_{\frac12}=q^{-\frac{1}{48}}\prod_{m\geq1}\bigl(1\pm q^{m-\frac12}\bigr),\qquad \chi_{\frac{1}{16}}=q^{\frac{1}{24}}\prod_{n\geq1}\bigl(1+q^n\bigr).\tag{3.97}$$

Proof. From (3.79) and Proposition 3.4.3, $[L_0,\psi_r]=-r\psi_r$ in either sector; the normal-ordering constant drops out of the commutator. Hence a monomial has L_0 equal to the weight of its ground state plus $r_1+\cdots+r_k$. The ground state weights are 0 and $\frac{1}{16}$ (Theorem 3.4.12). Summing over the finite sets $\{r_i\}$ gives $\operatorname{Tr}_{\mathcal{H}_{\mathrm{NS}}}q^{L_0}=\prod_m(1+q^{m-1/2})$ and $\operatorname{Tr}_{\mathcal{H}_{\mathrm{NS}}}(-1)^Fq^{L_0}=\prod_m(1-q^{m-1/2})$, whose half sum and half difference are the traces over $\mathcal{H}_{\mathrm{NS}}^\pm$. Likewise $\operatorname{Tr}_{\mathcal{H}_{\mathrm{R}}}q^{L_0}=q^{1/16}\prod_n(1+q^n)$, and $\frac{1}{16}-\frac{1}{48}=\frac{1}{24}$. □

Comparing (3.97) with (3.30) and $\eta=q^{1/24}\prod_m(1-q^m)$,

$$\bigl(\chi_0+\chi_{\frac12}\bigr)^2=\frac{\theta_3}{\eta},\qquad \bigl(\chi_0-\chi_{\frac12}\bigr)^2=\frac{\theta_4}{\eta},\qquad 2\chi_{\frac{1}{16}}^2=\frac{\theta_2}{\eta}.\tag{3.98}$$

The left sides are squares of the products (3.97), which are holomorphic and nowhere zero on $\mathbb{H}$.

Proposition 3.4.22 (Ising S-matrix from the fermion)**.** *Order the sectors as $(\mathbb{1},\sigma,\varepsilon)$, that is, as $(\chi_0,\chi_{\frac{1}{16}},\chi_{\frac12})$. Then*

$$\chi_i(-1/\tau)=\sum_j S_{ij}\chi_j(\tau),\qquad S=\begin{pmatrix}\frac12&\frac{1}{\sqrt2}&\frac12\\ \frac{1}{\sqrt2}&0&-\frac{1}{\sqrt2}\\ \frac12&-\frac{1}{\sqrt2}&\frac12\end{pmatrix}.\tag{3.99}$$

Proof. By (3.31) and Corollary 3.2.4, the factors $\sqrt{-i\tau}$ cancel, and under $\tau \mapsto -1/\tau$

$$\frac{\theta_3}{\eta} \mapsto \frac{\theta_3}{\eta}, \qquad \frac{\theta_4}{\eta} \mapsto \frac{\theta_2}{\eta}, \qquad \frac{\theta_2}{\eta} \mapsto \frac{\theta_4}{\eta}. \tag{3.100}$$

By (3.98),

$$\begin{aligned} (\chi_0 + \chi_{\frac{1}{2}})(-1/\tau) &= \epsilon_1\,(\chi_0 + \chi_{\frac{1}{2}})(\tau), \\ (\chi_0 - \chi_{\frac{1}{2}})(-1/\tau) &= \epsilon_2\sqrt{2}\,\chi_{\frac{1}{16}}(\tau), \\ \sqrt{2}\,\chi_{\frac{1}{16}}(-1/\tau) &= \epsilon_3\,(\chi_0 - \chi_{\frac{1}{2}})(\tau), \end{aligned} \tag{3.101}$$

with each $\epsilon_k(\tau) \in \{\pm 1\}$. Each ϵ_k is a quotient of two holomorphic functions without zeros, hence continuous; since $\mathbb{H}$ is connected it is constant. At the fixed point $\tau = i$ we have $q = e^{-2\pi} \in (0,1)$, every product in (3.97) is positive, and so $\epsilon_k = +1$. Solving the three relations for χ_0, $\chi_{\frac{1}{16}}$, $\chi_{\frac{1}{2}}$ at $-1/\tau$ gives the matrix stated. □

Remark 3.4.23 (what was used)**.** *Proposition 3.4.22 uses the mode algebra of the free fermion, Theorem 3.4.12, Lemma A.0.13, Theorem 3.2.2 and Corollary 3.2.4. It does not use the Kac determinant or the resolution* (3.113)*. The matrix coincides with* (3.144)*, which Example 3.6.8 obtains from the general formula.*

The Ising fusion rules also come without the Kac determinant: they follow from the level-two null vectors of $\sigma = \phi_{1,2}$ and $\varepsilon = \phi_{2,1}$ (Proposition 2.3.13, Theorem 2.4.17 and its counterpart under $t \mapsto 1/t$), and give $\sigma \times \sigma = \mathbb{1} + \varepsilon$, $\sigma \times \varepsilon = \sigma$, $\varepsilon \times \varepsilon = \mathbb{1}$. The fusion matrices

$$N_\sigma = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad N_\varepsilon = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \tag{3.102}$$

are diagonalized by the columns of S, with eigenvalues $S_{\sigma m}/S_{0m} = (\sqrt{2}, 0, -\sqrt{2})$ and $S_{\varepsilon m}/S_{0m} = (1, -1, 1)$. Thus the diagonalization hypothesis (3.148)*, which §3.6.2 assumes in general, holds for the Ising model by direct check.*

That the three sector traces equal the characters $\chi_{1,1}$, $\chi_{1,2}$, $\chi_{2,1}$ of Proposition 3.5.7 for $\mathcal{M}(4,3)$ (equivalently, that each sector is an irreducible Virasoro module) is consistent with this, and has been checked as q-series through order q^{58}. It is not used.

Remark 3.4.24 (sum over boundary conditions)**.** *The Remark after the identification with the Ising model and Remark 3.4.4 said that passing from the fermion to the Ising model requires all four boundary conditions on the torus. In terms of the full theory, by* (3.98)*,*

$$\mathrm{Tr}_{\mathrm{NS}}\, q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}} = \left|\frac{\theta_3}{\eta}\right|, \qquad \mathrm{Tr}_{\mathrm{NS}} (-1)^F q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}} = \left|\frac{\theta_4}{\eta}\right|, \tag{3.103}$$

$$\mathrm{Tr}_{\mathrm{R}}\, q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}} = \left|\frac{\theta_2}{\eta}\right|, \qquad \mathrm{Tr}_{\mathrm{R}} (-1)^F q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}} = 0. \tag{3.104}$$

Here F counts both chiralities, and the Ramond sector of the full theory has the two ground states of Remark 3.4.18, of opposite parity; this gives the factor 2 in the third trace and the

zero in the fourth. Keeping the states of even F in each sector,

$$\mathcal{Z}_{Ising} = |\chi_0|^2 + |\chi_{\frac{1}{2}}|^2 + |\chi_{\frac{1}{16}}|^2 = \frac{1}{2}\left(\left|\frac{\theta_3}{\eta}\right| + \left|\frac{\theta_4}{\eta}\right| + \left|\frac{\theta_2}{\eta}\right|\right). \tag{3.105}$$

This is the projection called GSO in that Remark. By (3.31), (3.32) and Corollary 3.2.4, both S and T permute the three terms, which proves the modular invariance of $\mathcal{Z}_{Ising}$ directly. No single term is invariant under both S and T: the fermion with fixed boundary conditions is not a modular invariant theory on its own.

3.5 Characters of minimal models

3.5.1 Virasoro characters

The character of a module is the generating function for the degeneracy of its states.

$$\chi_j(\tau) := \mathrm{Tr}_j\, q^{L_0 - \frac{c}{24}} = q^{h_j - \frac{c}{24}} \sum_{n=0}^{\infty} a(n) q^n, \qquad q = e^{2\pi i \tau} \tag{3.106}$$

Here τ is the modulus of §3.1.2. Since $\dim V^{(n)}(c,h) = P(n)$ (§2.3.2), the character of a Verma module is

$$\begin{aligned}
\chi_{c,h}(\tau) &= q^{h-\frac{c}{24}} \sum_{n=0}^{\infty} P(n) q^n \\
&= q^{h-\frac{c}{24}} (1 + q + q^2 + \cdots)(1 + q^2 + q^4 + \cdots)(1 + q^3 + q^6 + \cdots)\cdots \\
&= q^{h-\frac{c}{24}} \prod_{n=1}^{\infty} \frac{1}{1-q^n}.
\end{aligned} \tag{3.107}$$

3.5.2 Resolution of $M_{r,s}$

Before writing down the resolution let us explain where it comes from.

For generic (c,h) the Verma module $V(c,h)$ is already irreducible and there is nothing to do. If (c,h) is a root of the Kac determinant with multiplicity one, then V contains a single singular vector, generating a proper submodule $U \subset V$ which is itself a Verma module, and the irreducible quotient is computed by the short exact sequence

$$0 \longrightarrow U \longrightarrow V \longrightarrow M \longrightarrow 0. \tag{3.108}$$

What makes the minimal models different is that the resonance condition $p\alpha_- + p'\alpha_+ = 0$ makes *two* entries of the Kac determinant vanish simultaneously.

Observation 3.5.1. *For $\mathcal{M}(p,p')$ with $1 \le r \le p'-1$, $1 \le s \le p-1$, the maximal proper submodule of the Verma module $V_{r,s} := V(c, h_{r,s})$ is generated by two singular vectors, at levels*

$$rs \qquad \text{and} \qquad (p'-r)(p-s). \tag{3.109}$$

Indeed, by (2.138) one has $h_{r,-s} = h_{r,s} + rs$ and $h_{r,2p-s} = h_{r,s} + (p'-r)(p-s)$, and both differences are positive integers.

Caution 3.5.2. *$V_{r,s}$ contains infinitely many singular vectors, since a singular vector of a submodule is again one of $V_{r,s}$. The content of Observation 3.5.1 is that two of them* generate *the maximal proper submodule; the rest are accounted for by the resolution below.*

Write $U_A, U_B \subset V_{r,s}$ for the submodules generated by these two singular vectors. Each is isomorphic to a Verma module, and the maximal proper submodule of $V_{r,s}$ is their *sum*, so that

$$M_{r,s} = V_{r,s}/(U_A + U_B). \tag{3.110}$$

Here lies the difficulty. The sum $U_A + U_B$ is not direct: the two submodules overlap, and a naive count of characters would subtract the overlap twice. Inclusion–exclusion gives

$$\operatorname{ch}(U_A + U_B) = \operatorname{ch} U_A + \operatorname{ch} U_B - \operatorname{ch}(U_A \cap U_B), \tag{3.111}$$

and the alternating sign appearing here is precisely the sign $(-1)^k$ of the complex below. But $U_A \cap U_B$ is again a module of the same type, itself generated by two singular vectors whose submodules again overlap, so (3.111) must be applied once more — and the process never terminates. The resolution is nothing but the bookkeeping device for this infinite inclusion–exclusion.

This is visible already at the level of the maps. Writing ι_A, ι_B for the two embeddings and $f_1(x,y) = \iota_A(x) + \iota_B(y)$, we have $\operatorname{im} f_1 = U_A + U_B$, which is (3.110), while

$$\ker f_1 = \{\, (x,y) \,|\, \iota_A(x) = -\iota_B(y) \,\} \;\simeq\; U_A \cap U_B. \tag{3.112}$$

The kernel of one map *is* the overlap that the next step must account for. Covering $U_A \cap U_B$ by two Verma modules produces f_2, whose kernel is the next overlap, and so on. The resulting complex is

$$\cdots \begin{matrix} V_{kp'+r,(-1)^k s+[1-(-1)^k]p/2} \\ \oplus \\ V_{r,kp+(-1)^k s+[1-(-1)^k]p/2} \end{matrix} \xrightarrow{f_k} \cdots \xrightarrow{f_3} \begin{matrix} V_{2p'+r,s} \\ \oplus \\ V_{r,2p+s} \end{matrix} \xrightarrow{f_2} \begin{matrix} V_{p'+r,p-s} \\ \oplus \\ V_{r,2p-s} \end{matrix} \xrightarrow{f_1} V_{r,s} \tag{3.113}$$

where $f_1(x,y) = x + y$ and $f_k(x,y) = (x+y, -x-y)$ for $k \geq 2$, each entry of these matrices denoting the unique (up to normalization) embedding of Verma modules supplied by Observation 3.5.1.

Remark 3.5.3. *That (3.113) is a complex, $f_k f_{k+1} = 0$, is immediate from the stated form of the f_k. That it is* exact — *i.e.* $\ker f_k = \operatorname{im} f_{k+1}$ *for $k \geq 1$, and* $\operatorname{coker} f_1 = M_{r,s}$ — *is the theorem of Feigin and Fuks, and is what licenses the character computation below: only for an exact complex does the alternating sum of characters of the terms equal the character of $M_{r,s}$.*

The Virasoro character is given by

$$\begin{aligned} \chi_{r,s}(q) &= \sum_{k=0}^{\infty} \operatorname{Tr}(-1)^k q^{L_0 - \frac{c}{24}}\big|_{M_{r,s}^{(k)}} \\ &= \frac{q^{-c/24}}{\varphi(q)} \left[q^{h_{r,s}} + \sum_{k=1}^{\infty} (-1)^k \left\{ q^{h_{r+kp',(-1)^k s+[1-(-1)^k]p/2}} + q^{h_{r,kp+(-1)^k s+[1-(-1)^k]p/2}} \right\} \right] \end{aligned} \tag{3.114}$$

where $M_{r,s}^{(k)}$ is the degree k piece in (3.113).

Example 3.5.4 (Vacuum module of the Ising model). *For $\mathcal{M}(4,3)$ and $(r,s)=(1,1)$, Observation 3.5.1 gives singular vectors at levels $rs=1$ and $(p'-r)(p-s)=2\cdot 3=6$; the first is $L_{-1}|0\rangle$, which vanishes in any vacuum module. The complex begins*

$$\cdots \longrightarrow V_{h=13}\oplus V_{h=11} \longrightarrow V_{h=1}\oplus V_{h=6} \longrightarrow V_{h=0}. \tag{3.115}$$

The degree-2 term is the first overlap $U_A\cap U_B$ of (3.112): inside U_A (a Verma module of weight 1) its generators sit at levels 12 and 10, and inside U_B (weight 6) at levels 7 and 5, meeting at the absolute heights 13 and 11. Taking the Euler characteristic,

$$q^{1/48}\chi_{1,1}(q)=\frac{1-q-q^6+q^{11}+q^{13}-\cdots}{\varphi(q)}=1+q^2+q^3+2q^4+2q^5+3q^6+\cdots \tag{3.116}$$

The vanishing coefficient at level 1 records the null vector $L_{-1}|0\rangle$; at level 6 the four states $L_{-6}, L_{-4}L_{-2}, L_{-3}^2, L_{-2}^3$ acting on $|0\rangle$ are reduced to three.

3.5.3 Theta functions attached to the Kac table

Throughout this section we abbreviate

$$N:=pp',\qquad \lambda_{r,s}:=pr-p's,\qquad \lambda_{r,-s}:=pr+p's. \tag{3.117}$$

The point of this notation is the following consequence of Eq. (2.138): the conformal weight depends on the Kac labels *only* through λ, and moreover the combination $h-c/24$ takes a particularly simple form.

Lemma 3.5.5. *With $\lambda=pr-p's$ one has*

$$h_{r,s}-\frac{c}{24}=\frac{\lambda^2}{4N}-\frac{1}{24}. \tag{3.118}$$

Proof. Substituting (2.138),

$$\begin{aligned} h_{r,s}-\frac{c}{24} &= \frac{\lambda^2-(p-p')^2}{4pp'}-\frac{1}{24}\left[1-6\frac{(p-p')^2}{pp'}\right] \\ &= \frac{\lambda^2}{4N}-\frac{(p-p')^2}{4N}-\frac{1}{24}+\frac{(p-p')^2}{4N}, \end{aligned} \tag{3.119}$$

and the two $(p-p')^2$ terms cancel. □

By Lemma 3.5.5, $h_{r,s}-c/24+1/24=\lambda^2/4N$ is the leading exponent of the theta function Θ_λ of level $N=pp'$, (3.16).

3.5.4 Resummation of the resolution

We now rewrite the character formula obtained from the resolution (3.113). Recall that the degree-k term of the resolution is the direct sum of the two Verma modules

$$\text{(A)}\quad V_{kp'+r,\,(-1)^k s+\frac{1-(-1)^k}{2}p},\qquad \text{(B)}\quad V_{r,\,kp+(-1)^k s+\frac{1-(-1)^k}{2}p}, \tag{3.120}$$

for $k\geq 1$, while the degree-0 term is the single module $V_{r,s}$. Separating the two parities of k, the Kac labels read

k	(A)	(B)
$2m$	$(2mp'+r,\ s)$	$(r,\ 2mp+s)$
$2m+1$	$\big((2m{+}1)p'+r,\ p-s\big)$	$\big(r,\ (2m{+}2)p-s\big)$

Lemma 3.5.6. *Let λ be evaluated on the labels of Table 3.5.4. Then*

$$\text{(A),}\ k=2m: \quad \lambda=\lambda_{r,s}+2mN, \qquad \text{(B),}\ k=2m: \quad \lambda=\lambda_{r,s}-2mN, \tag{3.121a}$$

$$\text{(A),}\ k=2m{+}1: \quad \lambda=\lambda_{r,-s}+2mN, \qquad \text{(B),}\ k=2m{+}1: \quad \lambda=\lambda_{r,-s}-2(m{+}1)N. \tag{3.121b}$$

Consequently the even-degree terms of (3.113) realize each element of $\lambda_{r,s}+2N\mathbb{Z}$ exactly once, and the odd-degree terms realize each element of $\lambda_{r,-s}+2N\mathbb{Z}$ exactly once.

Proof. Direct substitution into (3.117), using $pp'=N$:

$$\begin{aligned}
&\text{(A),}\ k=2m: && p(2mp'+r)-p's = 2mN+(pr-p's),\\
&\text{(B),}\ k=2m: && pr-p'(2mp+s) = -2mN+(pr-p's),\\
&\text{(A),}\ k=2m{+}1: && p\big((2m{+}1)p'+r\big)-p'(p-s) = (2m{+}1)N+pr-N+p's,\\
&\text{(B),}\ k=2m{+}1: && pr-p'\big((2m{+}2)p-s\big) = pr+p's-(2m{+}2)N.
\end{aligned} \tag{3.122}$$

For the last statement: at $k=0$ the two families (A) and (B) would both return the label (r,s), and the resolution indeed has a single module in degree 0; for $m\geq 1$ the pair $\{+2mN,-2mN\}$ is covered by (A) and (B) respectively. In odd degree, (A) covers $+2mN$ for $m\geq 0$ and (B) covers $-2nN$ for $n\geq 1$. □

Proposition 3.5.7. *The character of the irreducible Virasoro module $M_{r,s}$ of the minimal model $\mathcal{M}(p,p')$ is*

$$\chi_{r,s}(\tau)=\frac{\Theta_{\lambda_{r,s}}(\tau)-\Theta_{\lambda_{r,-s}}(\tau)}{\eta(\tau)}, \qquad \eta(\tau)=q^{1/24}\varphi(q)=q^{1/24}\prod_{n\geq 1}(1-q^n). \tag{3.123}$$

Proof. By exactness of (3.113) the character is the Euler characteristic of the complex,

$$\chi_{r,s}=\sum_{k\geq 0}(-1)^k\ \mathrm{Tr}_{M_{r,s}^{(k)}}\, q^{L_0-\frac{c}{24}}. \tag{3.124}$$

Each summand is a Verma module, whose character is $q^{h-c/24}/\varphi(q)$, so the factor $1/\varphi(q)$ may be pulled out:

$$\chi_{r,s}=\frac{1}{\varphi(q)}\sum_{k\geq 0}(-1)^k\sum_{h\in S_k}q^{h-\frac{c}{24}}, \tag{3.125}$$

where S_k denotes the set of highest weights occurring in degree k. Lemma 3.5.5 converts the exponent to $\lambda^2/4N-1/24$, and Lemma 3.5.6 identifies the even-k sum with $\Theta_{\lambda_{r,s}}$ and the odd-k sum with $\Theta_{\lambda_{r,-s}}$, the sign $(-1)^k$ supplying the relative minus. Hence

$$\chi_{r,s}=\frac{q^{-1/24}}{\varphi(q)}\left[\Theta_{\lambda_{r,s}}-\Theta_{\lambda_{r,-s}}\right], \tag{3.126}$$

and $q^{-1/24}/\varphi(q)=1/\eta(\tau)$. □

Remark 3.5.8. *Both Θ_λ and η are modular forms of weight $\frac{1}{2}$. Their quotient (3.123) therefore has weight 0, which is what makes the characters close on themselves under $\mathrm{PSL}(2,\mathbb{Z})$; this is the content of the next subsection.*

Corollary 3.5.9. $\chi_{p'-r,\,p-s} = \chi_{r,s}$.

Proof. $\lambda_{p'-r,p-s} = p(p'-r)-p'(p-s) = -\lambda_{r,s}$ and $\lambda_{p'-r,-(p-s)} = p(p'-r)+p'(p-s) = 2N-\lambda_{r,-s}$. Now apply (3.17). □

This is the reflection symmetry of the Kac table already noted below (2.138); it will be responsible for the overall factor 2 in the modular S-matrix, since each primary is counted twice by the naive sum over $\lambda \bmod 2N$.

Example 3.5.10 (Ising model)**.** *For $\mathcal{M}(4,3)$ we have $N = 12$, $2N = 24$ and $\lambda_{r,s} = 4r - 3s$. The three primaries have*

(r,s)	$\lambda_{r,s}$	$\lambda_{r,-s}$	h
$(1,1) = \mathbb{1}$	1	7	0
$(1,2) = \sigma$	-2	10	1/16
$(1,3) = \varepsilon$	-5	13	1/2

$$\chi_{\mathbb{1}} = \frac{\Theta_1 - \Theta_7}{\eta}, \qquad \chi_\sigma = \frac{\Theta_2 - \Theta_{10}}{\eta}, \qquad \chi_\varepsilon = \frac{\Theta_5 - \Theta_{11}}{\eta}. \tag{3.127}$$

To verify Lemma 3.5.6 we evaluate λ on the Kac labels supplied by (3.113) for $M_{1,1}$, and compare with the closed formulae of the lemma:

k	*(A)*	λ	*(B)*	λ
0	$(1,1)$	1	—	—
1	$(4,3)$	7	$(1,7)$	$-17 = 7 - 24$
2	$(7,1)$	$25 = 1 + 24$	$(1,9)$	$-23 = 1 - 24$
3	$(10,3)$	$31 = 7 + 24$	$(1,15)$	$-41 = 7 - 48$

The even-degree entries exhaust $1+24\mathbb{Z}$ and the odd-degree entries exhaust $7+24\mathbb{Z}$, each value occurring exactly once; note that degree 0 carries a single module, which is what prevents $\lambda = 1$ from being counted twice.

3.5.5 Behavior under the modular group

We can now state the transformation law of the characters.

Proposition 3.5.11. *The minimal model characters are modular functions of weight 0:*

$$\chi_{r,s}(-1/\tau) = \frac{1}{\sqrt{2N}} \sum_{\lambda' \bmod 2N} \left(e^{i\pi\lambda_{r,s}\lambda'/N} - e^{i\pi\lambda_{r,-s}\lambda'/N} \right) \frac{\Theta_{\lambda'}(\tau)}{\eta(\tau)}, \tag{3.128a}$$

$$\chi_{r,s}(\tau+1) = e^{2\pi i\left(h_{r,s} - \frac{c}{24}\right)} \chi_{r,s}(\tau). \tag{3.128b}$$

Proof. For (3.128a), insert (3.20) and Corollary 3.2.4 into (3.123): the factors $\sqrt{-i\tau}$ in numerator and denominator cancel exactly, the two weight-$\frac{1}{2}$ objects compensating one another. For (3.128b), note from (3.117) that $\lambda_{r,-s}^2 - \lambda_{r,s}^2 = 4Nrs$, so by (3.18) the two theta functions in (3.123) acquire the *same* phase $e^{2\pi i\lambda_{r,s}^2/4N}$; dividing by the phase $e^{i\pi/12}$ of η and using Lemma 3.5.5 gives the result. □

Remark 3.5.12. *In the proof of Proposition 3.2.1, the exponents E_n are not integers in general, so $e^{2\pi i\tau E_n}$ must be read as a function of τ rather than of $q = e^{2\pi i\tau}$; it is precisely this multivaluedness in q that makes T act nontrivially. The integrality of $E_n - E_0$ has a representation-theoretic origin: by (3.16) and (3.123) each term of Θ_λ/η is the character of one Verma module of the resolution (3.113), and $E_n - E_0$ is the level of the singular vector generating it. Since all these modules are submodules of $V_{r,s}$, their highest weight vectors are descendants and therefore sit at integer level. Consequently the whole conformal family transforms under T with the single phase $e^{2\pi i(h-c/24)}$, which is what makes T diagonal in (3.128b) and Proposition 3.7.2 possible.*

Remark 3.5.13. *Equation (3.128b) shows that T acts diagonally on the characters, with eigenvalues fixed by the conformal weights; all the content of modular invariance therefore resides in S. The right-hand side of (3.128a) is not yet expressed in the basis $\{\chi_{r',s'}\}$: doing so requires reorganizing the sum over $\lambda' \bmod 2N$ into the Kac table, which is carried out in §3.6.1.*

3.6 The S-matrix and the Verlinde formula

3.6.1 The modular S-matrix of minimal models

Proposition 3.5.11 expresses $\chi_{r,s}(-1/\tau)$ in terms of the individual theta functions $\Theta_{\lambda'}$, whereas a character is a *difference* of two of them. The task of this subsection is purely one of reorganisation: no analysis is involved, only trigonometry and the arithmetic of the Kac table. We write

$$\mathcal{K} := \left\{ (r,s) \in \mathbb{Z}^2 \,\middle|\, 1 \le r \le p'-1,\ 1 \le s \le p-1 \right\} \tag{3.129}$$

for the Kac rectangle, on which the involution $\iota(r,s) := (p'-r,\, p-s)$ acts freely; by Corollary 3.5.9 the distinct primaries are indexed by $\mathcal{K}/\iota$, of which there are $\frac{1}{2}(p-1)(p'-1)$. Fix once and for all a set $\mathcal{E} \subset \mathcal{K}$ of representatives.

Step A: symmetrization

Lemma 3.6.1.

$$\chi_{r,s}(-1/\tau) = \frac{2}{\sqrt{2N}} \sum_{\lambda' \bmod 2N} \sin\frac{\pi r\lambda'}{p'} \sin\frac{\pi s\lambda'}{p} \frac{\Theta_{\lambda'}(\tau)}{\eta(\tau)}. \tag{3.130}$$

Proof. Denote by $c_{\lambda'}$ the bracket in (3.128a). Reindexing the full sum by $\lambda' \mapsto -\lambda'$ and using $\Theta_{-\lambda'} = \Theta_{\lambda'}$ from (3.17) shows that $\sum_{\lambda'} c_{\lambda'}\Theta_{\lambda'} = \sum_{\lambda'} c_{-\lambda'}\Theta_{\lambda'}$; hence the sum equals its own

average,

$$\sum_{\lambda'} c_{\lambda'}\Theta_{\lambda'} = \frac{1}{2}\sum_{\lambda'} (c_{\lambda'} + c_{-\lambda'})\,\Theta_{\lambda'}. \tag{3.131}$$

The symmetrized coefficient is a difference of cosines,

$$c_{\lambda'} + c_{-\lambda'} = 2\cos\frac{\pi\lambda_{r,s}\lambda'}{N} - 2\cos\frac{\pi\lambda_{r,-s}\lambda'}{N}, \tag{3.132}$$

and $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$ applies with

$$\frac{A+B}{2} = \frac{\pi\lambda'(\lambda_{r,s}+\lambda_{r,-s})}{2N} = \frac{\pi\lambda'\cdot 2pr}{2pp'} = \frac{\pi r\lambda'}{p'}, \qquad \frac{A-B}{2} = \frac{\pi\lambda'(-2p's)}{2pp'} = -\frac{\pi s\lambda'}{p}, \tag{3.133}$$

giving $c_{\lambda'} + c_{-\lambda'} = 4\sin\frac{\pi r\lambda'}{p'}\sin\frac{\pi s\lambda'}{p}$. Combining the factors $\frac{1}{2}\cdot 4 = 2$ yields (3.130). □

Remark 3.6.2. *The two sines of the final answer are already visible here. Note also that the coefficient in (3.130) vanishes whenever $p \mid \lambda'$ or $p' \mid \lambda'$, which restricts the sum to $2(p-1)(p'-1)$ of the $2pp'$ residues; this is the content of Step B.*

Step B: parametrization by the Kac table

Lemma 3.6.3. *The map*

$$\mathcal{K}\times\{+,-\} \longrightarrow \mathbb{Z}/2N\mathbb{Z}, \qquad \big((r',s'),\pm\big) \longmapsto \lambda_{r',\pm s'} = pr' \mp p's' \tag{3.134}$$

is a bijection onto $\{\,\lambda' \bmod 2N \mid p\nmid\lambda' \text{ and } p'\nmid\lambda'\,\}$.

Proof. Counting. The residues divisible by p number $2p'$, those divisible by p' number $2p$, and those divisible by both—equivalently by pp', since $\gcd(p,p') = 1$—number 2. By inclusion–exclusion the target has $2pp' - 2p' - 2p + 2 = 2(p-1)(p'-1)$ elements, which is $|\mathcal{K}\times\{\pm\}|$.

The image lies in the target. If $p' \mid pr' \mp p's'$ then $p' \mid pr'$, and $\gcd(p,p') = 1$ forces $p' \mid r'$, impossible for $1 \le r' \le p'-1$; similarly for p.

Injectivity. Suppose $pr'_1 \mp p's'_1 \equiv pr'_2 \mp p's'_2 \pmod{2pp'}$. Reducing mod p' gives $pr'_1 \equiv pr'_2$, hence $r'_1 \equiv r'_2 \pmod{p'}$ and so $r'_1 = r'_2 =: r'$. Reducing mod p gives $\mp s'_1 \equiv \mp s'_2 \pmod{p}$. If the two signs agree this forces $s'_1 = s'_2$. If they differ, then $s'_1 + s'_2 = p$, and the two images differ by

$$(pr' - p's'_1) - (pr' + p's'_2) = -p'(s'_1 + s'_2) = -pp' \not\equiv 0 \pmod{2pp'}, \tag{3.135}$$

a contradiction. Injectivity plus equality of cardinalities gives the bijection. □

Lemma 3.6.3 says precisely that the surviving theta functions in (3.130) are exactly those occurring in the characters $\chi_{r',s'}$, $(r',s') \in \mathcal{K}$, each exactly once. It remains to check that the two members of each pair enter with opposite coefficients, so that they assemble into a character rather than into some other combination.

Step C: signs and the double count

Lemma 3.6.4. *Put* $g(\lambda') := \sin\frac{\pi r\lambda'}{p'}\sin\frac{\pi s\lambda'}{p}$ *and*

$$G^{(r',s')}_{(r,s)} := (-1)^{1+rs'+r's}\sin\frac{\pi p\, rr'}{p'}\sin\frac{\pi p'\, ss'}{p}. \tag{3.136}$$

Then for $(r',s')\in\mathcal{K}$,

$$g(\lambda_{r',s'}) = G^{(r',s')}_{(r,s)}, \qquad g(\lambda_{r',-s'}) = -G^{(r',s')}_{(r,s)}. \tag{3.137}$$

Moreover $G^{\iota(r',s')}_{(r,s)} = G^{(r',s')}_{(r,s)}$.

Proof. Using $\sin(x\pm\pi n) = (-1)^n\sin x$ and $\sin(\pi n - x) = (-1)^{n+1}\sin x$ for $n\in\mathbb{Z}$,

$$\sin\frac{\pi r(pr'\mp p's')}{p'} = \sin\left(\frac{\pi prr'}{p'}\mp\pi rs'\right) = (-1)^{rs'}\sin\frac{\pi prr'}{p'}, \tag{3.138}$$

$$\sin\frac{\pi s(pr'\mp p's')}{p} = \sin\left(\pi sr'\mp\frac{\pi p'ss'}{p}\right) = \begin{cases}(-1)^{r's+1}\sin\frac{\pi p'ss'}{p}, & \text{upper sign},\\ (-1)^{r's}\sin\frac{\pi p'ss'}{p}, & \text{lower sign}.\end{cases} \tag{3.139}$$

Multiplying gives the two stated values. For the last claim, replacing (r',s') by $(p'-r',p-s')$ multiplies the two sines by $(-1)^{pr+1}$ and $(-1)^{p's+1}$ respectively, while the explicit sign is multiplied by $(-1)^{rp+sp'}$; the total is $(-1)^{2pr+2p's+2} = +1$. □

Theorem 3.6.5. *Let* $\mathcal{E}\subset\mathcal{K}$ *be a set of representatives for the primaries. Then*

$$\begin{aligned}\chi_{r,s}(-1/\tau) &= \sum_{(r',s')\in\mathcal{E}} S^{(r',s')}_{(r,s)}\,\chi_{r',s'}(\tau),\\ S^{(r',s')}_{(r,s)} &= 2\sqrt{\frac{2}{pp'}}\,(-1)^{1+rs'+r's}\sin\frac{\pi p\, rr'}{p'}\sin\frac{\pi p'\, ss'}{p}.\end{aligned} \tag{3.140}$$

Proof. By Lemma 3.6.3 the sum (3.130) may be reindexed over $\mathcal{K}\times\{\pm\}$, and by Lemma 3.6.4 the two members of each pair carry opposite coefficients, so they combine as

$$G^{(r',s')}_{(r,s)}\,\frac{\Theta_{\lambda_{r',s'}} - \Theta_{\lambda_{r',-s'}}}{\eta} = G^{(r',s')}_{(r,s)}\,\chi_{r',s'} \tag{3.141}$$

by (3.123). Hence

$$\chi_{r,s}(-1/\tau) = \frac{2}{\sqrt{2N}}\sum_{(r',s')\in\mathcal{K}} G^{(r',s')}_{(r,s)}\,\chi_{r',s'}. \tag{3.142}$$

Finally, $\mathcal{K}$ contains each primary twice, and by the last part of Lemma 3.6.4 the two occurrences carry equal coefficients; restricting to $\mathcal{E}$ therefore doubles the coefficient. With $\frac{4}{\sqrt{2pp'}} = 2\sqrt{\frac{2}{pp'}}$ this is (3.140). □

Corollary 3.6.6 (properties of S)**.** *The matrix* (3.140) *is real and symmetric, satisfies* $S^2 = \mathbb{1}$ *and is therefore orthogonal, and its vacuum row has no zero entry,* $S^{(r,s)}_{(1,1)} \neq 0$. *In particular every primary of* $\mathcal{M}(p,p')$ *is self-conjugate: the conjugation matrix* $C = S^2$ *is the identity.*

Proof. Reality and the symmetry $(r,s) \leftrightarrow (r',s')$ are visible in (3.140). Applying Theorem 3.6.5 twice gives $\chi_{r,s}(\tau) = \sum (S^2)_{(r,s)}^{(r',s')} \chi_{r',s'}(\tau)$. The characters of distinct primaries are linearly independent, because their leading exponents $h_{r,s} - c/24$ are distinct: $h_{r,s} = h_{r',s'}$ forces $pr - p's = \pm(pr' - p's')$. Since $\gcd(p,p') = 1$ and $1 \le r, r' < p'$, $1 \le s, s' < p$, this gives either $(r',s') = (r,s)$ or $(r',s') = (p'-r, p-s)$, the same primary. Hence $S^2 = \mathbb{1}$, and with $S = S^{\mathsf{T}}$ real, $SS^{\mathsf{T}} = \mathbb{1}$. Finally $S_{(1,1)}^{(r,s)} \propto \sin(\pi pr/p') \sin(\pi p's/p)$, which vanishes only if $p' \mid pr$ or $p \mid p's$, impossible for $0 < r < p'$, $0 < s < p$ with $\gcd(p,p') = 1$. □

Remark 3.6.7 (existence of S)**.** *For minimal models S is not assumed: Theorem 3.6.5 constructs it from the characters, and Corollary 3.6.6 derives its properties. In general three separate facts are involved. (i) The span of the characters need not be mapped into itself by $\tau \mapsto -1/\tau$; for minimal models it is, by Theorem 3.6.5. For the compact boson, (3.69) is modular invariant, yet the S-transform of a single character $q^{h-1/24}/\eta$ is a Gaussian integral over a continuum of weights; there is no S-matrix on the span of the characters, and only the full $\mathcal{Z}$ is invariant. (ii) The characters may not determine S. If $i \neq i^*$, then $\chi_i = \chi_{i^*}$ as functions of τ, and S has to be defined on the torus one-point blocks, or on characters refined by further variables. Relations such as $S_{i^*m} = \overline{S_{im}}$ and $S^2 = C$ live at that level. (iii) The diagonalization hypothesis (3.148) is assumed in this chapter (Remark 3.6.12) and checked by integrality. For the Ising model it can be verified directly (Example 3.6.15 and Remark 3.6.16).*

Example 3.6.8 (Ising model)**.** *For $\mathcal{M}(4,3)$ take $\mathcal{E} = \{(1,1),(1,2),(1,3)\}$, so that $2\sqrt{2/pp'} = 2/\sqrt{6}$. For instance*

$$S_{(1,1)}^{(1,2)} = \frac{2}{\sqrt{6}} (-1)^{1+2+1} \sin\frac{4\pi}{3} \sin\frac{3\pi}{2} = \frac{2}{\sqrt{6}} \cdot \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{2}}, \tag{3.143}$$

and evaluating the remaining entries in the order $(\mathbb{1}, \sigma, \varepsilon)$,

$$S = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix}. \tag{3.144}$$

One checks $S = S^{\mathsf{T}}$, $S^2 = \mathbb{1}$, and that $S_{0i} > 0$ with quantum dimensions $d_i = S_{0i}/S_{00} = (1, \sqrt{2}, 1)$ and total dimension $\mathcal{D} = 1/S_{00} = 2$. The same matrix was obtained from the free fermion in Proposition 3.4.22, without the Kac determinant or the resolution.

3.6.2 The Verlinde formula

The modular S-matrix was constructed from the characters, which know nothing about the operator product expansion. It is therefore remarkable that S determines the fusion rules. We first isolate the purely algebraic content of this statement, then record what must be supplied by conformal field theory.

Definition 3.6.9. *Let I index the primaries of a rational conformal field theory, with $0 \in I$ labeling the vacuum. The fusion algebra is the free $\mathbb{Z}$-module on I equipped with the product*

$$\phi_i \times \phi_j = \sum_{k \in I} N_{ij}^{\,k}\, \phi_k, \qquad N_{ij}^{\,k} \in \mathbb{Z}_{\geq 0}, \tag{3.145}$$

where $N_{ij}^{\,k}$ is the number of times the conformal family of ϕ_k appears in the operator product expansion of ϕ_i with ϕ_j. In the notation of (2.129) this means $N_{ij}^{\,k} \neq 0$ precisely when the structure constant C_{ijk} is nonzero, i.e. when the conformal block $\mathcal{F}(h_k|z)$ contributes; for Virasoro minimal models every multiplicity is 0 or 1. The fusion matrices are

$$(N_i)_{jk} := N_{ij}^{\,k}. \tag{3.146}$$

Lemma 3.6.10. *Commutativity and associativity of the OPE imply*

$$N_0 = \mathbb{1}, \qquad N_i N_j = N_j N_i = \sum_k N_{ij}^{\,k}\, N_k. \tag{3.147}$$

In particular $i \mapsto N_i$ is the regular representation of the fusion algebra, and the N_i form a commuting family.

Proof. $N_{0j}^{\,k} = \delta_{jk}$ because the vacuum is the unit. Expanding $(\phi_i \times \phi_j) \times \phi_l$ and $\phi_i \times (\phi_j \times \phi_l)$ and comparing coefficients of ϕ_k gives $\sum_m N_{ij}^{\,m} N_{ml}^{\,k} = \sum_m N_{jl}^{\,m} N_{im}^{\,k}$, which is the stated matrix identity; commutativity of the product gives $N_{ij}^{\,k} = N_{ji}^{\,k}$. □

Since the N_i commute they may in principle be simultaneously diagonalized. The content of Verlinde's theorem is that the diagonalising matrix is the modular S-matrix, and that the eigenvalues are the ratios S_{im}/S_{0m}.

Theorem 3.6.11 (Verlinde)**.** *Let S be unitary and suppose that it simultaneously diagonalizes the fusion matrices,*

$$N_i = S\, \Lambda_i\, S^{-1}, \qquad (\Lambda_i)_{mm} = \frac{S_{im}}{S_{0m}}. \tag{3.148}$$

Then

$$N_{ij}^{\,k} = \sum_{m \in I} \frac{S_{im} S_{jm} S_{km}^{*}}{S_{0m}}. \tag{3.149}$$

Proof. Reading off the (j,k) entry of (3.148),

$$N_{ij}^{\,k} = (N_i)_{jk} = \sum_m S_{jm}\, \frac{S_{im}}{S_{0m}}\, (S^{-1})_{mk}, \tag{3.150}$$

and $(S^{-1})_{mk} = (S^{\dagger})_{mk} = S_{km}^{*}$ by unitarity. □

Remark 3.6.12. *The hypothesis (3.148) is the substantive part, and its justification lies outside what has been developed so far. Verlinde's argument runs as follows. The space of conformal blocks on the torus is spanned by the characters χ_j. Inserting a chiral primary ϕ_i along a*

cycle of the torus defines an operator on this space; insertion along the spatial cycle fuses ϕ_i into the state, giving $\phi_i(b)\chi_j = \sum_k N_{ij}^k \chi_k$, while insertion along the time cycle acts diagonally, $\phi_i(a)\chi_j = \frac{S_{ij}}{S_{0j}}\chi_j$, the eigenvalue being read off from the modular transform of a one-point block. Since S exchanges the two cycles, $\phi_i(a) = S\phi_i(b)S^{-1}$, which is (3.148). Making this precise requires the theory of conformal blocks on surfaces with punctures; see the discussion in Chapter 5. In the present chapter we take (3.148) as given and verify its consequences.

Corollary 3.6.13. *The quantum dimensions $d_i := S_{0i}/S_{00}$ satisfy $d_i d_j = \sum_k N_{ij}^k d_k$, i.e. $(d_i)_{i\in I}$ is a one-dimensional representation of the fusion algebra.*

Proof. Multiply (3.149) by S_{0k}/S_{00} and sum over k. Since $\sum_k S_{0k}S_{km}^* = (SS^\dagger)_{0m} = \delta_{0m}$, only the term $m = 0$ survives, leaving $S_{i0}S_{j0}/S_{00}^2 = d_i d_j$. □

Caution 3.6.14. *Nothing in Theorem 3.6.11 guarantees that the right-hand side of (3.149) is a non-negative integer; this is a nontrivial property of the modular data of an actual conformal field theory, and it fails for a generic unitary symmetric matrix. Verifying integrality is therefore a genuine consistency check on the S-matrix computed in Theorem 3.6.5.*

Example 3.6.15 (Ising model)**.** *Take S from (3.144), which is real and symmetric with $S^2 = \mathbb{1}$, and label the primaries $(\mathbb{1}, \sigma, \varepsilon)$ by $m = 0, 1, 2$. Since S is real, (3.149) reads $N_{ij}^k = \sum_m S_{im}S_{jm}S_{km}/S_{0m}$.*

For $i = j = \sigma$ the weights are

$$\frac{S_{1m}^2}{S_{0m}} = \left(\frac{1/2}{1/2}, \frac{0}{1/\sqrt{2}}, \frac{1/2}{1/2}\right) = (1,\, 0,\, 1), \tag{3.151}$$

so that $N_{\sigma\sigma}^k = S_{k0} + S_{k2} = (1, 0, 1)$, i.e.

$$\sigma \times \sigma = \mathbb{1} + \varepsilon. \tag{3.152}$$

The same computation for the remaining products gives $\sigma \times \varepsilon = \sigma$ and $\varepsilon \times \varepsilon = \mathbb{1}$, so that ε generates a $\mathbb{Z}_2$ and σ has $d_\sigma = \sqrt{2}$, consistently with Corollary 3.6.13.

As a further check, the fusion matrix

$$N_\sigma = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{3.153}$$

is the adjacency matrix of the A_3 Dynkin diagram, with eigenvalues $\sqrt{2}$, 0, $-\sqrt{2}$; these agree with $S_{\sigma m}/S_{0m} = (\sqrt{2},\, 0,\, -\sqrt{2})$ as required by (3.148).

Remark 3.6.16. *The fusion rule $\sigma\times\sigma = \mathbb{1}+\varepsilon$ was already obtained by a completely independent route in Example 2.4.19: the level-two null vector of $V_{1,2}$ imposes a second order differential equation on $\langle\sigma\sigma\sigma\sigma\rangle$, and the roots of its indicial equation are $h_p = 0, \frac{1}{2}$. That an argument in the theory of ordinary differential equations and an argument in number theory produce the same integers is the deepest statement of this chapter.*

3.7 Modular invariant partition functions

We are finally in a position to impose (3.15). So far the Kac table has produced a *list* of admissible chiral representations; what has not yet been asked is which combinations of them assemble into a consistent theory.

Definition 3.7.1. *Let $\mathcal{H} \cong \bigoplus_{i,j} Z_{ij}\, M_i \otimes \bar{M}_j$ be the state space of a theory built from the chiral data of $\mathcal{M}(p,p')$. The multiplicity matrix Z is called a modular invariant if*

$$Z_{ij} \in \mathbb{Z}_{\geq 0}, \qquad Z_{00} = 1, \qquad [Z,S] = [Z,T] = 0. \tag{3.154}$$

The first condition expresses that Z_{ij} is a multiplicity, the second the uniqueness of the vacuum, and the third is (3.15) applied to $\mathcal{Z} = \sum_{i,j} Z_{ij}\chi_i\overline{\chi_j}$.

Proposition 3.7.2 (spin selection rule)**.** *$[Z,T] = 0$ if and only if*

$$Z_{ij} \neq 0 \quad \Longrightarrow \quad h_i - h_j \in \mathbb{Z}. \tag{3.155}$$

Proof. By (3.128b) the matrix T is diagonal with entries $e^{2\pi i(h_i - c/24)}$, so the term $Z_{ij}\chi_i\overline{\chi_j}$ of $\mathcal{Z}$ acquires the phase $e^{2\pi i(h_i - h_j)}$ under $\tau \to \tau + 1$. Invariance requires this phase to be trivial whenever $Z_{ij} \neq 0$. □

Proposition 3.7.3. *$Z = \mathbf{1}$ is a modular invariant for every minimal model.*

Proof. It commutes with everything, and $Z_{ii} = 1$ satisfies the first two conditions of (3.154). □

The corresponding partition function $\mathcal{Z} = \sum_i |\chi_i|^2$ is called the diagonal, or A-type, invariant. The substantive questions are whether it is unique, and if not what the alternatives mean physically. For the Ising model the answer is that it is unique, and the proof uses the S-matrix of Theorem 3.6.5 in an essential way.

Theorem 3.7.4. *For $\mathcal{M}(4,3)$ the only modular invariant is $Z = \mathbf{1}$; that is,*

$$\mathcal{Z}_{Ising} = |\chi_{\mathbb{1}}|^2 + |\chi_\sigma|^2 + |\chi_\varepsilon|^2 \tag{3.156}$$

is the unique conformal field theory with central charge $c = \frac{1}{2}$ built on this chiral data.

Proof. The weights are $h \in \{0, \frac{1}{16}, \frac{1}{2}\}$, whose pairwise differences $\pm\frac{1}{16}, \pm\frac{7}{16}, \pm\frac{1}{2}$ are never integers. By Proposition 3.7.2, Z is diagonal, $Z = \mathrm{diag}(Z_0, Z_1, Z_2)$ with $Z_0 = 1$. For diagonal Z,

$$(ZS - SZ)_{ij} = (Z_i - Z_j)S_{ij}, \tag{3.157}$$

so $[Z,S] = 0$ forces $Z_i = Z_j$ whenever $S_{ij} \neq 0$. From (3.144) we have $S_{01} = 1/\sqrt{2} \neq 0$ and $S_{02} = 1/2 \neq 0$, giving $Z_1 = Z_2 = Z_0 = 1$. □

Remark 3.7.5. *Uniqueness is special to small models. In general a chiral algebra admits several modular invariants, and they describe genuinely different physical theories. The smallest example is $\mathcal{M}(6,5)$ with $c = \frac{4}{5}$, which admits the diagonal invariant — the tetracritical Ising model — and one other, the three-state Potts model, whose state space has twelve sectors rather than ten and which carries an S_3 symmetry absent from the diagonal theory.*

Slogan 3.7.6. *Chiral data — central charge, representations, fusion rules, modular data — does not determine a conformal field theory. It determines the space of candidates, and modular invariance selects among them.*

3.8 Notes

These notes collect pointers to more advanced frameworks and to the literature. They are outside the logical line of the chapter: nothing in the main text depends on them. Each item records where it was moved from.

- *Existence of S* (moved from Remark 3.6.7(i)). That the span of the characters is mapped into itself by $\tau \mapsto -1/\tau$ is a theorem for C_2-cofinite rational vertex operator algebras [Zhu96].
- *The Verlinde conjecture* (moved from Remark 3.6.7(iii)). The diagonalization hypothesis (3.148) and the symmetry of S are theorems for the same class of vertex operator algebras [Hua08b].
- *Two routes* (author's sentence, moved from Remark 3.6.16). The agreement of the two routes is the elementary shadow of the Moore–Seiberg programme, in which crossing symmetry on the sphere and modular invariance on the torus are shown to generate all consistency conditions of the theory.
- *Classification of modular invariants* (author's sentence, moved from Remark 3.7.5). The complete classification for minimal models is due to Cappelli, Itzykson and Zuber [CIZ87a; CIZ87b], completed by Kato [Kat87]; it is labeled by pairs of simply-laced Dynkin diagrams, a correspondence we do not develop here.
- *String theory* (author's phrase, moved from the paragraph before the vertex operator $\mathcal{V}_\alpha$ of the free boson). In string theory, the state $|\alpha\rangle$ is interpreted as a state of definite center-of-mass momentum $\pi_0 = \alpha$.
- *The Casimir energy is observable* (moved from Remark 3.1.3). The finite-size correction to the free energy of a critical strip is fixed by c [BCN86]. Equivalently, the leading low-temperature correction to the free energy of a critical quantum chain is $-(\pi/6)\, c\, L\, (k_B T)^2/\hbar v$, with v the velocity of the gapless excitations [Aff86]. This is a standard way of measuring c numerically.

- *Uniqueness of the vacuum* (moved from the remark after Corollary 3.1.5). It is closely related to cluster decomposition; in the Wightman framework the two are equivalent.

- *The self-dual radius* (moved from §3.3.4). At $R_* = (2\pi g)^{-1/2}$, the states $(n, w) = (\pm 1, \mp 1)$ have $(h, \bar{h}) = (1, 0)$ and $(\pm 1, \pm 1)$ have $(0, 1)$. Together with $i\partial\varphi$ and $i\bar{\partial}\varphi$ they give three holomorphic and three antiholomorphic currents of weight one, the $\mathfrak{su}(2)$ current algebra at level one. For generic R the only weight-one currents are $i\partial\varphi$ and $i\bar{\partial}\varphi$.

Chapter 4

The disk and the annulus

A boundary identifies the holomorphic and antiholomorphic halves, so the disk and the annulus are governed by a single Virasoro algebra; this is the doubling trick of §4.1.1. Section 4.1 develops correlation functions on the half plane, and §4.2 encodes a boundary condition as a state of the closed theory. On the annulus the two readings of the same trace are related by the modular S-matrix, and the Cardy condition then fixes the boundary states of a diagonal minimal model (§4.3). Section 4.4 introduces the g-function and boundary renormalization group flows.

4.1 Conformal boundary conditions

4.1.1 The upper half plane

We take the worldsheet to be the closed upper half plane

$$\mathbb{H} = \{\, z \in \mathbb{C} \,|\, \operatorname{Im} z \geq 0 \,\}\,, \tag{4.1}$$

whose boundary is the real axis. Any surface with a boundary looks locally like this, and the disk is conformally equivalent to it, so nothing is lost.

The gluing condition

A boundary condition is a prescription for the fields at $\operatorname{Im} z = 0$. Most such prescriptions destroy conformal invariance; the ones that do not are singled out by a condition on the energy-momentum tensor.

Definition 4.1.1. *A boundary condition is conformal if*

$$T(z) = \bar{T}(\bar{z}) \qquad \text{for} \qquad z = \bar{z}. \tag{4.2}$$

Equation (4.2) is called the gluing condition.

Two things should be said about this definition: what it means physically, and why it deserves the name conformal.

Proposition 4.1.2. *On the real axis, (4.2) is equivalent to $T_{xy} = 0$, the vanishing of the flux through the boundary of momentum parallel to it.*

Proof. With $z = x + iy$ one has $\partial x/\partial z = \frac{1}{2}$ and $\partial y/\partial z = -\frac{i}{2}$, so that

$$T_{zz} = \tfrac{1}{4}(T_{xx} - T_{yy}) - \tfrac{i}{2}T_{xy}, \qquad T_{\bar z\bar z} = \tfrac{1}{4}(T_{xx} - T_{yy}) + \tfrac{i}{2}T_{xy}. \tag{4.3}$$

Since $T = -2\pi T_{zz}$ and $\bar T = -2\pi T_{\bar z\bar z}$, the condition $T = \bar T$ is exactly $T_{xy} = 0$. The component $T^{\mu y}$ is the flux of μ-momentum through a surface of normal $\hat y$, so T^{xy} is the flux of tangential momentum through the boundary. □

Remark 4.1.3. *$T^{xy} = 0$ is the statement that the associated charge is conserved. On $\mathbb{H}$ the charge generating x-translations is an integral $Q_C = \int_C T^{\mu x} n_\mu \mathrm{d}s$ along a curve running from boundary to boundary, and deforming C to C' picks up the flux through the boundary segment between them,*

$$Q_C - Q_{C'} = \int T^{yx}\,\mathrm{d}x. \tag{4.4}$$

So Q_C is independent of C — that is, translation along the boundary is a symmetry — precisely when T^{xy} vanishes there. This is the case $\epsilon = $ const of Proposition 4.1.6 below: on a closed surface the contour closes and the question does not arise.

Caution 4.1.4. *Equation (4.2) is not automatic. A boundary mass term, or a boundary magnetic field of generic strength, gives $T_{xy} \neq 0$; such boundary conditions exist and are perfectly consistent, but they are not conformal. They flow under the renormalization group, and (4.2) is the condition satisfied at the fixed points of that flow — see §4.4. Definition 4.1.1 is therefore a restriction we impose, not a consequence of anything proved so far.*

Why the surviving symmetry is conformal is best seen from the charges. In the bulk a conformal transformation $z \mapsto z + \epsilon(z)$, $\bar z \mapsto \bar z + \bar\epsilon(\bar z)$ is generated by

$$Q_\epsilon = \oint \frac{\mathrm{d}z}{2\pi i}\,\epsilon(z)T(z) \;-\; \oint \frac{\mathrm{d}\bar z}{2\pi i}\,\bar\epsilon(\bar z)\bar T(\bar z), \tag{4.5}$$

where the contours are closed curves. On $\mathbb{H}$ they cannot be closed: a contour must be cut off at the real axis, and the two halves are joined by a segment along it.

Proposition 4.1.5 (geometric condition)**.** *The infinitesimal transformation $z \mapsto z + \epsilon(z)$, $\bar z \mapsto \bar z + \bar\epsilon(\bar z)$ maps $\mathbb{H}$ to itself if and only if*

$$\epsilon(x) = \bar\epsilon(x) \in \mathbb{R} \qquad \textit{for real } x. \tag{4.6}$$

Proof. A boundary point has $z = \bar z = x$, and its image has $z' = x + \epsilon(x)$, $\bar z' = x + \bar\epsilon(x)$. It remains on the boundary precisely when $z' = \bar z'$ and this common value is real. □

This is a statement about the vector field only; whether such a transformation is a *symmetry* is a separate question, and this is where the boundary condition enters. The current associated with ϵ has components ϵT and $\bar\epsilon\bar T$, conserved by the equations of motion, and the corresponding charge is carried by a curve C running from one point of the boundary to another,

$$Q_C[\epsilon] = \frac{1}{2\pi i}\int_C \Big(\epsilon(z)T(z)\,\mathrm{d}z - \bar\epsilon(\bar z)\bar T(\bar z)\,\mathrm{d}\bar z\Big). \tag{4.7}$$

Proposition 4.1.6 (conservation)**.** *Let ϵ satisfy (4.6). Then $Q_C[\epsilon]$ is independent of C — that is, the transformation is a symmetry — if and only if the gluing condition (4.2) holds.*

Proof. Two curves C, C' with the same endpoints on the boundary, together with the segment $I \subset \mathbb{R}$ between their endpoints, bound a region of $\mathbb{H}$. Current conservation makes the integral over the closed boundary of that region vanish, so

$$Q_C[\epsilon] - Q_{C'}[\epsilon] = \frac{1}{2\pi i} \int_I \Big(\epsilon T - \bar{\epsilon} \bar{T} \Big) \mathrm{d}x, \tag{4.8}$$

since $\mathrm{d}z = \mathrm{d}\bar{z} = \mathrm{d}x$ on the real axis. By (4.6) the integrand is $\epsilon\,(T - \bar{T})$, which vanishes for every admissible ϵ exactly when $T = \bar{T}$ there. □

Remark 4.1.7. *The case $\epsilon = $ const, which generates translation along the boundary, is Remark 4.1.3. On a closed surface the curve C closes and there is no segment I, so no condition arises: this is a question that only a boundary can pose.*

Remark 4.1.8. *In the global case (4.6) restricts (1.12) to those Möbius transformations with real coefficients, so the global symmetry group of $\mathbb{H}$ is $\mathrm{PSL}(2,\mathbb{R}) \subset \mathrm{PSL}(2,\mathbb{C})$: exactly half the dimension.*

Proposition 4.1.9 (global symmetry of the half plane)**.** *The globally defined conformal transformations preserving $\mathbb{H}$ are exactly the Möbius transformations (1.12) with real coefficients,*

$$z \longmapsto \frac{az+b}{cz+d}, \qquad a,b,c,d \in \mathbb{R}, \qquad ad - bc = 1, \tag{4.9}$$

forming the group $\mathrm{PSL}(2,\mathbb{R})$, of real dimension three. For such a transformation

$$\operatorname{Im} \frac{az+b}{cz+d} = \frac{\operatorname{Im} z}{|cz+d|^2}. \tag{4.10}$$

Proof. Infinitesimally. By Proposition 1.1.2 at $d = 2$, a globally defined conformal vector field is at most quadratic, $\epsilon(z) = \alpha + \beta z + \gamma z^2$. For a genuine transformation of the plane the two components are conjugate, $\bar{\epsilon}(\bar{z}) = \overline{\epsilon(z)}$, so (4.6) states that $\epsilon(x) \in \mathbb{R}$ for all real x; since a polynomial real on $\mathbb{R}$ has real coefficients, this holds exactly when α, β, γ are real. The space of such fields is three dimensional over $\mathbb{R}$, half the six of Proposition 1.1.2.

Globally. A Möbius transformation is determined by the images of $0, 1, \infty$. If it maps $\hat{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$ to itself, those three images are real, and solving for (a,b,c,d) gives real coefficients up to an overall scale; conversely real coefficients clearly preserve $\hat{\mathbb{R}}$.

Which half. Writing $\operatorname{Im} w = \frac{1}{2i}(w - \bar{w})$ and using the identity

$$\frac{az+b}{cz+d} - \frac{a\bar{z}+b}{c\bar{z}+d} = \frac{(ad-bc)(z-\bar{z})}{(cz+d)(c\bar{z}+d)} \tag{4.11}$$

together with $(c\bar{z}+d) = \overline{(cz+d)}$ for real coefficients gives (4.10) when $ad - bc = 1$. Hence $\operatorname{Im} z > 0$ is preserved. Normalising instead to $ad - bc = -1$ would reverse the sign and exchange the two half planes, so (4.9) with $ad - bc = +1$ is the subgroup preserving $\mathbb{H}$ rather than merely $\hat{\mathbb{R}}$. □

Corollary 4.1.10. $\mathrm{PSL}(2,\mathbb{R})$ *acts transitively on* $\mathbb{H}$*, and the stabiliser of* i *consists of the rotations*

$$(a,b,c,d) = (\cos\theta,\ \sin\theta,\ -\sin\theta,\ \cos\theta), \tag{4.12}$$

isomorphic to $\mathrm{SO}(2)/\{\pm 1\}$.

Proof. For $z_0 = x_0 + iy_0$ with $y_0 > 0$, the map $z \mapsto y_0 z + x_0$ has real coefficients and unit determinant after normalization, and sends i to z_0. For the stabiliser, $(ai+b)/(ci+d) = i$ gives $ai + b = -c + di$, hence $b = -c$ and $a = d$; combined with $ad - bc = 1$ this is $a^2 + b^2 = 1$. □

The doubling trick

Propositions 4.1.5 and 4.1.6 say that one function's worth of symmetry survives, and Proposition 4.1.9 has identified its global part. The doubling trick makes this concrete by reflecting the anti-holomorphic field on $\mathbb{H}$ into the lower half plane, where it becomes holomorphic, so that a boundary problem becomes a chiral problem on the whole plane.

Theorem 4.1.11 (doubling)**.** *On* $\mathbb{H}$ *the field* $\bar{T}$ *of* (1.26) *is anti-holomorphic, since Theorem 1.2.7 gives* $\partial_z \bar{T} = 0$*. Extend it to the lower half plane by assigning to each point its value at the reflected point, and set*

$$\mathcal{T}(z) := \begin{cases} T(z), & \operatorname{Im} z > 0, \\ \bar{T}(\bar{z}), & \operatorname{Im} z < 0, \end{cases} \tag{4.13}$$

the second line meaning the value of $\bar{T}$ *at the point* $\bar{z} \in \mathbb{H}$*. If the gluing condition* (4.2) *holds, then* $\mathcal{T}$ *is holomorphic on all of* $\mathbb{C}$*.*

Proof. Since $\bar{T}$ is anti-holomorphic on $\mathbb{H}$ we may write its value at a point of coordinate ζ as $G(\bar{\zeta})$ with G holomorphic. For $\operatorname{Im} z < 0$ the reflected point has coordinate $\bar{z}$, so

$$\mathcal{T}(z) = G\big(\overline{\bar{z}}\big) = G(z), \tag{4.14}$$

which is holomorphic in z. Both branches of (4.13) are therefore holomorphic on their half planes, and on the real axis the point and its reflection coincide, so the two agree by (4.2) and $\mathcal{T}$ is continuous across the axis. A function continuous on a domain and holomorphic off a line segment is holomorphic throughout, by Morera's criterion applied to triangles straddling the segment. □

Caution 4.1.12. *In* (4.13) *the symbol* $\bar{T}(\bar{z})$ *denotes the value of the field* $\bar{T}$ *at the point labeled by* $\bar{z}$*, not a holomorphic function of one variable evaluated at* $\bar{z}$*. Under the second reading the lower branch would be anti-holomorphic and no gluing would be possible.*

Corollary 4.1.13. *For* ϵ *satisfying* (4.6)*, the charge* (4.7) *is a single contour integral of the doubled field,*

$$Q_C[\epsilon] = \oint \frac{\mathrm{d}z}{2\pi i}\, \epsilon(z)\, \mathcal{T}(z), \tag{4.15}$$

the contour being a closed curve in the full plane. Taking $\epsilon(z) = z^{n+1}$, which is real on the real axis and so satisfies (4.6), gives the Laurent modes of $\mathcal{T}$; the residual symmetry algebra is therefore a single Virasoro algebra, labeled by one holomorphic function rather than two.

Proof. Condition (4.6) says that ϵ and $\bar{\epsilon}$ agree on the real axis; since each is holomorphic in its own argument, they agree as functions, and we write ϵ for both.

Take C to be a semicircle in $\mathbb{H}$ running from $+R$ to $-R$. The first term of (4.7) is already an integral of $\epsilon\mathcal{T}$ along C, since $\mathcal{T} = T$ there. In the second term, substitute for the integration variable the reflected point: as the insertion runs along C, its reflection runs along the arc $\bar{C}$ in the lower half plane, and by (4.13) the value of $\bar{T}$ at the original point is the value of $\mathcal{T}$ at the reflected one. Hence

$$Q_C[\epsilon] = \frac{1}{2\pi i}\left[\int_C \epsilon\,\mathcal{T}\,\mathrm{d}z - \int_{\bar{C}} \epsilon\,\mathcal{T}\,\mathrm{d}z\right]. \tag{4.16}$$

The minus sign reverses the orientation of $\bar{C}$, which then runs from $-R$ to $+R$ and continues C; together the two arcs close. □

Slogan 4.1.14. *A closed conformal field theory carries two commuting copies of the Virasoro algebra, $\mathfrak{Vir} \oplus \overline{\mathfrak{Vir}}$, acting on $\mathcal{H} \otimes \bar{\mathcal{H}}$. A boundary ties them together: the two are realized as the modes of one holomorphic field on the doubled surface, and only a single copy survives as a symmetry.*

Remark 4.1.15. *In modes, (4.13) identifies the anti-holomorphic modes on $\mathbb{H}$ with the holomorphic modes of $\mathcal{T}$ on the reflected half plane, so that $\bar{L}_n$ and L_n are no longer independent. Corollary 4.1.13 is the statement that only the combinations surviving this identification generate symmetries. The precise form of the surviving combination depends on the geometry: on $\mathbb{H}$ it pairs $\bar{L}_n$ with L_n, whereas the conformal map to the disk reverses the orientation of one factor and produces $L_n - \bar{L}_{-n}$. It is the latter form that will define the Ishibashi states of §4.2, where the boundary is viewed as a state in the closed string Hilbert space rather than as a locus in the worldsheet.*

The same construction applies to any chiral field, not only to T, provided the boundary condition relates its two chiralities. An n-point function on $\mathbb{H}$ then becomes a chiral $2n$-point function on $\mathbb{C}$, with an image insertion at $\bar{z}_i$ for every insertion at z_i; correlation functions on the half plane are therefore computed by the conformal blocks of Chapter 2, with twice as many insertions.

4.1.2 Correlation functions on the half plane

The method of images

Theorem 4.1.11 concerns the stress tensor alone. Its consequence for correlation functions is that every insertion on $\mathbb{H}$ behaves as a pair of chiral insertions on $\mathbb{C}$, one at the point and one at its mirror image.

Before stating the consequence for correlation functions we fix what the image of a bulk field is. A bulk primary ϕ of weights $(h,\bar h)$ generates the irreducible module $M(c,h)$ under the holomorphic Virasoro algebra and $\bar M(c,\bar h)$ under the anti-holomorphic one. Since an irreducible highest weight module of the Virasoro algebra is determined by c and its highest weight, each half is labelled by a single number, and we write Φ_h for the chiral primary of $M(c,h)$. The reflection of Theorem 4.1.11 carries the anti-holomorphic half to the lower half plane and makes it holomorphic, so the image of ϕ is again a chiral field of the same chiral algebra:

$$\phi(z,\bar z) \quad\longmapsto\quad \Phi_h(z) \ \text{ at } z, \qquad \Phi_{\bar h}(\bar z) \ \text{ at } \bar z. \tag{4.17}$$

Nothing here is a choice: the weights $h,\bar h$ come with ϕ, and they determine the chiral fields.

Throughout the following it is convenient to write $T(p)$ and $\widetilde T(p)$ for the values of the two components of the stress tensor *at a point* $p\in\mathbb{H}$, reserving the notation of (1.26) for the dependence on coordinates. Theorem 1.2.7 then reads $\partial_{\bar p}T(p)=0$ and $\partial_p\widetilde T(p)=0$, and (4.13) becomes

$$\mathcal{T}(z)=\begin{cases} T(z), & \operatorname{Im} z>0,\\ \widetilde T(\bar z), & \operatorname{Im} z<0. \end{cases} \tag{4.18}$$

Lemma 4.1.16 (reflected operator product)**.** *Let ϕ be a primary of weights $(h,\bar h)$ inserted at $z_0\in\mathbb{H}$. Then, as $z\to\bar z_0$ from the lower half plane,*

$$\mathcal{T}(z)\,\phi(z_0,\bar z_0) \ \sim\ \frac{\bar h\,\phi(z_0,\bar z_0)}{(z-\bar z_0)^2}+\frac{1}{z-\bar z_0}\,\frac{\partial}{\partial\bar z_0}\,\phi(z_0,\bar z_0). \tag{4.19}$$

Proof. The antiholomorphic counterpart of (2.25) states that, for $q\in\mathbb{H}$ approaching z_0,

$$\widetilde T(q)\,\phi(z_0,\bar z_0) \ \sim\ \frac{\bar h\,\phi}{(\bar q-\bar z_0)^2}+\frac{1}{\bar q-\bar z_0}\,\frac{\partial\phi}{\partial\bar z_0}, \tag{4.20}$$

the poles occurring in $\bar q-\bar z_0$ rather than $q-z_0$; this is what $\partial_q\widetilde T=0$ means.

Set $q=\bar z$ with $\operatorname{Im} z<0$, so that $q\in\mathbb{H}$ and $\bar q=z$. Then $q\to z_0$ if and only if $z\to\bar z_0$, and substituting into (4.20) replaces $\bar q-\bar z_0$ by $z-\bar z_0$. By (4.18) the left-hand side is $\mathcal{T}(z)\phi$, which is (4.19). □

Remark 4.1.17. *The substitution $\bar q=z$ is the whole of the doubling trick at the level of operator products. A pole in $\bar q$, which is antiholomorphic in the point q, becomes a pole in z, holomorphic in the point z; and it has moved from z_0 to $\bar z_0$. Comparing (4.19) with (2.25), the singularity is exactly the one produced by a* chiral *primary of weight $\bar h$ inserted at $\bar z_0$, the role of ∂ being played by $\partial/\partial\bar z_0$.*

Proposition 4.1.18. *Let $\phi_1,\dots,\phi_n$ be primaries of weights $(h_i,\bar h_i)$ inserted at $z_i\in\mathbb{H}$, and set $G=\big\langle\prod_i\phi_i\big\rangle_{\mathbb{H}}$. Then*

$$F(z):=\Big\langle\mathcal{T}(z)\prod_i\phi_i(z_i,\bar z_i)\Big\rangle_{\mathbb{H}} \tag{4.21}$$

is holomorphic on $\mathbb{C}\setminus\{z_1,\dots,z_n,\bar z_1,\dots,\bar z_n\}$*, vanishes as* z^{-4} *at infinity, and has the expansions*

$$F(z)\sim\frac{h_i\,G}{(z-z_i)^2}+\frac{1}{z-z_i}\,\partial_{z_i}G,\qquad z\to z_i,\tag{4.22}$$

$$F(z)\sim\frac{\bar h_i\,G}{(z-\bar z_i)^2}+\frac{1}{z-\bar z_i}\,\partial_{\bar z_i}G,\qquad z\to \bar z_i.\tag{4.23}$$

These are precisely the singularities carried by the chiral $2n$*-point function*

$$\widetilde G(w_1,\dots,w_{2n}):=\Big\langle\Phi_{h_1}(w_1)\cdots\Phi_{h_n}(w_n)\,\Phi_{\bar h_1}(w_{n+1})\cdots\Phi_{\bar h_n}(w_{2n})\Big\rangle_{\mathbb{C}}\tag{4.24}$$

when its arguments are specialized to $w_i=z_i$ *and* $w_{n+i}=\bar z_i$*. Consequently* G *obeys the Virasoro Ward identities of* $\widetilde G$ *and lies in the corresponding space of conformal blocks.*

Proof. Holomorphy off the listed points is Theorem 4.1.11, the correlator being holomorphic in z wherever $\mathcal{T}$ is and no insertion sits. Equation (4.22) is (2.25) inserted in the correlator, valid for $\operatorname{Im} z>0$; equation (4.23) is Lemma 4.1.16. The decay at infinity holds because $\mathbb{H}\cup\{\infty\}$ compactifies to a disk whose boundary contains the point at infinity, so no insertion lies there and $\mathcal{T}$ is regular; a field of weight 2 regular at infinity vanishes as z^{-4}.

Now let ϵ be holomorphic on $\mathbb{C}$ and of degree at most two, and integrate ϵF over a circle $|z|=R$. The integral tends to zero as $R\to\infty$, since $\epsilon F=O(z^{-2})$, while by the residue theorem it equals the sum of the residues at the $2n$ points. Using (4.22) and (4.23),

$$\sum_i\Big[\epsilon(z_i)\partial_{z_i}+h_i\epsilon'(z_i)\Big]G+\sum_i\Big[\epsilon(\bar z_i)\partial_{\bar z_i}+\bar h_i\epsilon'(\bar z_i)\Big]G=0,\tag{4.25}$$

which is the global Ward identity of Theorem 2.2.8 for the $2n$ chiral insertions Φ_{h_i} at z_i and $\Phi_{\bar h_i}$ at $\bar z_i$. Allowing ϵ with poles at the insertions and deforming contours in the usual way produces the identities for all L_n, so G satisfies the full Virasoro Ward identities of (4.24). □

Remark 4.1.19. *The name is borrowed from electrostatics, where a charge in front of a grounded conducting plane is treated by placing a fictitious image charge behind it. Here the image at* $\bar z_i$ *carries the anti-holomorphic weight* $\bar h_i$*, and it encodes the boundary condition rather than a second physical excitation: only* n *fields have been inserted.*

Caution 4.1.20. *Proposition 4.1.18 constrains the correlator but does not determine it, and it is worth being precise about what is and is not fixed.* Which *chiral field sits at each of the* $2n$ *points is fixed by the bulk fields themselves, through* (4.17)*: an irreducible highest weight module of the Virasoro algebra is determined by* c *and its highest weight, and the weights* $h_i,\bar h_i$ *come with* ϕ_i*. Nothing is chosen there. What is not fixed is* which *element of the resulting block space the correlator equals: by* (2.129) *the chiral* $2n$*-point function* (4.24) *is a combination of conformal blocks whose coefficients conformal symmetry leaves free, and it is the boundary condition that selects the combination. For* $n=1$ *the block space is one dimensional and the freedom is the single constant* A_ϕ *of Proposition 4.1.26. The division of labor is the same as on the sphere, where the chiral algebra supplies the blocks and the structure constants select the combination.*

The counting is worth recording, since it determines how much conformal symmetry can do in each case.

on $\mathbb{H}$	chiral counterpart	determined up to
one point	two points	one constant A_ϕ
two points	four points	a function of one cross ratio

The first line is Proposition 4.1.26. The second is the first case in which a genuine function remains undetermined, and it is the natural home for the machinery of Chapter 2: with

$$\eta = \frac{(z_1 - \bar{z}_1)(z_2 - \bar{z}_2)}{(z_1 - \bar{z}_2)(z_2 - \bar{z}_1)} \tag{4.26}$$

the two-point function on $\mathbb{H}$ is a function of η alone, expandable in conformal blocks; and if one of the fields is degenerate in the sense of Definition 2.4.5, Theorem 2.4.7 applies to the doubled correlator and produces an ordinary differential equation for it.

Unlike its counterpart on the sphere, this cross ratio is real, and this is the trace left by the reduction of the symmetry group.

Proposition 4.1.21. *For $z_1, z_2 \in \mathbb{H}$ the quantity (4.26) satisfies*

$$\eta = \frac{4 \operatorname{Im} z_1 \operatorname{Im} z_2}{|z_1 - \bar{z}_2|^2} \in (0, 1], \tag{4.27}$$

with $\eta = 1$ if and only if $z_1 = z_2$, and $\eta \to 0$ when either insertion approaches the boundary.

Proof. The numerator of (4.26) is $(z_1 - \bar{z}_1)(z_2 - \bar{z}_2) = (2i \operatorname{Im} z_1)(2i \operatorname{Im} z_2) = -4 \operatorname{Im} z_1 \operatorname{Im} z_2$. For the denominator, $z_2 - \bar{z}_1 = -\overline{(z_1 - \bar{z}_2)}$, so

$$(z_1 - \bar{z}_2)(z_2 - \bar{z}_1) = -\,|z_1 - \bar{z}_2|^2, \tag{4.28}$$

and the two signs cancel, giving (4.27). Positivity is clear. Writing $z_j = x_j + iy_j$,

$$|z_1 - \bar{z}_2|^2 = (x_1 - x_2)^2 + (y_1 + y_2)^2 \;\geq\; (y_1 + y_2)^2 \;\geq\; 4y_1 y_2, \tag{4.29}$$

the second step by the arithmetic–geometric mean inequality, with equality throughout precisely when $x_1 = x_2$ and $y_1 = y_2$. □

Remark 4.1.22. *On the sphere the cross ratio of four points is a complex number, because $\mathrm{PSL}(2, \mathbb{C})$ has real dimension six and four points carry eight real coordinates. Here the four points are constrained to be two reflection pairs and the group is only $\mathrm{PSL}(2, \mathbb{R})$, of real dimension three; two points of $\mathbb{H}$ carry four real coordinates, leaving $4 - 3 = 1$. The single real modulus is η.*

Remark 4.1.23. *A two-point function on $\mathbb{H}$ admits two distinct degenerations, and this is what makes it more than a repetition of the sphere. By Proposition 4.1.21 they are the two endpoints of the interval $(0, 1]$. As $z_1 \to z_2$, that is $\eta \to 1$, the two physical insertions collide and the bulk operator product expansion applies. As $z_1 \to \bar{z}_1$, that is $\eta \to 0$, an insertion approaches*

the boundary and collides with its own image; the resulting expansion is in operators living on the boundary, and is called the boundary operator expansion. Requiring the two to agree is a consistency condition of the same kind as crossing symmetry in §2.5, imposed now on a surface with boundary. It will reappear, in the form adapted to the annulus, as the Cardy condition of §4.3.

The second of these is worth making explicit, since doubling turns it into an ordinary chiral operator product expansion.

Proposition 4.1.24 (boundary operator expansion)**.** *Let ϕ be a bulk primary of weights $(h,\bar h)$. As $\operatorname{Im} z \to 0$,*

$$\phi(z,\bar z) \;\sim\; \sum_k C_k^{\phi}\,(2\operatorname{Im} z)^{\,h_k-h-\bar h}\,\psi_k(x), \qquad x=\operatorname{Re} z, \tag{4.30}$$

where the ψ_k are chiral fields of weight h_k living on the boundary, and the sum runs over the chiral primaries occurring in the fusion of Φ_h with $\Phi_{\bar h}$.

Proof. By (4.17) the insertion is the pair $\Phi_h(z)$, $\Phi_{\bar h}(\bar z)$ of chiral fields on $\mathbb{C}$, separated by

$$z-\bar z = 2i\operatorname{Im} z. \tag{4.31}$$

As $\operatorname{Im} z \to 0$ the two collide, and the chiral operator product expansion of Chapter 2 applies:

$$\Phi_h(z)\,\Phi_{\bar h}(\bar z) \;\sim\; \sum_k C_k^{\phi}\,(z-\bar z)^{\,h_k-h-\bar h}\,\big[\Phi_k(\bar z)+\cdots\big], \tag{4.32}$$

the exponent being the one recorded in (2.129). The limit point $\bar z \to x$ lies on the real axis, so the fields Φ_k appearing on the right are evaluated on the boundary; absorbing the phase of $(2i)^{h_k-h-\bar h}$ into C_k^{ϕ} gives (4.30). □

Remark 4.1.25. *Two checks. First, the identity term $k=0$ has $h_0=0$ and contributes $C_0^{\phi}\,(2\operatorname{Im} z)^{-h-\bar h}$, which for $h=\bar h$ is exactly the one-point function (4.33) with $A_\phi = C_0^{\phi}$ up to the phase of $(2i)^{-2h}$: the leading behavior of a bulk field near the boundary is its own expectation value. Second, the boundary fields carry a single weight rather than a pair, as they must, since the boundary is one dimensional and Corollary 4.1.13 leaves only one Virasoro algebra to grade them.*

One-point functions and the disk amplitude

Proposition 4.1.26. *Let ϕ be a primary of weights $(h,\bar h)$ on $\mathbb{H}$ with a conformal boundary condition. Then*

$$\big\langle \phi(z,\bar z)\big\rangle_{\mathbb{H}} = \frac{A_\phi}{(z-\bar z)^{2h}}, \tag{4.33}$$

and $A_\phi = 0$ unless $h=\bar h$.

Proof. By Proposition 4.1.5 the residual global symmetry is $\mathrm{PSL}(2,\mathbb{R})$, acting by $w = (az+b)/(cz+d)$ with real coefficients and $ad-bc=1$. Reality of the coefficients means the same transformation acts on the anti-holomorphic coordinate, $\bar{w} = (a\bar{z}+b)/(c\bar{z}+d)$, so (1.18) gives

$$\big\langle \phi(z,\bar{z}) \big\rangle_{\mathbb{H}} = (cz+d)^{-2h}\,(c\bar{z}+d)^{-2\bar{h}}\,\big\langle \phi(w,\bar{w}) \big\rangle_{\mathbb{H}}. \tag{4.34}$$

The action is transitive: for $z_0 = x_0 + iy_0$ with $y_0 > 0$ the map $z \mapsto y_0 z + x_0$ lies in $\mathrm{PSL}(2,\mathbb{R})$ and sends i to z_0. Hence (4.34) determines the correlator everywhere from its value at $z = i$, which is a single constant.

That constant is constrained by the stabilizer of i, the rotations in Eq. (4.1.10), and $c\bar{z}+d = e^{i\theta}$ at $z = i$. Then (4.34) reads

$$\big\langle \phi(i,-i) \big\rangle_{\mathbb{H}} = e^{2i\theta(h-\bar{h})}\,\big\langle \phi(i,-i) \big\rangle_{\mathbb{H}} \qquad \text{for all } \theta, \tag{4.35}$$

so the value vanishes unless $h = \bar{h}$.

For $h = \bar{h}$ the form (4.33) satisfies (4.34), since a direct computation gives

$$w - \bar{w} = \frac{z-\bar{z}}{(cz+d)(c\bar{z}+d)}, \tag{4.36}$$

whose $(-2h)$-th power supplies exactly the two Jacobian factors. By transitivity it is the only such form. □

Remark 4.1.27. *The shape of (4.33) is that of a chiral two-point function of two fields of weight h, and this is not a coincidence. The field ϕ carries a holomorphic and an anti-holomorphic half, both attached to the same physical point z; the doubling of Theorem 4.1.11 moves the anti-holomorphic half to the reflected point $\bar{z}$, so that one insertion on $\mathbb{H}$ becomes two chiral insertions on $\mathbb{C}$, at z and at its image. Only one field has been inserted, and the second is an image in the sense of the method of images in electrostatics: it encodes the boundary, not a second excitation.*

Remark 4.1.28. *In a closed theory the one-point function of a non-identity primary vanishes, there being no invariant for a single insertion on the sphere. On $\mathbb{H}$ the residual symmetry is smaller and the correlator survives. The coefficients A_ϕ are not fixed by conformal symmetry; they are precisely the data distinguishing one conformal boundary condition from another, and §4.2 reorganizes them into a single state in the closed string Hilbert space.*

4.2 Boundary states

4.2.1 The disk as a state

By (4.33) a conformal boundary condition assigns a number A_ϕ to every bulk primary, and by Proposition 4.1.26 these numbers are not fixed by conformal symmetry. They are a great deal of data, one complex number for each primary, and it is convenient to package them into a single object.

The device is radial quantization applied to the disk rather than to the plane. Map $\mathbb{H}$ to the unit disk; the boundary becomes the unit circle, which in radial quantization is a circle of constant time. The path integral over the disk, with a conformal boundary condition a imposed on $|z| = 1$ and nothing inserted inside, therefore prepares a state on that circle. Equivalently, reading the radial evolution outward from the origin, the disk amplitude is the overlap of the closed string vacuum with a state determined by the boundary condition.

Definition 4.2.1. *The boundary state $|a\rangle\!\rangle$ associated with a conformal boundary condition a is the element of the closed string state space $\mathcal{H} \otimes \bar{\mathcal{H}}$ defined by*

$$\langle\!\langle a|\psi\rangle\!\rangle \;=\; \Big\langle \psi \Big\rangle_{\text{disk with boundary condition } a} \qquad \text{for every } |\psi\rangle \in \mathcal{H} \otimes \bar{\mathcal{H}}, \tag{4.37}$$

the right-hand side being the disk one-point function of the bulk field corresponding to $|\psi\rangle$ under the state–operator correspondence of §2.3.

Remark 4.2.2. *Taking $|\psi\rangle$ to be a primary recovers the coefficients of Proposition 4.1.26: $\langle\!\langle a|\phi\rangle\!\rangle = A^a_\phi$ up to the conformal factor relating the disk to $\mathbb{H}$. The boundary state is thus nothing more than the collection of all one-point functions, written as a single vector. Its value lies in the fact that the conditions (4.2) become linear conditions on that vector, which can then be solved.*

4.2.2 Ishibashi states

The gluing condition on the closed string Hilbert space

Proposition 4.2.3. *On the disk, the gluing condition (4.2) is equivalent to*

$$\Big(L_n - \bar{L}_{-n}\Big)|a\rangle\!\rangle = 0 \qquad \text{for all } n \in \mathbb{Z}. \tag{4.38}$$

Proof. The stress tensor has weight two, so under the map from $\mathbb{H}$ the condition $T = \bar{T}$ on the real axis becomes

$$z^2\, T(z) = \bar{z}^2\, \bar{T}(\bar{z}) \qquad \text{on } |z| = 1. \tag{4.39}$$

Insert the mode expansions $T(z) = \sum_n L_n z^{-n-2}$ and $\bar{T}(\bar{z}) = \sum_n \bar{L}_n \bar{z}^{-n-2}$, and use $\bar{z} = 1/z$ on the unit circle:

$$\sum_n L_n\, z^{-n} \;=\; \sum_n \bar{L}_n\, \bar{z}^{-n} \;=\; \sum_n \bar{L}_n\, z^{n} \;=\; \sum_n \bar{L}_{-n}\, z^{-n}, \tag{4.40}$$

the last step relabeling $n \mapsto -n$. Since the boundary state is defined by (4.37) to reproduce correlators on the disk, equality of the two sides as operators acting on $|a\rangle\!\rangle$ is (4.38), one condition for each power of z. □

Remark 4.2.4. *The reversal $\bar{L}_n \mapsto \bar{L}_{-n}$ is the effect of $\bar{z} = 1/z$, and is exactly the change of geometry anticipated in Remark 4.1.15: on $\mathbb{H}$ the gluing pairs $\bar{L}_n$ with L_n, while on the disk the inversion reverses one index. Only on the disk is the condition a linear condition on a* state, *which is what makes it solvable.*

Remark 4.2.5. *The case $n = 0$ reads $\big(L_0 - \bar{L}_0\big)|a\rangle\!\rangle = 0$: only states of zero spin contribute to a boundary state. This is consistent with Proposition 4.1.26, where A_ϕ was found to vanish unless $h = \bar{h}$.*

One Ishibashi state per spinless primary

Equation (4.38) is linear, so its solutions form a vector space, and the problem splits over the sectors of $\mathcal{H} \otimes \bar{\mathcal{H}}$. Within one sector the solution is essentially unique.

Definition 4.2.6. *Let $\mathcal{H}_i \otimes \bar{\mathcal{H}}_{i^*}$ be a sector of the closed string state space, let $\{|i, N\rangle\}$ be an orthonormal basis of $\mathcal{H}_i$ graded by L_0, and let $\overline{|i, N\rangle}$ denote the corresponding basis of $\bar{\mathcal{H}}_{i^*}$. The Ishibashi state of the sector is*

$$|i\rangle\!\rangle \;:=\; \sum_N |i, N\rangle \otimes \overline{|i, N\rangle}. \tag{4.41}$$

Proposition 4.2.7. *The state (4.41) satisfies (4.38), and up to scale it is the only solution inside $\mathcal{H}_i \otimes \bar{\mathcal{H}}_{\bar{\jmath}}$; in particular there are no solutions unless $\bar{\jmath} = i^*$. Consequently every boundary state is a combination*

$$|a\rangle\!\rangle \;=\; \sum_i A_i^a \, |i\rangle\!\rangle, \tag{4.42}$$

the sum running over the spinless sectors, with coefficients not determined by (4.38).

Proof sketch. Under the isomorphism $\mathcal{H}_i \otimes \bar{\mathcal{H}}_{\bar{\jmath}} \cong \mathrm{Hom}(\bar{\mathcal{H}}^*_{\bar{\jmath}}, \mathcal{H}_i)$ a state becomes a linear map, and (4.38) becomes the statement that the map commutes with the action of the Virasoro algebra. Both modules being irreducible, Schur's lemma gives a one dimensional space of such maps when they are isomorphic and none otherwise. The state (4.41) is the map corresponding to the identity. □

Example 4.2.8 (level one)**.** *That (4.41) is the right combination can be checked directly. Write $|i\rangle\!\rangle = |i\rangle \otimes \overline{|i\rangle} + \alpha\, L_{-1}|i\rangle \otimes \overline{L_{-1}|i\rangle} + \cdots$ and impose (4.38) with $n = 1$. The term $\bar{L}_{-1}$ acting on the level zero piece contributes with coefficient 1, while L_1 acting on the level one piece contributes $\alpha\, L_1 L_{-1}|i\rangle = 2h_i \alpha |i\rangle$, so*

$$\alpha = \frac{1}{2h_i} = \frac{1}{\big\| L_{-1}|i\rangle \big\|^2}. \tag{4.43}$$

The coefficient is the inverse Gram matrix at that level, which is precisely what (4.41) says once an orthonormal basis is used.

Caution 4.2.9. *Ishibashi states are not normalizable: $\langle\!\langle i|i\rangle\!\rangle = \sum_N 1$ diverges. They belong to a completion of $\mathcal{H} \otimes \bar{\mathcal{H}}$ rather than to the space itself, and expressions involving them are meaningful only when a damping factor is present. The overlaps appearing in §4.3 always carry such a factor, coming from propagation across the annulus.*

Remark 4.2.10. *The counting is now the exact analogue of the situation on the sphere and on the torus. The gluing condition is a linear, local requirement, and it leaves a continuum: any combination (4.42) solves it. Which combinations correspond to actual boundary conditions is a global question, answered in §4.3.*

4.3 The annulus

4.3.1 Open and closed channels

The annulus is the first surface on which two inequivalent quantizations of the same geometry are available, and comparing them is what turns the linear condition (4.38) into something with finitely many solutions.

Present the annulus as a strip of width L, with coordinate $0 \le \sigma \le L$ across it, carrying the conformal boundary condition a at $\sigma = 0$ and b at $\sigma = L$; the Euclidean time is compactified on a circle of circumference T. Either direction may be regarded as space.

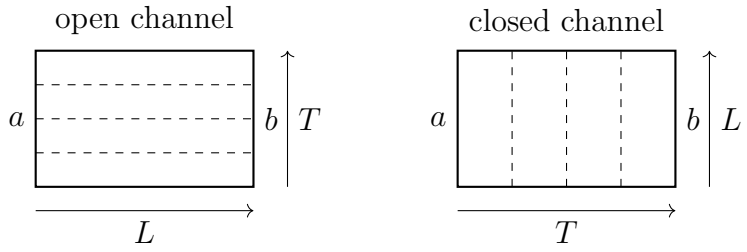


Figure 4.1: The same annulus read in two ways. On the left, time runs vertically and the state space is that of an open string stretched between the two boundaries. On the right, time runs horizontally and a closed string is emitted at a and absorbed at b; the dashed circles are its successive positions.

The open string spectrum $\mathcal{H}_{ab}$

Taking σ as space, the states live on an interval whose ends carry the two boundary conditions.

Definition 4.3.1. *The open string state space $\mathcal{H}_{ab}$ is the space of states on a strip of width L with boundary condition a at one end and b at the other. By Corollary 4.1.13 a single Virasoro algebra acts on it, so it decomposes into irreducible modules of that algebra,*

$$\mathcal{H}_{ab} \;\cong\; \bigoplus_j n_{ab}^{\,j}\, M_j, \qquad n_{ab}^{\,j} \in \mathbb{Z}_{\ge 0}. \tag{4.44}$$

Caution 4.3.2. *On a closed surface a state carries two weights $(h, \bar{h})$; here it carries one. This is not an approximation but a consequence of the gluing condition: the boundary identifies the two chiral algebras, and Corollary 4.1.13 leaves a single copy to grade the spectrum. The integers $n_{ab}^{\,j}$ are the analogue for a surface with boundary of the multiplicities Z_{ij} of Definition 3.7.1, and like them they are the data that a consistency condition will constrain.*

The generator of translations in Euclidean time on a strip of width L is obtained from the cylinder Hamiltonian by halving the circumference, since a strip of width L is half a cylinder of circumference $2L$:

$$H_{ab} = \frac{\pi}{L}\left(L_0 - \frac{c}{24}\right), \tag{4.45}$$

so that, tracing over $\mathcal{H}_{ab}$,

$$Z_{ab} = \mathrm{Tr}_{\mathcal{H}_{ab}}\, e^{-TH_{ab}} = \mathrm{Tr}_{\mathcal{H}_{ab}}\, q^{L_0 - \frac{c}{24}} = \sum_j n_{ab}^{\,j}\, \chi_j(q), \qquad q = e^{2\pi i \tau}, \quad \tau = \frac{iT}{2L}. \tag{4.46}$$

Remark 4.3.3. *The last equality is the definition of the characters in* (3.123) *applied to* (4.44)*. The whole content of the open channel is that Z_{ab} is a non-negative integer combination of characters, because it counts states.*

The modular transformation between channels

Taking instead the time circle as space, the annulus describes a closed string of circumference T propagating a distance L. Its initial and final states are the boundary states of Definition 4.2.1, and the generator of translations across the strip is the cylinder Hamiltonian for circumference T,

$$H_{\text{closed}} = \frac{2\pi}{T}\left(L_0 + \bar{L}_0 - \frac{c}{12}\right), \tag{4.47}$$

whence

$$Z_{ab} = \langle\!\langle a|\, e^{-LH_{\text{closed}}}\, |b\rangle\!\rangle = \langle\!\langle a|\, \tilde{q}^{\,\frac{1}{2}\left(L_0 + \bar{L}_0 - \frac{c}{12}\right)} |b\rangle\!\rangle, \qquad \tilde{q} = e^{2\pi i \tilde{\tau}}, \quad \tilde{\tau} = \frac{2iL}{T}. \tag{4.48}$$

Observation 4.3.4. *The two moduli are exchanged by the modular transformation S:*

$$\tilde{\tau} = -\frac{1}{\tau}. \tag{4.49}$$

Proof. $-1/\tau = -\big(iT/2L\big)^{-1} = 2iL/T = \tilde{\tau}$. □

Remark 4.3.5. *This is the same S that acted on the characters in Theorem 3.6.5, and it plays the same role: on the torus it exchanged the space and time cycles, and here it exchanges the two ways of reading the annulus. It is the reason a chapter about boundaries has to come after the torus.*

The closed channel can now be evaluated, because overlaps of Ishibashi states are immediate.

Proposition 4.3.6. *For the Ishibashi states of Definition 4.2.6,*

$$\langle\!\langle i|\, \tilde{q}^{\,\frac{1}{2}\left(L_0 + \bar{L}_0 - \frac{c}{12}\right)} |j\rangle\!\rangle = \delta_{ij}\, \chi_i(\tilde{q}). \tag{4.50}$$

Proof. Distinct sectors are orthogonal, which gives δ_{ij}. Within one sector, every state occurring in (4.41) has $L_0 = \bar{L}_0$, so that on $|i\rangle\!\rangle$

$$\tfrac{1}{2}\big(L_0 + \bar{L}_0 - \tfrac{c}{12}\big) = L_0 - \tfrac{c}{24}. \tag{4.51}$$

The sum (4.41) is diagonal and its terms are orthonormal, so the overlap counts each state of $\mathcal{H}_i$ once with weight $\tilde{q}^{\,h_i + N - c/24}$, which is the character (3.123). □

Remark 4.3.7. *The damping anticipated in Caution 4.2.9 is now visible. The right-hand side of* (4.50) *diverges as $\tilde{q} \to 1$, that is as $L \to 0$: the overlap of two boundary states is finite only when they are separated by an annulus of nonzero width.*

4.3.2 The Cardy condition

Inserting the expansion (4.42) into (4.48), using (4.50), and converting to the open channel by (4.49) and Theorem 3.6.5,

$$Z_{ab} = \sum_i \left(A_i^a\right)^* A_i^b \, \chi_i(\tilde{q}) = \sum_i \left(A_i^a\right)^* A_i^b \, \chi_i(-1/\tau)$$
$$= \sum_j \left[\sum_i \left(A_i^a\right)^* A_i^b \, S_{ij} \right] \chi_j(q). \tag{4.52}$$

Comparison with (4.46) identifies the bracket with the multiplicities of (4.44).

Definition 4.3.8 (Cardy condition)**.** *A family of boundary states $\{|a\rangle\rangle\}$ is consistent if*

$$n_{ab}^{\ j} = \sum_i \left(A_i^a\right)^* A_i^b \, S_{ij} \tag{4.53}$$

is a non-negative integer for every a, b and j.

Remark 4.3.9. *The asymmetry between the two channels is the whole point. In the closed channel the coefficients A_i^a may be arbitrary complex numbers: the gluing condition (4.38) is linear and leaves the continuum recorded in Proposition 4.2.7. In the open channel Z_{ab} is a trace over an honest Hilbert space, so by (4.44) the coefficient of χ_j counts states, and a count is a non-negative integer. A discrete condition imposed on a continuous family leaves a discrete set.*

This is the same movement as in Chapter 3. There the characters were combined into a partition function, modular invariance related the coefficients, and integrality of the multiplicities Z_{ij} in Definition 3.7.1 cut the solutions down to finitely many. Here the roles of T and S are played by the two readings of the annulus, and the multiplicities being constrained are the $n_{ab}^{\ j}$.

Caution 4.3.10. *Nothing so far shows that (4.53) has any solution, still less that the solutions can be written down. Both are true, and both rest on the Verlinde formula.*

4.3.3 Cardy states

Solution in terms of the modular S-matrix

We take the theory to be one of the minimal models with its diagonal modular invariant, Proposition 3.7.3, so that the spinless sectors are labeled by the primaries and the S-matrix of Theorem 3.6.5 is real and symmetric.

Theorem 4.3.11 (Cardy)**.** *For each primary a the state*

$$|\tilde{a}\rangle\rangle := \sum_i \frac{S_{ai}}{\sqrt{S_{0i}}} \, |i\rangle\rangle \tag{4.54}$$

satisfies the Cardy condition, and

$$n_{\tilde{a}\tilde{b}}{}^{j} = N_{ab}{}^{j}, \tag{4.55}$$

the fusion coefficients of Definition 3.6.9.

Proof. The coefficients are real, so (4.53) gives

$$n_{\tilde{a}\tilde{b}}{}^{j} = \sum_i \frac{S_{ai}\,S_{bi}}{S_{0i}}\,S_{ij} = \sum_i \frac{S_{ai}\,S_{bi}\,S^*_{ji}}{S_{0i}}, \tag{4.56}$$

using $S_{ij} = S_{ji}$ and $S^* = S$. By the Verlinde formula (3.149) the right-hand side is $N_{ab}{}^{j}$, a non-negative integer. □

Remark 4.3.12. *The square root in* (4.54) *is forced. The closed channel produces a* product $A_i^a A_i^b$ *of two coefficients, while the Verlinde formula* (3.149) *carries a single factor* S_{0m} *in the denominator; the two match only if each boundary state supplies* $S_{0i}^{-1/2}$*. The square root is real because* $S_{0i} = S_{00}\,d_i$ *with* $d_i > 0$*, by Corollary 3.6.13.*

Remark 4.3.13. *Conformal boundary conditions are therefore labeled by the primaries, one for each, and there are as many of them as there are spinless sectors. For a non-diagonal modular invariant the counting changes, since the sectors that admit Ishibashi states are no longer in bijection with the primaries; the classification is then more intricate and we do not pursue it.*

Verlinde revisited: $n_{ab}^{c} = N_{ab}^{c}$

Equation (4.55) deserves reading twice, because its two sides were defined by unrelated procedures.

Slogan 4.3.14. *The number of times the module* M_j *occurs in the spectrum of an open string stretched between the boundary conditions* a *and* b *equals the number of times* ϕ_j *occurs in the fusion of* ϕ_a *with* ϕ_b*.*

Remark 4.3.15. *The fusion coefficients have now been obtained three times, by computations with no step in common. In §2.4.5 they were the characteristic exponents of a differential equation, forced by a null vector. In §3.6.2 they came from the modular S-matrix, that is from the behavior of characters under* $\tau \mapsto -1/\tau$*. Here they count states of an open string on a strip. The first two agreements were the subject of Remark 3.6.16; the third extends the statement to surfaces with boundary, and is the first sign that the fusion ring is an invariant of the theory rather than an artifact of the sphere.*

Remark 4.3.16. *Two special cases are worth recording. Setting* $b = a$ *and* $j = 0$ *gives* $n_{\tilde{a}\tilde{a}}{}^{0} = N_{aa}{}^{0} = 1$*: an open string with the same boundary condition at both ends contains the identity module exactly once, which is the statement that the vacuum of the strip is unique. Setting* $a = 0$ *gives* $n_{\tilde{0}\tilde{b}}{}^{j} = \delta_b{}^{j}$*, so the boundary condition labeled by the identity is the one whose strip spectrum is a single module.*

By contrast the coefficient of the vacuum Ishibashi state,

$$g_a := A_0^{\tilde{a}} = \frac{S_{a0}}{\sqrt{S_{00}}}, \tag{4.57}$$

is not an integer. By (4.37) *it is the disk partition function with boundary condition a; it is the subject of §4.4.*

4.3.4 Example: boundary conditions of the Ising model

Take $\mathcal{M}(4,3)$ with the S-matrix (3.144), in the order $(\mathbb{1}, \sigma, \varepsilon)$. Theorem 4.3.11 produces three boundary conditions, and (4.57) with $S_{00} = \frac{1}{2}$ gives

$$g_{\tilde{\mathbb{1}}} = \frac{1/2}{\sqrt{1/2}} = \frac{1}{\sqrt{2}}, \qquad g_{\tilde{\sigma}} = \frac{1/\sqrt{2}}{\sqrt{1/2}} = 1, \qquad g_{\tilde{\varepsilon}} = \frac{1}{\sqrt{2}}. \tag{4.58}$$

The strip spectra follow from (4.55) and the fusion rules of Example 3.6.15.

boundary conditions	$\mathcal{H}_{ab}$	weights present
$\tilde{\mathbb{1}}$, $\tilde{\mathbb{1}}$	M_0	0
$\tilde{\varepsilon}$, $\tilde{\varepsilon}$	M_0	0
$\tilde{\mathbb{1}}$, $\tilde{\varepsilon}$	$M_{1/2}$	$\frac{1}{2}$
$\tilde{\mathbb{1}}$, $\tilde{\sigma}$	$M_{1/16}$	$\frac{1}{16}$
$\tilde{\sigma}$, $\tilde{\sigma}$	$M_0 \oplus M_{1/2}$	0, $\frac{1}{2}$

The lattice Ising model admits three natural boundary conditions: the edge spin may be held up, held down, or left free. The correspondence is

$$|\tilde{\mathbb{1}}\rangle\rangle \leftrightarrow \text{fixed } +, \qquad |\tilde{\varepsilon}\rangle\rangle \leftrightarrow \text{fixed } -, \qquad |\tilde{\sigma}\rangle\rangle \leftrightarrow \text{free}. \tag{4.59}$$

Remark 4.3.17. *Every line of the table can be checked against this dictionary. With both edges fixed the same way the ground state is unique and no domain wall is available, so only the identity module occurs; the first two lines agree, as they must, since $+$ and $-$ are related by the symmetry $\sigma \mapsto -\sigma$. With the edges fixed oppositely a domain wall is forced, and the lowest state of the strip is the one of weight $\frac{1}{2}$, namely ε; accordingly $M_{1/2}$ appears and M_0 does not. Mixing a fixed edge with a free one produces $\frac{1}{16}$, the weight of σ, which is the field that changes the boundary condition. Finally the free–free strip contains both M_0 and $M_{1/2}$, and the corresponding fusion is $\sigma \times \sigma = \mathbb{1} + \varepsilon$.*

Remark 4.3.18. *The ratio*

$$\frac{g_{\tilde{\sigma}}}{g_{\tilde{\mathbb{1}}}} = \sqrt{2} = d_\sigma \tag{4.60}$$

is the quantum dimension of Example 3.6.15. The free boundary condition has the larger g, and §4.4 will show that this is the direction in which boundary renormalization group flow runs: a free edge can be driven to a fixed one, but not the reverse. The same $\sqrt{2}$ appeared as the degeneracy of the Ramond ground state in Remark 3.4.18, which is not an accident: the free boundary condition of the Ising model is the one on which the fermion of §3.4 has a zero mode.

4.4 The g-function

4.4.1 Boundary entropy

The number (4.57) attached to each Cardy state is not a multiplicity, and it is worth asking what it measures. The answer is read off from the free energy of a strip.

Proposition 4.4.1. *For a strip of width L with boundary conditions a and b, the free energy at inverse temperature T behaves for $T \gg L$ as*

$$\ln Z_{ab} \;=\; \frac{\pi c\, T}{24\, L} \;+\; \ln g_a + \ln g_b \;+\; O\big(e^{-\pi T/L}\big)\,. \tag{4.61}$$

Proof. Evaluate (4.46) in the limit $q \to 0$. The trace is dominated by the lowest state of $\mathcal{H}_{ab}$, whose weight is $h_{\min}$, so $Z_{ab} \simeq q^{\,h_{\min}-c/24}$; with $q = e^{-\pi T/L}$ this gives the first term when $h_{\min} = 0$. To identify the constant, evaluate instead (4.48) in the same limit, which by (4.49) is $\tilde q \to 1$ and hence is *not* dominated by a single state. Expanding the closed channel and keeping the vacuum Ishibashi state,

$$Z_{ab} \;\simeq\; g_a\, g_b\, \tilde q^{\,-c/24} \;=\; g_a\, g_b\, e^{\pi c T/24L}, \tag{4.62}$$

since $\tilde q^{-c/24} = e^{2\pi L c/(24T)\cdot(T/L)^2}$ reduces to $e^{\pi cT/24L}$ on using $\tilde\tau = -1/\tau$. Taking logarithms gives (4.61), and the corrections are suppressed by the gap in the open channel. □

Definition 4.4.2. *The boundary entropy of a conformal boundary condition a is $\ln g_a$.*

Remark 4.4.3. *The name is justified by (4.61). The extensive term is the Casimir energy of the strip and belongs to the bulk; it is proportional to c and knows nothing about the boundaries. What the boundaries contribute is a constant, independent of L and of T, and additive over the two ends. A constant term in a free energy is an entropy, and $\ln g_a$ is the amount each boundary condition carries.*

Caution 4.4.4. *Unlike a conventional entropy, $\ln g$ need not be the logarithm of an integer, and is often negative: for the Ising model (4.58) gives $\ln g_{\mathbb{I}} = -\frac{1}{2}\ln 2$ while $\ln g_{\tilde\sigma} = 0$. It is a ground state degeneracy only in a generalized sense; the quantity g itself is the natural object, and by Corollary 3.6.13 it is a ratio of quantum dimensions rather than a count.*

4.4.2 Boundary renormalization group flow

Boundary conformal perturbation theory

Caution 4.1.4 observed that the gluing condition (4.2) is a restriction, satisfied at fixed points of a flow rather than automatically. We can now say what that flow is.

A boundary condition may be deformed by adding to the action an integral over the boundary of a field living there,

$$S \;\longrightarrow\; S + \lambda \int \mathrm{d}x\; \psi(x), \tag{4.63}$$

where ψ is one of the boundary operators of Proposition 4.1.24, of weight h_ψ. Since the integral is one dimensional, the coupling has mass dimension $1 - h_\psi$, and the leading term of the beta function is

$$\beta_\lambda = \big(1 - h_\psi\big)\lambda + O(\lambda^2). \tag{4.64}$$

Observation 4.4.5. *A boundary operator is relevant when $h_\psi < 1$, marginal when $h_\psi = 1$, irrelevant when $h_\psi > 1$. The threshold is 1 rather than the 2 of the bulk, because the perturbation is integrated over a line rather than a surface.*

Remark 4.4.6. *The spectrum of available perturbations is exactly $\mathcal{H}_{aa}$, whose decomposition (4.44) is known from Theorem 4.3.11. A conformal boundary condition is thus stable precisely when $n_{aa}^{\,j} = 0$ for every j with $0 < h_j < 1$, and this is a finite check.*

Example 4.4.7 (the Ising free boundary)**.** *By the table of §4.3.4, $\mathcal{H}_{\tilde\sigma\tilde\sigma} = M_0 \oplus M_{1/2}$, so the free boundary condition admits a perturbation by a boundary operator of weight $\frac{1}{2}$, which is relevant. Physically it is the boundary magnetic field: adding $\lambda \int \mathrm{d}x\, \psi(x)$ with ψ the boundary spin drives the free edge towards a fixed one, the sign of λ selecting $+$ or $-$. By contrast $\mathcal{H}_{\tilde{\mathbb{1}}\tilde{\mathbb{1}}} = M_0$ contains no operator of weight below 1 other than the identity, so the fixed boundary conditions are stable. The flow runs*

$$\textit{free} \longrightarrow \textit{fixed}, \tag{4.65}$$

and not the reverse.

The g-theorem

Example 4.4.7 exhibits a flow from $g = 1$ to $g = 1/\sqrt{2}$, that is towards smaller g. This is general.

Theorem 4.4.8 (Affleck–Ludwig)**.** *Along a boundary renormalization group flow connecting two conformal boundary conditions of the same bulk theory,*

$$g_{\mathrm{UV}} \;\geq\; g_{\mathrm{IR}}, \tag{4.66}$$

with equality only if the flow is trivial.

Remark 4.4.9. *We do not prove this. The original argument establishes it to leading order in perturbation theory around a fixed point, and later work exhibits $\ln g$ as a monotone of the flow in the same sense in which the central charge is a monotone of bulk flows. The parallel is close: c decreases along bulk flows and is a property of the fixed point theory, while $\ln g$ decreases along boundary flows and is a property of the fixed point boundary condition. The bulk theory is unchanged along a boundary flow, so c is constant and g is the quantity that moves.*

Remark 4.4.10. *The statement has a direct physical reading. A boundary condition is an impurity,* $\ln g$ *is the entropy it carries, and the theorem says that this entropy is screened as the temperature is lowered. In the Kondo problem the impurity is a magnetic moment coupled to conduction electrons; at high temperature it is free and contributes* $\ln 2$*, while at low temperature it is screened and contributes nothing. In the language of this chapter the flow runs from a boundary condition with larger* g *to one with smaller* g*, and the Ising flow of Example 4.4.7 is the simplest instance of the same phenomenon.*

Remark 4.4.11. *This closes the chapter where it began. Definition 4.1.1 imposed* $T = \bar{T}$ *on the boundary and Caution 4.1.4 warned that the condition is a choice; the* g*-theorem is the statement that the choice is not arbitrary but is where a generic boundary condition ends up, and that the possible endpoints are ordered by* g*. Together with Theorem 4.3.11, which enumerates them, this is a complete qualitative picture of what a boundary can do to a conformal field theory.*

Chapter 5

Higher genus

This chapter asks what genus 0 and genus 1 determine on a closed surface of higher genus. The note does not construct a conformal field theory there, so the answer is conditional on one hypothesis, factorization (§5.2), whose instances at genus 0 and 1 are proved earlier. Granted it, the number of conformal blocks on any closed surface is fixed by the modular S-matrix (§5.3). For the Ising model every other input is proved in the note.

5.1 Cutting along a circle

Two constructions of the earlier chapters are instances of one operation: cutting a surface along a circle and summing over what flows through it.

(i) *The sphere.* In (2.129), the circle $|w| = r$ with $|z| < r < 1$ separates the insertions at 0 and z from those at 1 and ∞. Inserting a complete set of states on this circle (Theorem 2.2.1) and grouping the states into conformal families turns the four-point function into a sum over primaries p. A holomorphic block $\mathcal{F}_p$ appears only if p occurs in the fusion of the two fields on each side. The number of s-channel blocks of $\langle\phi_l\phi_k\phi_j\phi_i\rangle$ is therefore

$$\sum_x N_{ij}^{\ x}\, N_{kl}^{\ x^*}, \tag{5.1}$$

where x^* is the conjugate label defined below. For $\langle\sigma\sigma\sigma\sigma\rangle$ in the Ising model it is 2, from $x \in \{\mathbb{1}, \varepsilon\}$ (Example 2.4.19).

(ii) *The torus.* The torus is the annulus $|q| \le |z| \le 1$ with its two boundary circles identified, and $\mathcal{Z}$ is the trace over the states on the circle (§3.1.2). Grouping the states into conformal families gives $\mathcal{Z} = \sum_{i,j} Z_{ij}\chi_i\overline{\chi_j}$. The holomorphic factor is spanned by the characters, one for each label x that can flow around the cycle.

In both cases the cut creates two new boundary circles. Shrinking each to a marked point, the label x on one side is met by x^* on the other, because a two-point function pairs a field only with its conjugate. On the torus both new points lie on the same, now cut, surface: the cut torus is a sphere with two marked points labelled x and x^*.

5.2 The factorization hypothesis

Fix a rational chiral theory with label set I, vacuum $0 \in I$ and fusion numbers N_{ab}^c (Definition 3.6.9). The conjugate a^* of a is the unique label with $N_{aa^*}^0 = 1$; in minimal models $a^* = a$ (Corollary 3.6.6). For a closed oriented surface Σ_g of genus g with n marked points labelled $a_1, \dots, a_n$, let

$$d_g(a_1, \dots, a_n) \in \mathbb{Z}_{\geq 0} \tag{5.2}$$

be the number of linearly independent holomorphic conformal blocks. The note gives this number a meaning only at genus 0, where it counts the blocks of an n-point function, and at genus 1 without marked points, where it counts characters.

Definition 5.2.1 (hypothesis (F)). *The counts d_g satisfy:*

(F1) $d_0(a) = \delta_{a0}$, $d_0(a,b) = \delta_{b,a^*}$ *and* $d_0(a,b,c) = N_{ab}^{c^*}$;

(F2) for every simple closed curve γ on Σ_g, cutting along γ gives

$$d(\Sigma; \boldsymbol{a}) = \sum_{x \in I} d(\Sigma_\gamma; \boldsymbol{a}, x, x^*), \tag{5.3}$$

where Σ_γ is the cut surface with the two new points labelled x and x^*. If γ separates, Σ_γ has two components, the marked points are distributed between them, and the right side is the sum over x of the product of the two counts.

Remark 5.2.2 (what is proved and what is assumed). *(F1) restates earlier results. On the sphere, Möbius covariance (G4) forces the one-point function of a quasi-primary of nonzero weight to vanish, and the two-point function of quasi-primaries of different weights (Theorem 2.1.6); the three-point count is the definition of N. (F2) holds in the two cases of §5.1. For a separating curve on the four-pointed sphere, (5.3) with (F1) is the block count (5.1). For a non-separating curve on the torus it gives $d_1 = \sum_x d_0(x, x^*) = |I|$, the number of characters. Every other instance of (F2) is a hypothesis.*

Cutting along different curves must give the same count. On the four-pointed sphere the s- and t-channel counts agree by associativity of fusion (Lemma 3.6.10). In general, consistency follows from Proposition 5.3.1 below, whose answer does not refer to the curves used.

5.3 Counting conformal blocks

By (F2), any count of blocks reduces to three-point counts on the sphere, i.e. to fusion numbers, and the Verlinde formula (3.149) diagonalizes those. The result depends on S alone.

Proposition 5.3.1 (dimension of conformal blocks). *Assume (F) and the Verlinde formula (3.149), with S unitary and symmetric, $S^2 = C$ (the conjugation $C_{ab} = \delta_{b,a^*}$), $S_{a^*m} = \overline{S_{am}}$ and $S_{0m} \neq 0$. Then for a closed surface of genus g with n marked points,*

$$d_g(a_1, \dots, a_n) = \sum_m S_{0m}^{2-2g-n} \prod_{i=1}^n S_{a_i m}. \tag{5.4}$$

In particular, without marked points,

$$d_g = \sum_m S_{0m}^{2-2g}. \tag{5.5}$$

Proof. We use the identity

$$\sum_x S_{xm} S_{x^* m'} = \sum_x S_{mx} \overline{S_{m'x}} = (SS^\dagger)_{mm'} = \delta_{mm'}. \tag{5.6}$$

Genus 0, $n = 3$. Since $S^*_{c^* m} = S_{cm}$, the Verlinde formula gives $N_{ab}^{c^*} = \sum_m S_{am} S_{bm} S_{cm} / S_{0m}$, which is (5.4).

Genus 0, $n \leq 2$. By unitarity and $S^2 = C$, $\sum_m S_{0m}^2 = 1$, $\sum_m S_{0m} S_{am} = \delta_{a0}$ and $\sum_m S_{am} S_{bm} = \delta_{b,a^*}$, in agreement with (F1).

Genus 0, $n \geq 4$. Cut along a curve separating $k \geq 2$ points from $l \geq 2$ points. Apply (F2) and the induction hypothesis to the two pieces, which have $k+1$ and $l+1$ points. The identity above collapses the double sum over (m, m') to $m = m'$, and the exponents of S_{0m} add to $(2-k-1) + (2-l-1) = 2-n$.

Raising the genus. Cut Σ_{g+1} along a non-separating curve, which gives Σ_g with two extra points labelled x, x^*. By (F2) and the induction hypothesis,

$$d_{g+1}(\boldsymbol{a}) = \sum_m S_{0m}^{2-2g-n-2} \prod_i S_{a_i m} \sum_x S_{xm} S_{x^* m} = \sum_m S_{0m}^{2-2(g+1)-n} \prod_i S_{a_i m}. \quad \square \tag{5.7}$$

Remark 5.3.2 (status for minimal models)**.** *For minimal models the properties of S assumed in Proposition 5.3.1 are Corollary 3.6.6, with $C = \mathbb{1}$. The Verlinde formula rests on the diagonalization hypothesis of Remark 3.6.12. For the Ising model that hypothesis is verified directly (Example 3.6.15), and S itself is computed from the free fermion in §3.4.4, without the Kac determinant or the Feigin–Fuks resolution. For the Ising model, then, (5.4) depends only on (F).*

Example 5.3.3. *For the Ising model $\mathcal{M}(4,3)$ the vacuum row of (3.144) is $(S_{0\mathbb{1}}, S_{0\sigma}, S_{0\varepsilon}) = (\frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2})$, so*

$$d_g = 2 \cdot \left(\frac{1}{2}\right)^{2-2g} + \left(\frac{1}{\sqrt{2}}\right)^{2-2g} = 2^{2g-1} + 2^{g-1}, \tag{5.8}$$

which is $1, 3, 10, 36, 136$ for $g = 0, \ldots, 4$. At $g = 1$ the value 3 is the number of characters, as it must be. With one marked point on the torus, (5.4) gives $d_1(\sigma) = 0$ and $d_1(\varepsilon) = 1$, in agreement with $d_1(a) = \sum_x N_{ax}^x$ obtained from (F2) directly: $\sigma \times x$ never contains x, while $\varepsilon \times \sigma = \sigma$. For the Lee–Yang model $\mathcal{M}(5,2)$ the same formula gives $1, 2, 5, 15, 50$, although there $S_{00} < 0$. Integrality is not automatic from (5.5), and checking it is a useful test of a proposed S-matrix.

Remark 5.3.4. *Equation (5.5) closes the circle opened in §3.6.2. There, fusion numbers obtained from null vectors on the sphere were recovered from the torus S-matrix. Here, granted (F), the same S-matrix counts blocks on every closed surface.*

5.4 Notes

These notes collect pointers to frameworks in which the hypothesis (F) is part of a larger structure. They are outside the logical line of the chapter: nothing in the main text depends on them. Each part records where it was moved from.

5.4.1 Segal's axioms

(Moved from §5.1 of the previous draft, where it stood as a definition, a table and a remark.)

Let $\mathcal{C}$ be the category whose objects are finite families of parametrized circles, each marked incoming or outgoing. Its morphisms are Riemann surfaces Σ whose boundary is identified with these circles by the parametrizations, taken up to conformal equivalence relative to the boundary. Composition is sewing along matching outgoing and incoming circles, and disjoint union makes $\mathcal{C}$ symmetric monoidal. A conformal field theory of central charge $(c, \bar{c})$ consists of:

(S1) a Hilbert space $\mathcal{H}$ attached to the circle, and $\mathcal{H}^{\otimes k}$ to k circles;

(S2) for each Σ, a trace-class operator $U_\Sigma \colon \mathcal{H}^{\otimes \mathrm{in}} \to \mathcal{H}^{\otimes \mathrm{out}}$, defined up to a scalar that is fixed by a point of a determinant line over the moduli of Σ, whose curvature is governed by $(c, \bar{c})$;

subject to:

(S3) $U_{\Sigma' \circ \Sigma} = U_{\Sigma'} U_\Sigma$ and $U_{\Sigma \sqcup \Sigma'} = U_\Sigma \otimes U_{\Sigma'}$, with the scalars multiplying accordingly;

(S4) $U_{\bar{\Sigma}} = U_\Sigma^\dagger$, where $\bar{\Sigma}$ is Σ with the opposite complex structure and in- and out-circles exchanged.

The definition is due to Segal [Seg04]. In the foreword to the published version, Segal names §9 of the paper, where local fields are defined and studied, as its crucial part: it shows that the definition encodes the usual structure of local field operators and their vacuum expectation values. Here we only record how the objects of the earlier chapters arise as values of U.

Surface Σ	U_Σ	In this note
disk, one outgoing circle	the vacuum $\lvert 0\rangle \in \mathcal{H}$	Remark 2.3.1
annulus $\lvert q\rvert \le \lvert z\rvert \le 1$	$q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24}$	Proposition 3.1.2
sphere with n holes	a functional on $\mathcal{H}^{\otimes n}$	Definition 2.1.1
torus (annulus, ends sewn)	$\operatorname{Tr} U_{\mathrm{annulus}} = \mathcal{Z}(\tau)$	§3.1.2
annulus with one free boundary a	the boundary state $\lvert B_a\rangle$	Chapter 4
Σ versus $\bar{\Sigma}$	$U_{\bar{\Sigma}} = U_\Sigma^\dagger$	Definition 2.3.2

Table 5.1: Earlier chapters as values of the Segal functor. The last two rows use the extensions to surfaces with free boundary and to the Hermitian structure.

(i) *Sphere.* Shrinking a parametrized hole of a sphere to a point with a chosen local coordinate turns U into a field insertion. This is the state–operator correspondence, which in §2.2 became the identity map of F. The values of U on spheres with holes are thus a genus-zero correlation system. In this reading, (G1) is the independence of U_Σ from the labelling of holes, (G4) is functoriality under biholomorphisms, and (G5) is the factorization obtained by cutting along a circle that encloses two of the insertions. For chiral theories the equivalence between vertex operator algebras and the genus-zero part of such geometric axioms is a theorem [Hua97].

(ii) *Cylinder and torus.* The projective scalar of (S2) is the anomaly. On the flat cylinder it is the Casimir energy of Remark 3.1.3, which is why the annulus gives $q^{L_0-c/24}$ rather than q^{L_0}. Sewing the two ends of an annulus is a trace, and modular invariance (§3.7) is the statement that U depends on the torus and not on the chosen annulus.

(iii) *Reflection.* (S4) is the global form of the reflection of Definition 2.3.2. Gluing a surface to its mirror image along a circle gives $U_\Sigma^\dagger U_\Sigma \geq 0$. For a disk with insertions this is the positivity of the form (2.83).

5.4.2 Chiral versus full conformal field theory

(The author's section, moved from §5.2 together with the draft text that stood under it.)

[The two consistency conditions established in Chapters 2 and 3 — crossing symmetry on the sphere and modular invariance on the torus — are the genus-0 and genus-1 shadows of a single structure. The Moore–Seiberg programme shows that they generate all higher-genus consistency conditions, and that the resulting algebraic data is a modular tensor category.]

Object	Chiral CFT	Full CFT
Approach	bottom-up	top-down
Flavor	complex analytic	algebraic
Input	subset of correlators	conformal blocks
Output	conformal blocks	correlators

Table 5.2: Comparison of two approaches to rational conformal field theory (RCFT). In the algebraic approach, the properties of conformal blocks are provided in the form of modular tensor category (MTC).

A chiral theory assigns no number to a surface. It assigns a finite-dimensional space of conformal blocks $\mathcal{V}(\Sigma; a_1, \dots, a_n)$, one for each closed surface with n marked points labelled by primaries. These spaces satisfy:

(M1) they form a vector bundle over the moduli of $(\Sigma;\text{points})$ with a projectively flat connection, so the mapping class group acts projectively on each fibre;

(M2) for a simple closed curve γ, cutting along γ gives

$$\mathcal{V}(\Sigma; \boldsymbol{a}) \cong \bigoplus_x \mathcal{V}(\Sigma_\gamma; \boldsymbol{a}, x, x^*), \tag{5.9}$$

where Σ_γ is the cut surface with the two new points labelled x and its conjugate x^*. If γ separates, Σ_γ is disconnected and the right side is the tensor product of the spaces of its two components;

(M3) on the sphere, $\dim \mathcal{V}(S^2; a) = \delta_{a0}$, $\dim \mathcal{V}(S^2; a, b) = \delta_{b,a^*}$ and $\dim \mathcal{V}(S^2; a, b, c) = N_{ab}^{c^*}$.

A structure with these properties is a modular functor. At genus zero its vectors are the blocks (2.129), and the action of (M1) is generated by braiding and by the crossing matrix of §2.5. On the torus without insertions a basis is given by the characters χ_i, and the mapping class group acts through S and T (§3.7).

A full theory assigns to each surface a correlator, a sesquilinear combination $\sum_{i,j} M_{ij} \mathcal{F}_i \overline{\mathcal{F}_j}$ of blocks that is single-valued, i.e. invariant under the mapping class group. At genus 0 this is Proposition 2.5.3 together with the crossing equation; at genus 1 it is the modular invariance of Z_{ij} in §3.7. In the terms of Table 5.2, the full theory takes blocks as input and produces correlators. The hypothesis (F) of §5.2 is the statement about dimensions contained in (M2) and (M3).

Moore–Seiberg theorem (quoted). For a rational chiral theory, the projective action (M1) at every genus is generated by the fusing and braiding matrices on four-point blocks of the sphere, together with S and T on one-point blocks of the torus. All relations among them follow from finitely many polynomial equations: the pentagon and hexagon at genus 0, and at genus 1 the relations $(ST)^3 = S^2$ and $S^2 = C$, together with one relation that ties S on the one-point torus to fusing and braiding.

These are the polynomial equations of Moore and Seiberg [MS89]. In mathematical language they are the axioms of a modular tensor category, a name that grew out of exactly this work [Hua08a]. The status of the theorem should be stated precisely. In [MS89] it is derived within a physical framework that assumes the existence of the chiral theory on all surfaces. For the module category of a simple C_2-cofinite vertex operator algebra with completely reducible modules (and the further conditions stated there), the modular tensor category structure is a theorem [Hua08a].

Appendix A

Mathematical formulas

Analytic and number-theoretic facts used in the text, with proofs. The last two sections summarize the axioms of the note and list the results it uses without proof.

Theorem A.0.1 (Cauchy–Pompeiu)**.** *Let U be a domain in the complex plane with piecewise C^1 boundary ∂U. If $f : \overline{U} \to \mathbb{C}$ is a C^1 function which is not necessarily holomorphic, the following holds for every $\zeta \in U$.*

$$f(\zeta, \bar{\zeta}) = \frac{1}{2\pi i} \int_{\partial U} \frac{f(z, \bar{z})}{z - \zeta} dz + \frac{1}{2\pi i} \int_U \frac{\bar{\partial} f(z, \bar{z})}{z - \zeta} dz \wedge d\bar{z} \tag{A.1}$$

where the singularity in the last term is understood via principal value, i.e.

$$\frac{1}{2\pi i} \int_U \mathbf{p.v.} \left(\frac{1}{z - \zeta} \right) \frac{\partial f}{\partial \bar{z}}(z, \bar{z}) dz \wedge d\bar{z}. \tag{A.2}$$

Corollary A.0.2. *The Cauchy kernel $\mathbf{p.v.}(\pi z)^{-1}$, considered as a distribution, is a fundamental solution of the Cauchy-Riemann operator $\partial/\partial\bar{z}$.*

$$\frac{\partial}{\partial \bar{z}} \frac{1}{\pi(z - \zeta)} = \delta(z - \zeta). \tag{A.3}$$

Proposition A.0.3. *The coefficients of the following formal power series*

$$\left(\frac{1}{1 - x} \right)^n = (1 + x + x^2 + \cdots)^n = \sum_{m=0}^{\infty} a_m x^m \tag{A.4}$$

is given by

$$a_m = \left(\!\!\binom{n}{m} \!\!\right) = \binom{n + m - 1}{m} \tag{A.5}$$

Theorem A.0.4 (Poisson summation)**.** *Let $f \in \mathcal{S}(\mathbb{R})$ be a Schwartz function and let*

$$\hat{f}(\xi) := \int_{\mathbb{R}} f(x)\, e^{-2\pi i \xi x}\, \mathrm{d}x \tag{A.6}$$

denote its Fourier transform. Then for every $x \in \mathbb{R}$

$$\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{k\in\mathbb{Z}} \hat{f}(k)\, e^{2\pi i k x}, \tag{A.7}$$

both sides converging absolutely. In particular, at $x = 0$,

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{k\in\mathbb{Z}} \hat{f}(k). \tag{A.8}$$

Proof. Define the *periodisation* of f,

$$F(x) := \sum_{n\in\mathbb{Z}} f(x+n). \tag{A.9}$$

Step 1: F is a well-defined continuous function of period 1. Since f is Schwartz there is a constant C with $|f(y)| \le C(1+|y|)^{-2}$. For x in a compact interval $[-R, R]$ and $|n| > 2R$ we have $|x+n| \ge |n|/2$, so $|f(x+n)| \le C(1+|n|/2)^{-2}$, which is summable. Hence the series converges absolutely and uniformly on compacta, and F is continuous. Periodicity $F(x+1) = F(x)$ is immediate from reindexing $n \mapsto n-1$.

Step 2: the Fourier coefficients of F are the values $\hat{f}(k)$. By Step 1 we may interchange summation and integration:

$$\begin{aligned} c_k(F) := \int_0^1 F(x)\, e^{-2\pi i k x}\, \mathrm{d}x &= \sum_{n\in\mathbb{Z}} \int_0^1 f(x+n)\, e^{-2\pi i k x}\, \mathrm{d}x \\ &= \sum_{n\in\mathbb{Z}} \int_n^{n+1} f(y)\, e^{-2\pi i k (y-n)}\, \mathrm{d}y \\ &= \sum_{n\in\mathbb{Z}} \int_n^{n+1} f(y)\, e^{-2\pi i k y}\, \mathrm{d}y \;=\; \hat{f}(k), \end{aligned} \tag{A.10}$$

where the substitution $y = x + n$ was used and $e^{2\pi i k n} = 1$ because $k, n \in \mathbb{Z}$. The last step is the decomposition of $\mathbb{R}$ into unit intervals.

Step 3: the Fourier series of F converges to F. The Fourier transform maps $\mathcal{S}(\mathbb{R})$ to itself, so $|\hat{f}(\xi)| \le C'(1+|\xi|)^{-2}$ and $\sum_k |\hat{f}(k)| < \infty$. Therefore the series $\sum_k \hat{f}(k) e^{2\pi i k x}$ converges absolutely and uniformly, and its sum is a continuous 1-periodic function whose Fourier coefficients are, by Step 2, those of F. Two continuous periodic functions with the same Fourier coefficients coincide, whence (A.7). Evaluating at $x = 0$ gives (A.8). □

Remark A.0.5. *The Schwartz hypothesis is far stronger than necessary; it suffices that f be continuous with $|f(x)| + |\hat{f}(x)| \le C(1+|x|)^{-1-\delta}$ for some $\delta > 0$, which is exactly what Steps 1 and 3 use. The applications in this text involve Gaussians $f(x) = e^{-a(x+c)^2}$ with $\operatorname{Re} a > 0$ and $c \in \mathbb{R}$, which are Schwartz.*

Corollary A.0.6 (Rescaled form)**.** *For $\alpha > 0$ and $f \in \mathcal{S}(\mathbb{R})$,*

$$\sum_{n\in\mathbb{Z}} f(\alpha n) = \frac{1}{\alpha} \sum_{k\in\mathbb{Z}} \hat{f}\left(\frac{k}{\alpha}\right). \tag{A.11}$$

Proof. Apply Theorem A.0.4 to $g(x) := f(\alpha x)$, whose Fourier transform is $\hat{g}(\xi) = \alpha^{-1}\hat{f}(\xi/\alpha)$. □

Definition A.0.7 (Principal square root)**.** *Write* $H_+ := \{\, z \in \mathbb{C} \mid \operatorname{Re} z > 0 \,\}$ *for the open right half-plane. On the cut plane* $\mathbb{C} \setminus (-\infty, 0]$ *define*

$$\sqrt{z} := \exp\left(\tfrac{1}{2}\operatorname{Log} z\right), \qquad \operatorname{Log} z = \ln|z| + i \operatorname{Arg} z, \qquad \operatorname{Arg} z \in (-\pi, \pi). \tag{A.12}$$

This is the unique holomorphic branch with $\sqrt{z} > 0$ *for* $z > 0$*; its values satisfy* $\operatorname{Re}\sqrt{z} > 0$*.*

Lemma A.0.8. *The principal square root is multiplicative on the right half-plane: if* $z, w \in H_+$ *then* $zw \notin (-\infty, 0]$ *and*

$$\sqrt{z}\,\sqrt{w} = \sqrt{zw}. \tag{A.13}$$

Proof. $z, w \in H_+$ means $\operatorname{Arg} z, \operatorname{Arg} w \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so $\operatorname{Arg} z + \operatorname{Arg} w \in (-\pi, \pi)$. Hence this sum is itself the principal argument of zw, so $\operatorname{Log} z + \operatorname{Log} w = \operatorname{Log}(zw)$; exponentiating half of this identity gives the claim. □

Caution A.0.9. *Multiplicativity* fails *in general: for* $z = w = -1 + i\epsilon$ *one has* $\sqrt{z}\sqrt{w} = -\sqrt{zw}$*. Lemma A.0.8 is exactly what makes the modular transformation factors* $\sqrt{-i\tau}$ *compose without sign ambiguities, since all such arguments lie in* H_+*.*

Theorem A.0.10 (Gaussian integral with complex parameters)**.** *Let* $a \in H_+$ *and* $b \in \mathbb{C}$*. Then*

$$\int_{\mathbb{R}} e^{-ax^2 - bx}\, \mathrm{d}x = \sqrt{\frac{\pi}{a}}\, e^{\frac{b^2}{4a}}, \tag{A.14}$$

the square root being principal. The expression is unambiguous because $a \in H_+$ *implies* $\pi/a \in H_+$*, so the cut is never met.*

Proof. First note that H_+ is stable under inversion: if $a = re^{i\theta}$ with $|\theta| < \frac{\pi}{2}$ then $\pi/a = (\pi/r)e^{-i\theta}$, again in H_+. The right-hand side of (A.14) is therefore defined, and is holomorphic in a on H_+ and entire in b.

Step 1: $a > 0$ *and* $b \in \mathbb{R}$. Completing the square, $-ax^2 - bx = -a\left(x + \frac{b}{2a}\right)^2 + \frac{b^2}{4a}$, and the translation $x \mapsto x - \frac{b}{2a}$ is by a real number, so

$$\int_{\mathbb{R}} e^{-ax^2 - bx}\, \mathrm{d}x = e^{\frac{b^2}{4a}} \int_{\mathbb{R}} e^{-ax^2}\, \mathrm{d}x = \sqrt{\frac{\pi}{a}}\, e^{\frac{b^2}{4a}}. \tag{A.15}$$

Step 2: $a > 0$ *and* $b \in \mathbb{C}$. Fix $a > 0$. For b ranging over a compact set, say $|b| \le M$, the integrand is bounded by $e^{-ax^2 + M|x|}$, which is integrable; since $b \mapsto e^{-ax^2 - bx}$ is entire for each x, Fubini together with Morera's theorem shows that the left-hand side is an entire function of b. The right-hand side is entire in b as well, and by Step 1 the two agree on $\mathbb{R}$, a set with limit points. By the identity theorem they agree for all $b \in \mathbb{C}$.

Step 3: $a \in H_+$. Fix $b \in \mathbb{C}$. Let $K \subset H_+$ be compact and put $\delta := \min_{a \in K} \operatorname{Re} a > 0$. Then $\left|e^{-ax^2 - bx}\right| = e^{-(\operatorname{Re} a)x^2 - (\operatorname{Re} b)x} \le e^{-\delta x^2 + |b||x|}$, which is integrable and independent of $a \in K$. The same argument as in Step 2 shows the left-hand side is holomorphic in a on H_+. By Step 2 the two sides agree on $(0, \infty)$, which has limit points in the connected open set H_+, so they agree throughout H_+. □

Corollary A.0.11 (Normalization used for theta functions)**.** *Let $\tau \in \mathbb{H}$ and $N > 0$, and set $a := 2\pi i N/\tau$. Then $a \in H_+$ and*

$$\sqrt{\frac{\pi}{a}} = \sqrt{\frac{-i\tau}{2N}}, \qquad \text{with} \qquad \sqrt{-i\tau}\Big|_{\tau=i} = 1. \tag{A.16}$$

Proof. Writing $\tau = t_1 + it_2$ with $t_2 > 0$,

$$a = \frac{2\pi i N\bar{\tau}}{|\tau|^2} = \frac{2\pi N\,(t_2 + it_1)}{|\tau|^2}, \qquad \operatorname{Re} a = \frac{2\pi N t_2}{|\tau|^2} > 0. \tag{A.17}$$

Moreover $\pi/a = \tau/(2iN) = -i\tau/(2N)$, and $\operatorname{Re}(-i\tau) = t_2 > 0$ confirms $-i\tau \in H_+$. At $\tau = i$ we get $-i\tau = 1$, so the principal branch gives $+1$ there. □

Example A.0.12 (Consistency of $S^2 = 1$)**.** *The transformation law $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ applied twice gives*

$$\eta(\tau) = \sqrt{\tfrac{i}{\tau}}\,\sqrt{-i\tau}\,\eta(\tau). \tag{A.18}$$

Both i/τ and $-i\tau$ lie in H_+ for $\tau \in \mathbb{H}$, so Lemma A.0.8 applies and $\sqrt{i/\tau}\sqrt{-i\tau} = \sqrt{(i/\tau)(-i\tau)} = \sqrt{1} = 1$, as required. Had the two factors been allowed to leave H_+, this computation could have produced a spurious sign.

The next lemma supplies the product formulas for theta functions used in §3.4.4, and Euler's pentagonal number theorem used in Lemma 3.2.3. Write $P := \mathbb{Z} + \frac{1}{2}$ for the half-integers and $P_{>0}$, $P_{<0}$ for its positive and negative parts. For $q = e^{2\pi i\tau}$, $\tau \in \mathbb{H}$, powers are $q^a := e^{2\pi i a\tau}$.

Lemma A.0.13 (Jacobi triple product)**.** *For $\tau \in \mathbb{H}$ and $z \in \mathbb{C} \setminus \{0\}$,*

$$\prod_{m\geq 1} \left(1 - q^m\right)\left(1 + z\,q^{m-\frac{1}{2}}\right)\left(1 + z^{-1}q^{m-\frac{1}{2}}\right) = \sum_{n\in\mathbb{Z}} z^n q^{\frac{n^2}{2}}. \tag{A.19}$$

Proof. All products and sums below converge absolutely, so they may be expanded and rearranged freely. Expanding,

$$\prod_{m\geq 1} \left(1 + z\,q^{m-\frac{1}{2}}\right)\left(1 + z^{-1}q^{m-\frac{1}{2}}\right) = \sum_{A,B} z^{|A|-|B|}\,q^{\Sigma A + \Sigma B}, \tag{A.20}$$

where A and B run over the finite subsets of $P_{>0}$ and ΣA is the sum of the elements of A. Comparing coefficients of z^n, it suffices to show that for every $n \in \mathbb{Z}$

$$\sum_{|A|-|B|=n} q^{\Sigma A + \Sigma B} = q^{\frac{n^2}{2}} \prod_{m\geq 1} \frac{1}{1 - q^m}. \tag{A.21}$$

Encoding. To (A, B) assign $M := \big(P_{<0} \setminus (-B)\big) \cup A \subset P$. This is a bijection onto the subsets $M \subset P$ that differ from $P_{<0}$ in finitely many elements, with inverse $A = M \cap P_{>0}$, $B = -(P_{<0} \setminus M)$. Put $Q(M) := |A| - |B|$ and $E(M) := \Sigma A + \Sigma B$. Choose $K \in \mathbb{N}$ so large that M contains every $s \in P$ with $s < -K$. Then

$$Q(M) = \big|M \cap [-K, \infty)\big| - K, \qquad E(M) = \sum_{s\in M,\ s\geq -K} s \;-\; \sum_{s\in P,\ -K\leq s<0} s, \tag{A.22}$$

since the elements of M in $[-K, 0)$ cancel between the two sums.

Ground states. For $n \in \mathbb{Z}$ let $M_n := \{s \in P : s < n\}$. Then $Q(M_n) = n$ and $E(M_n) = \frac{1}{2} + \frac{3}{2} + \cdots + (|n| - \frac{1}{2}) = n^2/2$; for $n \geq 0$ the added elements are $\frac{1}{2}, \ldots, n - \frac{1}{2}$, and for $n < 0$ the removed ones are $-\frac{1}{2}, \ldots, -(|n| - \frac{1}{2})$.

Partitions. Let $Q(M) = n$ and list the elements of M in decreasing order, $s_1 > s_2 > \cdots$. By (A.22), $M \cap [-K, \infty) = \{s_1, \ldots, s_{K+n}\}$, and $s_k = n - k + \frac{1}{2}$ for $k > K + n$. Put $\lambda_k := s_k - (n - k + \frac{1}{2})$. Since $s_k - s_{k+1} \geq 1$, we have $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$, and $\lambda_k = 0$ for $k > K + n$, so λ is a partition. Conversely, every partition λ defines a strictly decreasing sequence $s_k := n - k + \frac{1}{2} + \lambda_k$, hence a set M of charge n. This is a bijection between sets of charge n and partitions. Applying (A.22) to M and M_n with the same K,

$$E(M) - E(M_n) = \sum_{k=1}^{K+n} \Big(s_k - \big(n - k + \tfrac{1}{2}\big) \Big) = |\lambda|. \tag{A.23}$$

Hence the left side of (A.21) is $q^{n^2/2} \sum_\lambda q^{|\lambda|}$. A partition is determined by how often each $m \geq 1$ occurs as a part, so $\sum_\lambda q^{|\lambda|} = \prod_{m \geq 1} \sum_{k \geq 0} q^{mk} = \prod_{m \geq 1} (1 - q^m)^{-1}$. □

Remark A.0.14. *The proof is the Fock-space count for a pair of fermions with half-integer modes: A and B are the occupied particle and hole modes, M is the filled sea $P_{<0}$ changed in finitely many places, Q is the charge, and E is the eigenvalue of L_0. The bijection with partitions says that the states of charge n are the excitations of the lowest one, M_n.*

Corollary A.0.15 (Euler's pentagonal number theorem)**.**

$$\prod_{m \geq 1} (1 - q^m) = \sum_{k \in \mathbb{Z}} (-1)^k q^{\frac{k(3k-1)}{2}}. \tag{A.24}$$

Proof. In (A.19) replace τ by 3τ and set $z = -q^{-1/2}$. The factors become $(1 - q^{3m})(1 - q^{3m-2})(1 - q^{3m-1})$, whose product over $m \geq 1$ is $\prod_{m \geq 1}(1 - q^m)$, and the right side becomes $\sum_n (-1)^n q^{(3n^2 - n)/2}$. □

A.1 Axioms at a glance

The table lists every axiom introduced in the draft treatment, where it is stated, and which results use it. A result not listed under an axiom does not use it. In particular, nothing before Definition 2.3.2 uses positivity.

Axiom	Content	Used in
(G1)	locality (Koszul signs for fermions, Remark 2.1.3)	Thm 2.1.6 ($C_{12} = C_{21}$), Thm 2.2.1, Rem. 2.1.10
(G2)	identity field $\mathbb{I}$	Thm 2.2.1 ($\lvert 0\rangle = \mathbb{I}$), Rem. 2.2.4, Rem. 2.3.1
(G3)	derivatives $\partial, \bar{\partial}$	Lemma 2.1.4, Rem. 2.2.4, Cor. 2.3.5
(G4)	Möbius covariance of quasi-primaries	Lemma 2.1.4, Thms 2.1.6, 2.1.8, Cor. 2.1.12, Thm 2.2.8 (decay at ∞)
(G5)	convergent operator product expansion	Thm 2.2.1, Cor. 2.2.3, (2.118), normal ordering
(T)	$T \in F^{\mathrm{qp}}_{(2,0)}$, $\bar{\partial}T = 0$; $L_{-1} = \partial$, $L_0 = h$	Def. 2.2.6, Thm 2.2.8, Prop. 3.1.2, Prop. 2.4.1
(U1)	reflection positivity, single insertions	Prop. 2.3.4, Cor. 2.3.5
(U2)	adjoint of $\phi(z)$ is (2.82)	Thm 2.3.9 ($L_n^\dagger = L_{-n}$), Cor. 3.1.5
(U3)	positive definiteness	Cor. 2.3.5, Cor. 3.1.5
(F1)–(F2)	factorization of block counts (Def. 5.2.1)	Prop. 5.3.1, Ex. 5.3.3

A.2 Results used without proof

Every statement of the note is meant to be one of three things: proved in the note, an axiom or hypothesis listed in §A.1, or one of the results below. For each result we record where it is stated, what uses it, and whether an argument of the note depends on it.

- **Kac determinant** (Theorem 2.3.12). *Used for* the Kac table and the spectrum of minimal models (§2.4.4), hence the resolution below and everything built on it. *Load-bearing* for general minimal models. *Checked in the note* at level 2 (Proposition 2.3.13, Corollary 2.3.15) and level 3 (Proposition 2.3.18, Corollary 2.3.19). *Not needed* for the Ising model (Remark 3.4.23).

- **Exactness of the resolution** (3.113) (Feigin–Fuks). *Used for* the characters (Proposition 3.5.7), hence Theorem 3.6.5, Corollary 3.6.6, the Verlinde formula and the Cardy states (4.54) for minimal models, and Proposition 5.3.1 for minimal models other than Ising. *Load-bearing*. *Not needed* for the Ising model, whose characters and S-matrix come from the free fermion (Lemma 3.4.21, Proposition 3.4.22).

- **Diagonalization hypothesis** (3.148). Not a quoted theorem but an assumption (Remark 3.6.12). *Used for* the Verlinde formula (3.149) as a statement about a given theory, hence the Cardy condition and Proposition 5.3.1. *Checked* by integrality for minimal models and *verified* for the Ising model (Remark 3.4.23).

- **Friedan–Qiu–Shenker** (Remark 2.4.13). Classification of unitary minimal models. *Not used* in any argument. The part needed to exclude a negative weight, as for Lee–Yang, is Corollary 2.3.5.

- **Classification of modular invariants** (Notes to Chapter 3). *Not used.* The uniqueness statement actually used, Theorem 3.7.4, is proved in the note.
- **Affleck–Ludwig g-theorem** (Theorem 4.4.8). *Not used* in any argument; it is stated for interpretation.

Results quoted in the Notes at the end of each chapter are outside the logical line and are not listed here.

Bibliography

[Aff86] Ian Affleck. "Universal term in the free energy at a critical point and the conformal anomaly". In: *Physical Review Letters* 56.7 (1986), pp. 746–748. DOI: 10.1103/PhysRevLett.56.746.

[AMT24] Maria Stella Adamo, Yuto Moriwaki, and Yoh Tanimoto. *Osterwalder–Schrader axioms for unitary full vertex operator algebras*. 2024. arXiv: 2407.18222 [math-ph].

[BCN86] H. W. J. Blöte, John L. Cardy, and M. P. Nightingale. "Conformal invariance, the central charge, and universal finite-size amplitudes at criticality". In: *Physical Review Letters* 56.7 (1986), pp. 742–745. DOI: 10.1103/PhysRevLett.56.742.

[BLT13] Ralph Blumenhagen, Dieter Lüst, and Stefan Theisen. *Basic Concepts of String Theory*. Springer Berlin Heidelberg, 2013. ISBN: 9783642294976. DOI: 10.1007/978-3-642-29497-6.

[CIZ87a] A. Cappelli, C. Itzykson, and J.-B. Zuber. "Modular invariant partition functions in two dimensions". In: *Nuclear Physics B* 280 (1987), pp. 445–465. ISSN: 0550-3213. DOI: 10.1016/0550-3213(87)90155-6.

[CIZ87b] A. Cappelli, C. Itzykson, and J.-B. Zuber. "The A-D-E classification of minimal and $A_1^{(1)}$ conformal invariant theories". In: *Communications in Mathematical Physics* 113.1 (1987), pp. 1–26. ISSN: 1432-0916. DOI: 10.1007/BF01221394.

[DMS97] Philippe Di Francesco, Pierre Mathieu, and David Sénéchal. *Conformal Field Theory*. Springer New York, 1997. ISBN: 9781461222569. DOI: 10.1007/978-1-4612-2256-9.

[GG00] Matthias R. Gaberdiel and Peter Goddard. "Axiomatic Conformal Field Theory". In: *Communications in Mathematical Physics* 209 (2000), pp. 549–594. DOI: 10.1007/s002200050031. arXiv: hep-th/9810019.

[HK07] Yi-Zhi Huang and Liang Kong. "Full Field Algebras". In: *Communications in Mathematical Physics* 272 (2007), pp. 345–396. DOI: 10.1007/s00220-007-0224-4. arXiv: math/0511328.

[Hua08a] Yi-Zhi Huang. "Rigidity and modularity of vertex tensor categories". In: *Communications in Contemporary Mathematics* 10 (2008), pp. 871–911. arXiv: math/0502533.

[Hua08b] Yi-Zhi Huang. "Vertex operator algebras and the Verlinde conjecture". In: *Communications in Contemporary Mathematics* 10 (2008). arXiv: math/0406291.

[Hua97] Yi-Zhi Huang. *Two-Dimensional Conformal Geometry and Vertex Operator Algebras*. Vol. 148. Progress in Mathematics. Birkhäuser, 1997.

[Kat87] Akishi Kato. "Classification of modular invariant partition functions in two dimensions". In: *Modern Physics Letters A* 2.8 (1987), pp. 585–600. ISSN: 1793-6632. DOI: 10.1142/S0217732387000745.

[Kus25] Yuya Kusuki. *Modern Approach to 2D Conformal Field Theory*. 2025. arXiv: 2412.18307 [hep-th]. URL: https://arxiv.org/abs/2412.18307.

[Mac77] Gerhard Mack. "Convergence of operator product expansions on the vacuum in conformal invariant quantum field theory". In: *Communications in Mathematical Physics* 53 (1977), pp. 155–184. DOI: 10.1007/BF01609130.

[Mor20] Yuto Moriwaki. *Full vertex algebra and bootstrap – consistency of four point functions in 2d CFT*. 2020. arXiv: 2006.15859 [math.QA].

[MS89] Gregory Moore and Nathan Seiberg. "Classical and quantum conformal field theory". In: *Communications in Mathematical Physics* 123 (1989), pp. 177–254. DOI: 10.1007/BF01238857.

[OS73] Konrad Osterwalder and Robert Schrader. "Axioms for Euclidean Green's functions". In: *Communications in Mathematical Physics* 31 (1973), pp. 83–112.

[Pap+12] Duccio Pappadopulo et al. "OPE Convergence in Conformal Field Theory". In: *Physical Review D* 86 (2012), p. 105043. DOI: 10.1103/PhysRevD.86.105043. arXiv: 1208.6449.

[Sch08] Martin Schottenloher. *A Mathematical Introduction to Conformal Field Theory*. 2nd ed. Vol. 759. Lecture Notes in Physics. Springer Berlin Heidelberg, 2008. DOI: 10.1007/978-3-540-68628-6.

[Seg04] Graeme Segal. "The definition of conformal field theory". In: *Topology, Geometry and Quantum Field Theory*. Ed. by Ulrike Tillmann. Vol. 308. London Mathematical Society Lecture Note Series. Cambridge University Press, 2004, pp. 421–577. DOI: 10.1017/CBO9780511526398.019.

[Ver88] Erik Verlinde. "Fusion rules and modular transformations in 2D conformal field theory". In: *Nuclear Physics B* 300 (1988), p. 360.

[Zhu96] Yongchang Zhu. "Modular invariance of characters of vertex operator algebras". In: *Journal of the American Mathematical Society* 9.1 (1996), pp. 237–302. DOI: 10.1090/S0894-0347-96-00182-8.